

Credit Card Customer NPV Estimation Component: Final Submission Proposal for a Replacement Valuation Module

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Abstract

This proposal asks the panel to approve the design and phase-1 build of a replacement credit-card customer NPV component. The component is a callable valuation module used by existing marketing, underwriting, line, retention, product-conversion, finance, and risk systems. It does not own those decision systems. It owns the value calculation: estimating the incremental, risk-adjusted value of a card action relative to a declared baseline, under explicit assumptions about scenario, funding, capital, rewards, losses, cannibalization, terminal value, relationship value, and downstream account-management policy.

The replacement is needed because common proxies are not enough. Response, approval, activation, first-card spend, and booked-account risk each capture one piece of the economics, but none is the same as all-bank incremental NPV. The literature supports the mechanisms that make this true: employment, liquidity, credit terms, rewards, payment choice, customer duration, and credit risk affect future spend, balances, payments, attrition, and losses. Public literature and supervisory sources establish structure, but they do not supply this bank's activation lift, cannibalization rate, wallet-capture elasticity, loss response, funding and capital economics, cost-to-serve, or relationship spillover. Those magnitudes must be estimated and governed using the bank's own data, experiments, quasi-experiments, finance/risk reconciliations, and approved external benchmarks.

The proposed design is modular but not fragmented. It decomposes value into funnel response, activation, use, wallet routing, balances, payments, PPNR, PD/LGD/EAD, rewards, servicing, fraud, funding, capital, tax, attrition, terminal value, and separately labeled relationship value. It assembles those modules into a monthly state-space valuation engine that can simulate a declared downstream policy and return decomposed NPV, uncertainty, status, lineage, and allowed-use labels. The dynamic model is presented as a disciplined forward-simulation object for fixed-policy valuation first. Offline policy evaluation and constrained reinforcement-learning research are later extensions, not day-one production claims.

The proposal also states what evidence the component may consume. A statistically clean experiment is not admissible production evidence unless the population, action set, customer-impact controls, compliance constraints, fair-lending and adverse-action boundaries, privacy limits, servicing capacity, complaint monitoring, and reputation guardrails are approved for the decision context. Predictive, observationally adjusted, quasi-experimental, randomized, and scenario-only evidence receive different output labels. The first production slice should therefore be deliberately narrow: a controlled batch acquisition or prescreen use case with one offer family, frozen feature snapshots, preserved holdouts or randomized contact-suppress cells, shadow scoring against the current component, and finance/risk reconciliation before cutover.

How to Read This Proposal

This document is long because the problem is not a single-model exercise. A panel review has to decide whether the proposed component is economically well-defined, statistically identifiable where

it claims to be causal, financially reconcilable, implementable inside the bank’s existing systems, and safe to introduce through a controlled replacement path. The proposal is therefore written as one connected argument: define the valuation object, ground the mechanisms in literature and bank evidence, specify the dynamic state-space assembly, state the identification and experimental evidence requirements, define validation and model-risk controls, and then specify how existing systems call the component.

The main text follows a repeated pattern. It starts with the business problem, states what the literature or bank evidence supports, translates that evidence into a model requirement, states the data or experiment needed to estimate the bank-specific magnitude, and then gives the resulting component design choice. Material whose primary purpose is reference, auditability, or contract testing is written as reviewable contract prose in the appendices. The appendices are human-facing parts of the proposal, not storage areas for dense material.

The proposal deliberately avoids reducing the problem to a short executive story. Difficult material is made explicit because omission would create false simplicity. The figures and reader boxes are included to make the structure easier to hold in mind: component boundary, valuation decomposition, data sources, model dependencies, dynamic belief-state simulation, causal identification, experimental admissibility, evidence grading, uncertainty reporting, first-slice rollout, and the staged path from fixed-policy valuation to possible later optimization.

Visual Reader Guide

The figures in this proposal are not decorative. Each one answers a concrete reviewer question.

Component boundary and valuation object

- Figure 1 shows what the replacement component owns and what caller systems still own.
- Figure 2 shows how incremental NPV decomposes into PPNR, losses, costs, capital, tax, terminal value, and relationship value.
- Figure 4 gives a numerical-looking waterfall so readers can see why high spend can still be low value.

Data, modules, and dynamic state

- Figure 3 shows how internal data, external public/licensed data, and assumption bundles enter the component.
- Figure 6 gives the high-level component architecture.
- Figure 7 shows the directional dependencies among submodels before cash-flow compilation.
- Figure 8 shows the dynamic belief-state loop used for fixed-policy simulation.
- Figure 15 shows how parameter, scenario, policy, and terminal-value uncertainty become reported intervals, rank-stability diagnostics, and use limits.

Funnel, evidence, and causal validity

- Figure 11 explains why prospect, applicant, booked, and active-account models are not interchangeable.
- Figure 10 shows the main endogeneity channels in targeting and new-card valuation.
- Figure 13 and Figure 14 show how evidence strength affects output labels.

Experiment, macro scenario, and optimization roadmap

- Figure 12 shows why a statistically clean experiment can still be inadmissible for production use.
- Figure 5 shows how macro, employment, and rate conditions move spend, balances, payments, losses, funding, capital, and attrition jointly.
- Figure 16 shows the proposed first production slice from caller request through governed NPV response.
- Figure 9 shows why fixed-policy valuation comes before offline policy evaluation and any later optimization.

The proposal is still for one component, not for a new bank platform. Marketing, underwriting, account management, finance, and risk systems keep their decision rights. The replacement component owns the valuation calculation: it estimates incremental NPV under a declared decision context, baseline, scenario, downstream-policy assumption, and allowed-use scope.

The intended first slice is deliberately concrete: a controlled batch acquisition or prescreen use case with one offer family, frozen feature snapshots, holdouts or randomized contact-suppress cells, shadow scoring against the current component, and finance/risk reconciliation before cutover. Later uses in underwriting, line assignment, existing-account treatment, account closure, CECL, stress testing, or official finance reporting require separate decision-context approval.

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1 Executive Proposal and Decision Request

The bank already makes many credit-card decisions that implicitly depend on customer value. Marketing teams decide whom to contact, what offer to present, and how much budget to spend. Existing-client programs decide whether a new card is incremental to the relationship or mostly cannibalizes the bank’s old card. Line, pricing, retention, and conversion programs decide whether a customer action creates enough additional value to justify incremental risk, funding, rewards, servicing cost, and capital. The current valuation component is therefore not a peripheral score. It sits inside a chain of decisions where small errors in estimated value can change campaign selection, portfolio composition, loss exposure, and reported business economics.

This proposal asks the panel to approve the design and phase-1 build of a replacement NPV valuation component. The component will sit behind existing bank systems. It will not choose campaigns, approve applications, assign credit lines, book finance results, or execute account-management treatments. Those systems retain their decision rights. The replacement component’s job is to produce a disciplined valuation object that those systems can call: incremental NPV relative to the relevant baseline, decomposed into the cash-flow and risk drivers that make the number explainable, reconcilable, and testable.

The phase-1 request is intentionally concrete. We propose one controlled batch acquisition or prescreen use case, one offer family, one approved valuation semantics bundle, frozen feature snapshots, shadow scoring against the current component, portfolio-level reconciliation, and an experiment design with campaign holdouts or randomized contact-suppress cells. This is the right first slice because it is valuable to the business, does not require real-time underwriting dependency, and gives the bank a clean place to compare the new component against the old one before expanding to higher-stakes uses.

The design principle is straightforward: caller systems own actions; the NPV component owns valuation. A caller may ask, for example, whether contacting a prospect has positive incremental value, whether an existing client should be offered a second card, or which line option changes expected value. The component should return the value estimate, its decomposition, the policy and scenario assumptions used to produce it, and any use limits attached to that estimate. Separate approval should be required before the same component output is used for underwriting, adverse-action reasoning, account closure, collections, CECL, stress testing, official finance reporting, broad account-management automation, or any other decision context with a different population, baseline, regulation, or consequence.

Each valuation call must therefore answer three questions:

1. **What decision context is being valued?** Targeting, existing client cross-sell, underwriting, line assignment, offer choice, product conversion, retention, and portfolio analysis have different populations, data availability, action sets, and counterfactuals.
2. **What valuation semantics and assumption bundle are being used?** A number called “NPV” is not useful unless horizon, discounting, funding, capital, tax, rewards, costs, terminal value, relationship value, cannibalization, macro scenario, and downstream policy assumptions are explicit and versioned.
3. **What can the caller safely do with the output?** The component may return values, decompositions, warnings, statuses, and non-binding option rankings. It should not itself mail a customer, approve an application, choose a line, close an account, or book a finance result.

The central valuation object is incremental value:

$$\Delta\text{NPV}_i(a; d, s, \pi) = \mathbb{E}[\text{NPV}_i(a; d, s, \pi) - \text{NPV}_i(a_0; d, s, \pi) \mid \mathcal{I}_{id}], \quad (1)$$

where i is the customer, prospect, account, household, application, or account-action pair being scored; a is a candidate action; a_0 is the decision-specific baseline; d is the decision context; s is the scenario and valuation-assumption bundle; π is the assumed downstream policy; and $\mathcal{I}_{id}$ is the information legally and operationally available at that decision point.

This definition is more than notation. It is the difference between valuing a bank action and ranking customers by a convenient proxy. A high response score is not necessarily high value. A high first-spend account may simply move spend from another card already issued by the bank. A high-line action may raise interchange and finance charges while also raising exposure, funding cost, capital, and expected loss. A retention or product-conversion offer may save value that would otherwise leave, or it may give rewards and pricing concessions to customers who would have stayed. For this reason, the replacement component should return a tagged and decomposed value object rather than a bare scalar.

Card wallet accounting is the second core idea. The bank observes its own new-card transactions, old-card transactions, deposits, debit activity, balances, payments, fees, and losses. It observes competitor-card spend, competitor revolving balances, card-on-file position, and outside payment routing only through partial signals such as bureau attributes, balance-transfer records, payoff records, merchant or network aggregates, account-aggregation samples, and experiments. The modeling question is therefore not simply “how much will the customer spend on the new card?” The question is how the bank action changes the allocation of spend, revolving balances, payments, losses, rewards cost, and relationship activity across the new card, any old bank card, other bank products, and competitors. That distinction is what separates outside wallet size, movable wallet, causal response, cannibalization, and true all-bank NPV.

The proposal is organized around five empirical and operational realities. First, NPV is latent and assumption-sensitive: it depends on long-horizon spend, revolve, attrition, losses, funding, capital, cost, and terminal-value assumptions. Second, card value is behaviorally heterogeneous: spend, revolve, utilization, payment behavior, rewards response, attrition, losses, funding, and capital interact rather than collapsing to one response or risk score. Third, the population changes through the funnel from prospect to applicant to approved account to booked and active account, so the available data and valid estimand change. Fourth, the objective changes by decision: contact-versus-suppress, approve-versus-decline, line-option choice, product conversion, and retention are different counterfactuals. Fifth, decisions are sequentially interdependent: upstream acquisition and approval choices change downstream account populations, while future line, pricing, retention, collections, and conversion policies change the economics assumed upstream. These realities drive the component architecture: decision-context declarations, valuation-semantics identifiers, decomposed outputs, statuses, lineage, and use-case-specific validation.

2 The Valuation Problem and Decision Semantics

The first source of confusion in card NPV is that several different questions look similar in operational systems. A response model asks who is likely to respond. A spend model asks how much activity may appear on the new card. A risk model asks how likely the account is to default. The valuation component has to ask a harder question: what incremental value does the bank create by taking this action rather than the relevant baseline action, after accounting for spend source, revolve, payments, losses, funding, capital, rewards, costs, attrition, and downstream policy?

This section defines the valuation object before the proposal turns to literature and architecture. The point is to give every later module the same contract: a module matters only insofar as it changes the path of incremental cash flows or the uncertainty around that path.

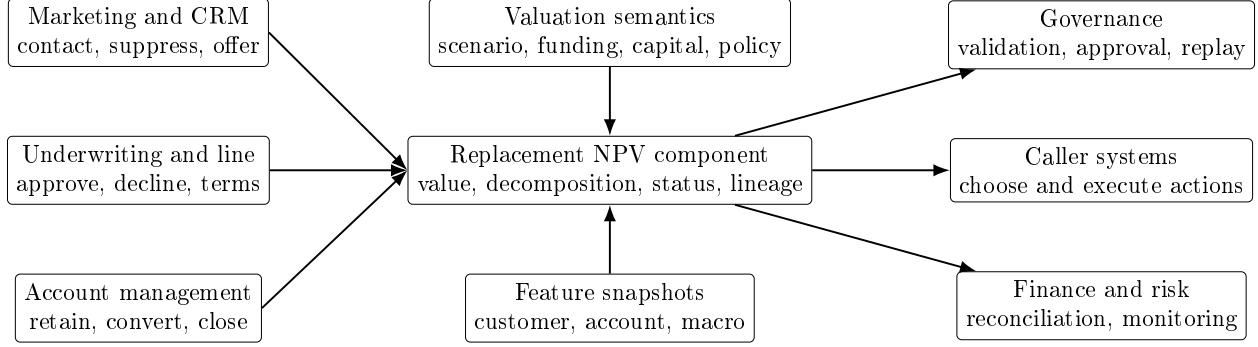


Figure 1: System context: the proposal replaces the valuation component, not the caller systems.

2.1 From Evidence to Component Design

The replacement effort starts with a practical inventory of the current component. Before cutover, the bank needs to know who consumes the current value measure, which fields and thresholds they use, how fallbacks work, which model versions and assumptions are live, where manual overrides occur, what latency classes are required, and which downstream reports depend on the old output. That inventory is not clerical work. It defines the migration surface and prevents the new component from silently changing a business rule that lives in another system.

The inventory should also surface the economic weaknesses the replacement is meant to fix. The most important candidates are:

- response, approval, activation, or sales volume used as value proxies;
- unclear counterfactuals for contact, approval, line, retention, or product-conversion decisions;
- weak cannibalization treatment for existing-client new-card offers;
- one value definition reused across different funnel stages;
- missing decomposition into spend, balances, payments, PPNR, losses, funding, capital, rewards, fraud, servicing, and terminal value;
- stale discount, funding, capital, tax, cost, or reward assumptions;
- output fields that are difficult to reconcile to finance, risk, or model-risk views.

The literature survey enters the proposal at this point. It is not included as background decoration. Each literature block is used to force a design choice. Evidence on unemployment, income shocks, liquidity, and credit limits says that purchase volume, payments, balances, attrition, and losses should be modeled as state-dependent paths rather than fixed averages [41, 42, 50]. Evidence on payment choice, rewards, and card usage says that issuing a card, activating it, using it once, using it repeatedly, and shifting wallet share are different outcomes [2, 53, 67]. Customer lifetime value and duration models say that dormancy, attrition, reactivation, and terminal value cannot be added as an afterthought [20, 35, 36, 72]. Credit-risk, accounting, and supervisory sources say that value must be decomposed into balances, PPNR, PD, LGD, EAD, expected loss, funding, capital, costs, and scenario assumptions [9, 15, 16]. Causal-inference literature says that a predictive score and an incremental treatment effect are different objects [5, 45, 47].

That is the organizing spine of the proposal. For every material mechanism, the document asks five questions. What does the literature or bank evidence support? What model requirement follows

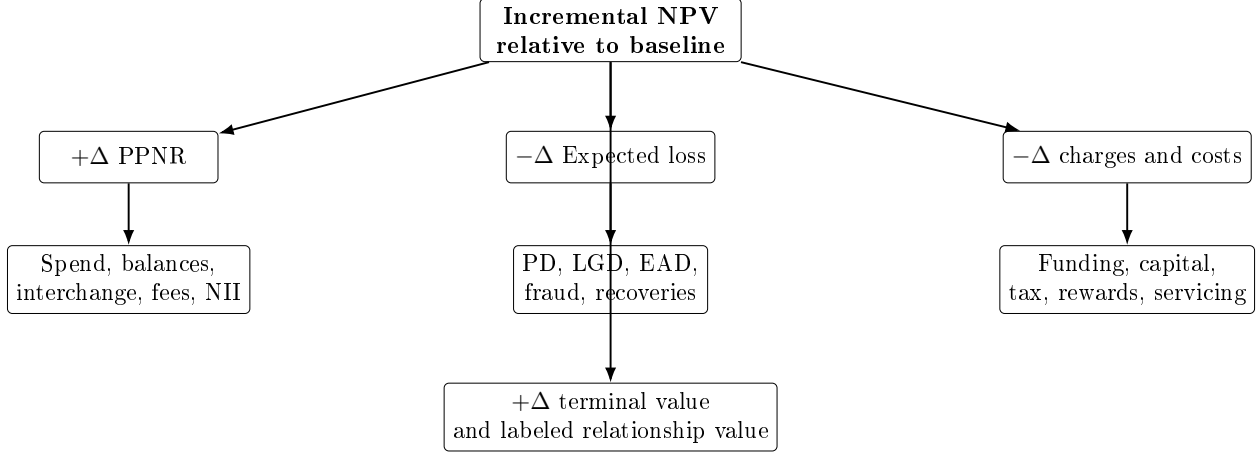


Figure 2: Cash-flow decomposition: every model module must enter NPV through an explicit contribution, loss, cost, capital, tax, terminal, or separately labeled relationship-value path.

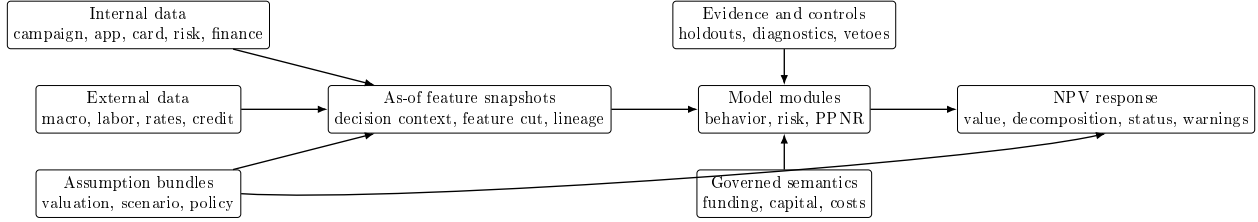


Figure 3: Data and source map. Internal data identify issuer-specific behavior and economics; external data support scenarios, controls, benchmarks, and priors; assumption bundles define the valuation convention. All three must be preserved as replayable, as-of inputs.

from it? What bank data, experiment, or quasi-experiment is needed to estimate the magnitude for this issuer? Which cash-flow or risk decomposition item changes because of it? What validation or monitoring result would stop the component from being used in production?

This distinction matters because public evidence and bank evidence have different jobs. Public literature can justify mechanisms and modeling structure; public and licensed data can support scenarios, controls, benchmarks, priors, and monitoring; regulatory and supervisory sources can discipline accounting, loss, PPNR, capital, and scenario language. None of those sources can tell this bank its own activation lift, wallet capture, cannibalization, reward elasticity, line-response curve, payment response, loss elasticity, servicing cost, or terminal relationship value. Those quantities must come from the bank’s own historical data, experiments, quasi-experiments, finance/risk reconciliations, and controlled migration.

The later specification sections translate this evidence into component interfaces. The component must expose decision context because the counterfactual changes by use case. It must expose valuation semantics because two NPV numbers are not comparable unless horizon, discounting, funding, capital, tax, rewards, cost, terminal value, scenario, and downstream policy are the same. It must expose decomposition because a profitable-looking customer can hide loss, funding, capital, rewards, or cannibalization costs. It must expose status and lineage because callers need to know when a value is causal, predictive, scenario-only, stale, out of support, or unsupported for the requested use. These are engineering requirements derived from the economics, not paperwork appended after the model is built.

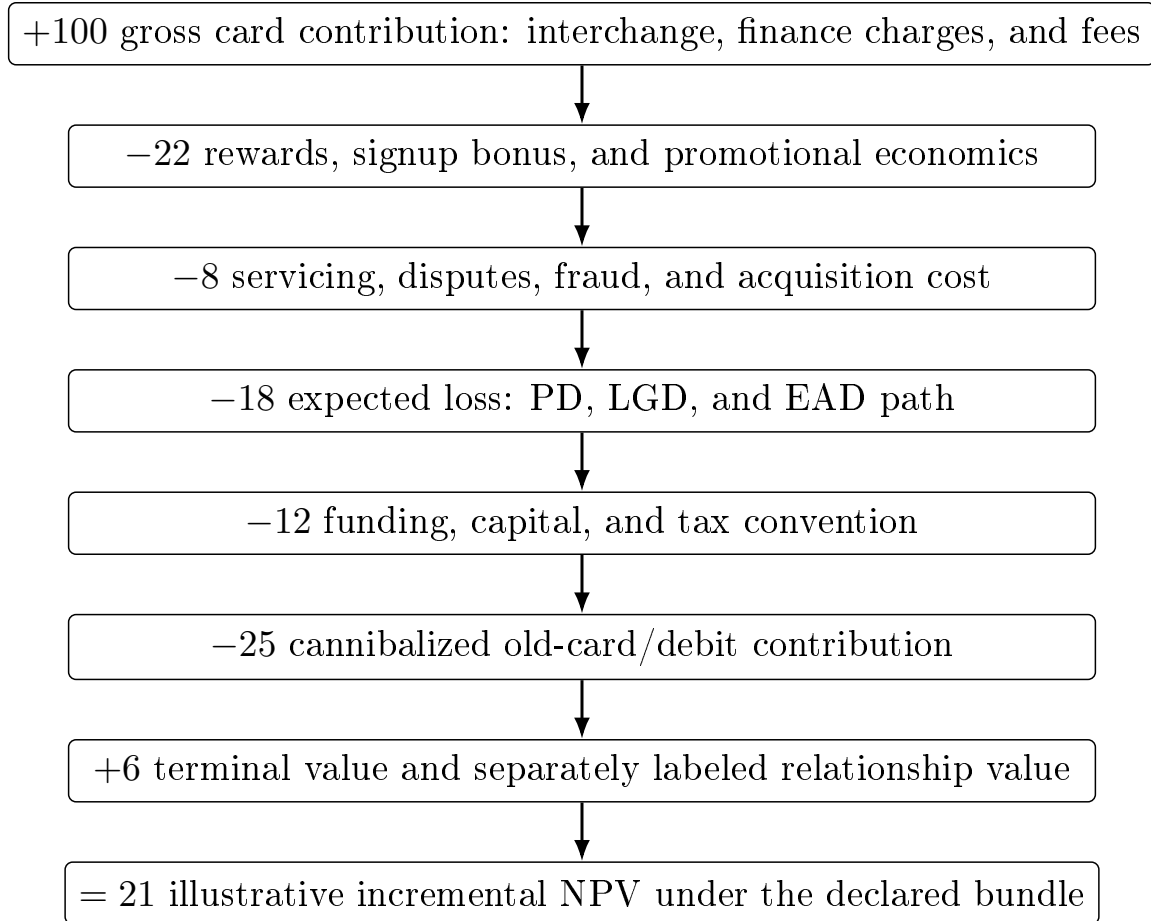


Figure 4: Illustrative NPV waterfall. The numbers are placeholders, not estimates. The purpose is to show the accounting logic: high observed spend can produce modest incremental value after rewards, losses, funding, capital, costs, cannibalization, and terminal assumptions.

2.2 Reader Map for the Proposal

The document is long because the problem is not a single-model exercise. A panel review has to decide whether the proposed component is economically well-defined, statistically identifiable where it claims to be causal, financially reconcilable, implementable inside the bank’s systems, and safe to introduce through a controlled replacement path. The proposal is therefore written as one connected argument: define the valuation object, justify the state and cash-flow modules, specify how the component will be called, and state what evidence is required before each output may be used.

The first pass through the document should focus on the decision request and the valuation object. Sections 1 and 2 explain why the bank needs a replacement component, what incremental NPV means, why the baseline and downstream policy must be declared, and how wallet accounting separates all-bank value from new-card activity. This pass answers the question: what exactly is the component estimating?

The second pass should focus on mechanisms and architecture. Sections 3 and 4 connect the literature to the component design: macro state, employment, liquidity, payment choice, rewards, balances, losses, PPNR, attrition, promotion decay, product conversion, and relationship value. This

pass answers the question: why do these modules belong in the valuation component, and how do they connect to cash flows?

The third pass should focus on dynamics, identification, and validation. Sections 5, 6, and 7 specify the stochastic state-space backbone, selection and endogeneity controls, experimental and quasi-experimental evidence program, uncertainty propagation, finance/risk reconciliation, and model-risk gates. This pass answers the question: which parts are causal, which are forecasts, which are scenario assumptions, and what would veto use?

The fourth pass should focus on replacement and caller integration. Section 8 explains how existing bank systems should call the component, what the component returns, how the first production slice should be staged, and how cutover should be controlled. The appendices are then reference tools: notation, data checklists, operating contracts, source-use boundaries, and worked evidence records. They are useful when a detailed reviewer presses a claim, but the main argument should be readable without reconstructing it from dense tables.

2.3 Valuation Object, State Space, and Wallet Accounting

This section fixes the notation used throughout the proposal. The document combines valuation, lifecycle modeling, causal inference, and operating contracts, so some local symbols in the literature review are inevitably local. The global convention is as follows. Customer, account, prospect, household, application, or account-action unit is indexed by i ; time is monthly unless stated otherwise and indexed by t ; action is a ; the decision-specific baseline is $a_0(d)$; decision context is d ; scenario and valuation-assumption bundle is s ; downstream account-management policy is π ; and the decision-time information set is $\mathcal{I}_{it}$ or X_{it}^d . When the experimental section uses assignment variables, those variables are local to that section and should not be confused with lifecycle state variables. In a final model specification the component team should use distinct implementation names such as `assignment_flag`, `treatment_received`, `lifecycle_state`, `delinquency_state`, `valuation_horizon`, and `observed_horizon`.

The notation is intentionally economic rather than database-specific. For implementation, every symbol must resolve to a field, model output, assumption bundle, or lineage object. For example, π is not a regression coefficient; it is the named future operating policy for line increases, repricing, retention, collections, hardship, closure, and conversion. Similarly, s is not a single macro variable; it is the versioned bundle containing macro path, funding, capital, tax, reward, cost, terminal-value, and scenario conventions.

The central object in this proposal is not a propensity, a response score, or observed new-card spend. It is a decision-conditional value functional. Let X_{it}^d denote the information available for customer or account i at decision context d and time t . Let $a \in \mathcal{A}_d(X_{it}^d)$ denote a feasible action and $a_0(d)$ the baseline action for that decision. Let S_{it} denote the economic and behavioral state of the relationship:

$$S_{it} = (Z_{it}, B_{it}, L_{it}, P_{it}, D_{it}, Rwd_{it}, Rel_{it}, M_{gt}),$$

where Z is lifecycle state, B the balance subledger, L the line and term state, P payment behavior, D delinquency/default state, Rwd reward and promotion state, Rel relationship state, and M macro/local state. A downstream policy π maps future states into account-management actions: line changes, repricing, retention, hardship, collections, closure, and conversion. The one-period transition kernel is

$$\Pr(S_{i,t+1} \mid S_{it}, X_{it}^d, a, \pi, M_{g,t+1}), \quad (2)$$

and the valuation layer maps the state path into cash flows.

This decomposition separates four objects that are often conflated. The *identity object* defines how accounting flows add up. The *causal object* asks what changes when action a replaces baseline $a_0(d)$. The *forecast object* projects state paths under a scenario and policy. The *policy-value object* discounts the resulting cash flows. A module can be a good forecast of $S_{i,t+h}$ under the historical policy while being invalid as a causal estimate for changing a . Conversely, a randomized experiment may identify 90-day activation lift but not by itself identify five-year NPV under future line and retention policy. The component should state which object each module supports.

The one-period incremental cash-flow primitive is

$$\begin{aligned} \Delta CF_{i,t+h}(a, \pi; s) = & \Delta PPNR_{i,t+h}(a, \pi; s) - \Delta EL_{i,t+h}(a, \pi; s) - \Delta Kchg_{i,t+h}(a, \pi; s) \\ & - \Delta Tax_{i,t+h}(a, \pi; s) + \Delta RelValue_{i,t+h}(a, \pi; s), \end{aligned} \quad (3)$$

where every term is a contrast against the baseline action under the same scenario s and downstream policy π . The discounted value is

$$\Delta NPV_i(a; d, s, \pi) = -C_i^{\text{acq}}(a) + \mathbb{E} \left[\sum_{h=0}^H \delta_h \Delta CF_{i,t+h}(a, \pi; s) + \delta_H \Delta TV_{i,t+H}(a, \pi; s) \mid X_{it}^d \right]. \quad (4)$$

Equation (4) is deliberately explicit about scenario, downstream policy, acquisition cost, terminal value, and information set. Removing any of those tags makes the number easier to print but harder to interpret.

Wallet accounting requires another distinction. Let total eligible cardable spend for customer i in period t be decomposed as

$$W_{it}^{\text{cardable}} = S_{it}^{\text{newbank}} + S_{it}^{\text{oldbank}} + S_{it}^{\text{outsidecard}} + S_{it}^{\text{debitdeposit}} + S_{it}^{\text{cashother}}. \quad (5)$$

The bank observes the first two components well, observes the debit/deposit component only where it owns the relevant deposit relationship, and often observes outside-card and cash/other components only through proxies. The offer-specific spend source contrast is

$$\begin{aligned} \Delta S_{it}^{\text{newbank}}(a) = & \Delta S_{it}^{\text{newconsumption}}(a) + \Delta S_{it}^{\text{outsidecard} \rightarrow \text{newbank}}(a) \\ & + \Delta S_{it}^{\text{oldbank} \rightarrow \text{newbank}}(a) + \Delta S_{it}^{\text{debitdeposit} \rightarrow \text{newbank}}(a) + \Delta S_{it}^{\text{timing}}(a). \end{aligned} \quad (6)$$

Only some of these terms create issuing-bank value. Outside-card shift and durable incremental consumption are the cleanest positive sources before costs and losses. Old-bank-card shift is internal cannibalization. Debit or deposit shift can be positive, neutral, or negative depending on interchange, deposit economics, rewards, funding, credit losses, and durability. Temporary promotion timing is not durable wallet capture unless it persists after the promotion and remains profitable after rewards and funding.

The proposal therefore uses four wallet terms carefully. *Outside wallet* is the customer's competitor-card or non-bank spend opportunity, whether or not the bank can move it. *Movable wallet* is the part of that outside wallet whose allocation can plausibly be changed by the bank's action. *Captured wallet* is the observed or estimated shift onto the bank's card caused by the action. *Profitable captured wallet* is captured spend or balance after interchange, rewards, funding, losses, capital, servicing, fraud, attrition, and cannibalization. The NPV component should optimize the last object, not the first three.

2.4 Scored Unit, Aggregation, and Relationship Boundary

The notation above deliberately allows i to denote a prospect, account, customer, household, application, or account-action pair. That flexibility is necessary because the replacement component will be called at different stages of the card funnel. It also creates a modeling responsibility: the component must declare the scored unit for each use case and must not silently move value between account, customer, household, and relationship levels.

For pure acquisition targeting, the natural unit may be a prospect or household-prescreen record. For application underwriting, it may be an application with an applicant and product offer attached. For line assignment, it is usually an account-action pair. For existing-client new-card cross-sell, retention, product conversion, or relationship-value reporting, the relevant unit is often customer or household, because the economics include old-bank-card activity, debit or deposit substitution, and cross-product spillovers. The request contract should therefore carry both `scored_unit_type` and `value_scope`: for example, account-level card NPV, customer-level all-bank card NPV, or relationship-level associated value.

This is not only a database convention. It is a state-aggregation choice in a dynamic valuation problem. Dynamic programming and approximate dynamic programming literatures treat aggregation as a necessary modeling device when the full state is too large to model directly [10, 68]. Customer-base, CLV, and duration models provide useful examples of economically meaningful life-cycle states [20, 35, 72], and dynamic customer or credit-card decision models show why the chosen state abstraction must match the reward and transition problem [61, 76]. Those literatures support disciplined aggregation; they do not identify this bank’s uniquely correct unit.

The implementation implication is straightforward. If the component receives account-level states but returns customer-level value, the aggregation rule must state how multiple cards, old-card activity, deposits, household linkage, relationship products, and shared costs are handled. If the component returns relationship value, it must say whether that value is included in core card NPV, reported as associated value, or excluded. Relationship-value inclusion should remain conservative: CLV and bank-value literatures justify why relationship economics matter [11, 33, 71, 87], but they do not prove that a card offer caused the deposit, loan, or other-product value for this issuer. Without randomized or credible quasi-experimental evidence, relationship value should be reported separately or scenario-tagged rather than folded into causal card NPV.

3 Literature-Grounded Economic Mechanisms

The literature review is not a decorative preface. It determines which behavioral and financial mechanisms must appear in the component. Each subsection below should be read through the same lens: what the evidence makes plausible, how that mechanism enters a card NPV model, and what must still be estimated from this bank’s own data.

What this section establishes

- The proposal is not asking the reader to accept a black-box NPV score. It identifies the economic mechanisms that must move the simulated cash-flow path: market state, liquidity, rewards, payment choice, balances, losses, attrition, and relationship behavior.
- The literature is used for structure, not for issuer-specific calibration. Public papers can justify including unemployment, credit limits, rewards, duration, and selection mechanisms; this bank must estimate the magnitudes from its own data and admissible experiments.

- Each mechanism is translated into a later component requirement: a state variable, a transition, a cash-flow term, a validation gate, or a use-limit on the returned NPV.

3.1 Literature Review Introduction

A large U.S. retail bank considering a credit card campaign faces two related business cases. First, the bank may market a card to a prospect who is not yet a card customer. Second, it may offer a new card to an existing client, including an existing bank customer with deposits or loans, or an existing cardholder who could add another card. In both cases, the operational decision is often framed as a propensity problem: who will respond, who will be approved, who will activate, and who will spend. That framing is useful for funnel management, but it is incomplete for capital allocation. The economically relevant object is a profit or value objective, not a response score alone. This is consistent with the direct-marketing literature on optimal selection, which ranks customers by expected net return rather than by response probability alone [21], and with the customer lifetime value (CLV) and customer-equity literatures, which define customer value as a discounted stream of expected future contribution [11, 71, 87]. The credit-card setting adds bank-specific components—losses, capital, funding, fraud, rewards, and cannibalization—to this familiar marketing objective.

Let i index customers, t monthly time, and $a \in \mathcal{A}$ a marketing or product action, such as a direct-mail offer, digital preapproval, branch offer, balance-transfer campaign, rewards bonus, or product upgrade. In the direct-mail formulation, the bank would mail customer i when expected margin times response probability exceeds contact cost:

$$d_i^* = \mathbf{1}\{p_i m_i - c_i > 0\}, \quad (7)$$

where p_i is response probability, m_i is conditional gross margin, and c_i is contact cost. In a CLV formulation, customer value is instead a discounted contribution stream:

$$\text{CLV}_i = \mathbb{E} \left[\sum_{t=0}^T \delta_t \{R_{it} - C_{it}\} \middle| \mathcal{I}_{i0} \right]. \quad (8)$$

The card-acquisition problem combines these ideas with treatment effects:

$$\begin{aligned} \Delta CF_{i,t+h}(a, \pi; s) &= \Delta PPNR_{i,t+h}(a, \pi; s) - \Delta EL_{i,t+h}(a, \pi; s) - \Delta Kchg_{i,t+h}(a, \pi; s) \\ &\quad - \Delta Tax_{i,t+h}(a, \pi; s) + \Delta RelValue_{i,t+h}(a, \pi; s), \end{aligned} \quad (9)$$

$$\Delta \text{NPV}_i(a; d, s, \pi) = -C_i^{\text{acq}}(a) + \mathbb{E} \left[\sum_{h=0}^H \delta_h \Delta CF_{i,t+h}(a, \pi; s) + \delta_H \Delta TV_{i,t+H}(a, \pi; s) \middle| \mathcal{I}_{id} \right]. \quad (10)$$

Here ΔCF is the one-period incremental cash-flow primitive, $\Delta PPNR$ is pre-provision net revenue, ΔEL is expected credit loss net of recovery, $\Delta Kchg$ is a priced capital or balance-sheet charge, ΔTax is the tax effect under the declared valuation convention, $\Delta RelValue$ is approved relationship-value spillover, C_i^{acq} is one-time acquisition or setup cost, and ΔTV is terminal value. The increments are measured against the decision-specific baseline action a_0 , under scenario and valuation bundle s , downstream policy π , and the information set $\mathcal{I}_{id}$ available at decision context d .

Thus, Equation (10) is not claimed to be a single canonical equation from one credit-card paper. It is the synthesis needed for a bank: Equation (7) contributes the expected-profit targeting logic, Equation (8) contributes the discounted lifetime stream, and causal/uplift methods contribute the requirement that all terms be incremental to the no-offer counterfactual.

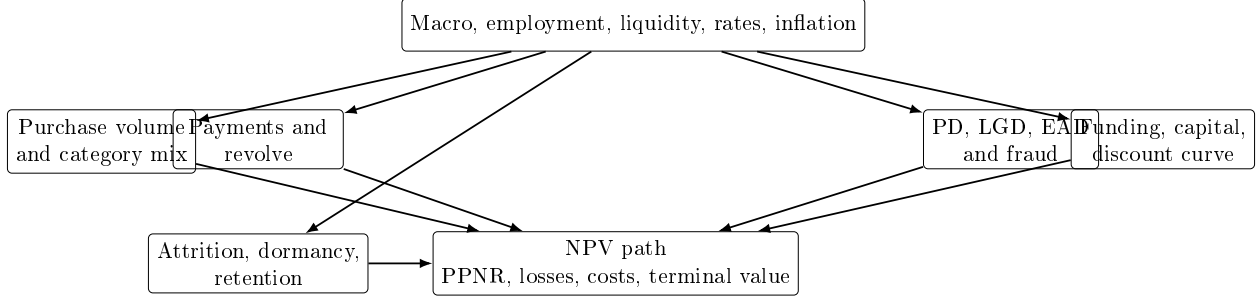


Figure 5: Macro transmission into card NPV. Market conditions do not affect only spend; they also affect payment capacity, revolving balances, losses, funding, capital, attrition, and terminal value.

Four Layers of the Valuation Problem

The proposal uses four distinct layers. The *identity layer* defines cash-flow and stock-flow accounting objects such as Equation (9); it must reconcile to finance and risk definitions. The *causal layer* estimates the effect of actions on observable short- and medium-horizon outcomes, such as response, activation, spend source, balance build, payment behavior, attrition, and loss. The *forecast layer* projects state transitions and cash-flow drivers under a declared scenario and current state. The *policy-value layer* evaluates the discounted path under downstream account-management policy π , including future line, repricing, retention, collections, closure, hardship, and conversion rules. A module may be predictive without being causal, or causal over a short horizon without by itself identifying long-horizon policy value; the component contract should say which layer each output supports.

This counterfactual is crucial. For an existing client, observed post-campaign spend on a new card can be high even when incremental NPV is low: the card may shift spend from the bank’s old card, pull forward spend through a costly promotion, or create balances whose losses exceed finance charges. Conversely, a moderate activation probability can be valuable if it creates durable incremental purchase volume, profitable revolving balances, and low servicing and loss costs. The bank’s literature problem is therefore not a search for a single “credit-card NPV paper.” It is a synthesis problem across household finance, payment choice, credit-card economics, causal marketing, and customer lifetime value.

The rest of this chapter follows the same pattern. Each mechanism is stated as a business problem, tied to the literature or to a project derivation, translated into a model implication, and connected to the data that the bank must supply. The literature can justify why a mechanism belongs in the component; it usually cannot identify this issuer’s activation lift, revolving-balance elasticity, cannibalization rate, reward cost, or terminal relationship value. Those issuer-specific quantities are carried forward into the architecture, dynamic model, and identification sections rather than left as loose survey observations.

3.2 Market Conditions, Employment, and Propensity to Spend

3.2.1 Income, unemployment, and liquidity

The first component of credit-card NPV is the customer’s capacity and propensity to spend. The relevant empirical papers do not estimate the bank’s Equation (10) directly. They estimate causal or quasi-causal effects of income, employment, and liquidity variables on spending or debt. Their contribution to our model is a set of state variables and elasticities for the spend, balance, and risk

modules.

Ganong and Noel [41] use de-identified bank account data to study spending around unemployment and the predictable income drop at unemployment insurance exhaustion. A stylized event-study version of the object is

$$y_{i\tau} = \alpha_i + \lambda_\tau + \sum_{k \neq -1} \beta_k \mathbf{1}\{\tau_i = k\} + \varepsilon_{i\tau}, \quad (11)$$

where $y_{i\tau}$ is spending for household i at event time τ relative to an unemployment or benefit-exhaustion event, α_i absorbs household fixed effects, and the β_k trace the spending path around the event. The important modeling implication is not merely “unemployment matters.” It is that the card spend forecast should include event-time variables such as job loss, payroll interruption, benefit receipt, and benefit exhaustion, because spend can move non-smoothly around predictable income events.

Johnson et al. [50] study the 2001 tax rebates using variation in rebate timing. A stylized specification is

$$\Delta c_{it} = \alpha_i + \lambda_t + \gamma \text{Rebate}_{it} + \eta' H_i \text{Rebate}_{it} + u_{it}, \quad (12)$$

where γ is a marginal propensity to consume out of the transfer and H_i contains heterogeneity variables such as income or liquid-wealth status. For a credit-card model, this motivates transfers, payroll changes, and liquid wealth proxies as pre-treatment controls or scenario variables in purchase-volume models, with heterogeneous effects. It also motivates hypotheses about payment behavior, but it does not by itself identify this bank’s payment-rate response, loss response, attrition response, or the share of extra consumption that will route through our card.

Gross and Souleles [42] provide the closest direct credit-card evidence. They study account-level card and bureau data and estimate how debt responds to credit limits and interest rates. A stylized account-level equation is

$$\Delta B_{it} = \alpha_i + \lambda_t + \theta \Delta \text{Limit}_{it} + \rho \Delta \text{APR}_{it} + \psi' X_{it} + e_{it}, \quad (13)$$

with stronger responses among initially high-utilization or liquidity- constrained borrowers. This directly motivates a balance-response module in which line assignment and APR are policy variables rather than passive controls. Extensions from balance response to payment behavior, expected loss, funding, and capital are necessary for NPV, but they require this bank’s own risk, payment, and account-management evidence.

3.2.2 Rates, regulation, and product terms

Credit-card NPV is also shaped by prices, fees, disclosures, repayment rules, and regulatory constraints. Agarwal et al. [3] study the Credit Card Accountability Responsibility and Disclosure Act using a large account-level card panel. The paper is not a spend-propensity model, and this survey should not use it as if it supplied a universal activation or balance elasticity. Its modeling value is different: it shows that product terms are governed economic objects. Fees, disclosures, and repayment nudges can change both issuer revenue components and consumer repayment behavior, so the NPV model cannot treat APRs, fees, and promotional terms as free scalar revenue add-ons.

Formally, the campaign action should be represented as a vector of product terms, not as a binary contact flag:

$$a_{it} = \left(\ell_{it}, r_{it}^{\text{purchase}}, r_{it}^{\text{cash}}, r_{it}^{\text{BT}}, r_{it}^{\text{promo}}, T_{it}^{\text{promo}}, f_{it}, q_{it}, b_{it}, d_{it} \right) \in \mathcal{A}_{it}(X_{it}, M_{gt}, G_t). \quad (14)$$

Here ℓ is the credit line; the r terms are purchase, cash-advance, balance-transfer, and promotional APRs; T^{promo} is promotional duration; f is the vector of annual, late, balance-transfer, cash-advance, foreign-transaction, and waiver terms; q is the reward earn or redemption schedule; b is a sign-up or statement-credit bonus; and d is the disclosure, minimum-payment, autopay, and digital-wallet provisioning design. The feasible set $\mathcal{A}_{it}$ is indexed by customer state, macro state, and governance state G_t because underwriting policy, compliance rules, disclosure requirements, risk appetite, pricing floors, and operational constraints limit which offers can be made.

This notation connects the CARD Act evidence to the later NPV architecture in two bounded ways. First, the evidence supports treating fees, disclosures, repayment allocation, and regulated terms as governed economic objects that can change the feasible set and cash-flow mapping:

$$a_{it} \in \mathcal{A}_{it}(G_t), \quad (15)$$

$$CF_{i,t+1} = c(Z_{it}, X_{it}, M_{gt}, a_{it}; G_t). \quad (16)$$

A rule change, fee cap, disclosure change, or repayment-allocation rule is therefore not merely an external dummy variable. It changes either the offers available to the bank or the function that maps behavior into interest, fees, payments, and balances. Second, as a project design convention, product terms should be represented as treatments that may enter the behavioral transition system. A reduced-form version is

$$\Pr(Z_{i,t+1} = z' \mid Z_{it}, X_{it}, M_{gt}, a_{it}, G_t), \quad (17)$$

$$\mathbb{E}[Y_{i,t+1} \mid Z_{i,t+1}, X_{it}, M_{gt}, a_{it}, G_t], \quad Y \in \{S, B, Pay, D, Q\}. \quad (18)$$

Thus APR, line, fees, rewards, and disclosures should be tested as potential drivers of activation, purchase volume, revolving balances, payments, delinquency, rewards liability, servicing cost, and attrition. The CARD Act evidence directly supports the fee, disclosure, repayment, and revenue/payment channels; the broader transition-kernel mapping is a bank-model requirement that must be validated with issuer data.

The balance-modeling implication of Gross and Souleles [42] is direct. Equation (13) says that limit and APR changes can move card debt, especially for liquidity-constrained accounts. In the NPV engine, the same terms should feed revenue and risk calculations, while the loss, funding, and capital extensions require bank-specific validation:

$$B_{i,t+1} = g_B(B_{it}, Pay_{it}, S_{it}, \ell_{it}, r_{it}, f_{it}, X_{it}, M_{gt}) + \varepsilon_{i,t+1}^B, \quad (19)$$

$$Int_{i,t+1} = r_{it}^{\text{revolve}} B_{i,t+1}^{\text{revolve}} + r_{it}^{\text{cash}} B_{i,t+1}^{\text{cash}} + r_{it}^{\text{BT}} B_{i,t+1}^{\text{BT}}, \quad (20)$$

$$EL_{i,t+1} = PD_{i,t+1}(Z_{it}, Util_{it}, X_{it}, M_{gt}, a_{it}) LGD_{i,t+1} EAD_{i,t+1}(B_{i,t+1}, \ell_{it}, a_{it}). \quad (21)$$

Raising APR can mechanically increase the rate applied to revolving balances, but it can also change borrowing, payment, attrition, and default behavior. Increasing the line can raise usage and finance revenue, but it can also raise utilization dynamics, exposure at default, capital, and downside risk. The correct object for decisioning is therefore a total derivative of value with respect to the governed offer vector, not a partial revenue calculation holding behavior fixed. Let $\mu_Y(a) = \mathbb{E}[Y_i(a) \mid \mathcal{I}_{it}]$ for $Y \in \{Z, S, B, Pay, D, Q\}$. For any feasible local perturbation v of the offer vector,

$$\begin{aligned} D_v \mathbb{E}[\Delta \text{NPV}_i(a)] &= \nabla_a \mathbb{E}[\Delta \text{NPV}_i(a, \mu)]^\top v \Big|_{\mu=\mu(a)} \\ &+ \sum_{Y \in \{Z, S, B, Pay, D, Q\}} \frac{\partial \mathbb{E}[\Delta \text{NPV}_i(a, \mu)]}{\partial \mu_Y} \Big|_{\mu=\mu(a)} D_v \mu_Y(a). \end{aligned} \quad (22)$$

The first term is the mechanical cash-flow effect of the term sheet; the second term is the behavioral response channel. The second term is precisely why the rates and regulation literature belongs before the usage, CLV, loss, balance, and PPNR sections.

This also clarifies how product terms should be validated. The promotion criterion is not whether a high-APR or high-fee offer raises modeled revenue conditional on the same balance path. It is whether the feasible offer $a \in \mathcal{A}_{it}(G_t)$ increases counterfactual value after behavioral response, regulation, cannibalization, expected loss, capital, funding, rewards, and servicing:

$$\Delta NPV_i(a; G_t) = -\Delta C_i^{\text{acq}}(a) + \sum_{h=0}^H \delta_h \mathbb{E}[\Delta CF_{i,t+h}^{\text{terms}}(a; G_t) \mid \mathcal{I}_{it}], \quad (23)$$

$$\Delta CF_{i,t+h}^{\text{terms}}(a; G_t) = \Delta PPNR_{i,t+h}(a; G_t) - \Delta EL_{i,t+h}(a; G_t) - \Delta K_{i,t+h}(a; G_t) + \Delta RelValue_{i,t+h}(a). \quad (24)$$

Equation (23) is the handoff to the component decomposition in Section 4.3: promotional APRs and balance-transfer terms become promo states and balance subledgers; fees and waivers become non-interest income; rewards become expense and liability; APR/line choices affect balances, EAD, and capital; disclosure and repayment design enter payment and attrition transitions; and all terms remain subject to the feasible governed action set.

3.2.3 Macro shocks and category substitution

Market shocks can change not only the level of spending but also the composition of card spending. Baker et al. [8] use transaction-level household financial data to study consumption during the COVID-19 shock. Their evidence documents large changes in spending levels and categories during the epidemic. For card NPV, category shifts matter because merchant-category mix affects interchange, rewards expense, fraud risk, dispute rates, and charge-off correlation. A recession, health shock, inflation shock, or local labor-market shock can lower discretionary purchase volume while increasing borrowing demand and credit risk.

The practical lesson is that spend models should be time varying and category-aware. A customer with high pre-campaign grocery and travel spend is not equally valuable in every macro state. A useful reduced-form component is

$$\log(1 + S_{ij,t+1}) = \mu_i + \lambda_t + \kappa_j + \beta'_j M_{gt} + \chi'_j X_{it} + \omega'_j (M_{gt} \times X_{it}) + \epsilon_{ij,t+1}, \quad (25)$$

where $S_{ij,t+1}$ is spend in merchant category j , M_{gt} is a local or national macro vector for geography g , and X_{it} includes customer liquidity, payroll, utilization, and relationship variables. Equation (25) is not lifted from one paper; it is the model slot suggested by the income, liquidity, and macro-shock evidence.

The evidence-to-model translation is best read as a set of modeling obligations rather than as a compressed table.

Ganong and Noel [41]: unemployment and benefit timing

- Mathematical object: event-time spending paths around unemployment and benefit exhaustion.
- Model implication: include job-loss, payroll interruption, benefit receipt, and benefit-exhaustion indicators in spend, payment, attrition, and loss modules.
- Boundary: the paper motivates the state variable; it does not tell this bank how much spend will route to its card.

Johnson et al. [50]: transfer timing and liquidity

- Mathematical object: heterogeneous marginal propensity to consume out of a timed transfer.
- Model implication: include income-shock, transfer, liquidity, and interaction terms in spend and payment equations.
- Boundary: liquidity response is not automatically card-routing response, so routing must be estimated separately.

Gross and Souleles [42]: line, APR, and borrowing response

- Mathematical object: account-level response of card debt to limits and interest rates.
- Model implication: treat line and APR as treatment components that affect balances, finance revenue, utilization, EAD, and losses.
- Boundary: balance response is not full NPV without funding, capital, losses, rewards, and attrition.

Agarwal et al. [3]: regulated terms and revenue channels

- Mathematical object: changes in fees, disclosures, repayment behavior, and issuer revenue components around a regulatory intervention.
- Model implication: APRs, fees, rewards, disclosures, and repayment rules enter as governed terms $a_{it} \in \mathcal{A}_{it}(G_t)$ feeding transition, balance, loss, PPNR, and promotion-state equations.
- Boundary: regulated terms are not unconstrained revenue levers; they belong inside an approved action set.

Baker et al. [8]: macro shocks and category mix

- Mathematical object: transaction-level spending response by level and category during a macro shock.
- Model implication: forecast category-level spend and the resulting interchange, rewards, fraud, dispute, and loss mix under scenarios.
- Boundary: category response should be estimated and monitored on this bank’s portfolio before it drives production valuation.

3.2.4 Implication for the NPV component

The macro and employment evidence should be incorporated as a dynamic state requirement, not as a cosmetic scenario overlay. The replacement component should carry macro, local labor-market, rate, inflation, payroll, benefit, and liquidity variables into the spend, payment, balance, attrition, and loss modules. It should also return the scenario identifier and sensitivity of major NPV components to macro assumptions. For marketing and cross-sell, this means the model must distinguish customers whose expected value comes from durable purchase volume from customers whose value depends on fragile revolve or liquidity stress. For line and offer decisions, it means the line, APR, and promo terms must be modeled as actions that can change both revenue and EAD, not as exogenous descriptors.

The public literature does not supply this bank’s elasticity of card spend, payment rate, or default to unemployment and liquidity shocks. Those parameters require issuer transaction data, bureau refreshes, payroll or deposit signals where permitted, campaign holdouts, and vintage backtests.

The output implication is concrete: the response object should expose category spend, balances, payments, utilization, PD, LGD, EAD, expected loss, and PPNR by scenario and horizon. The validation implication is equally concrete: macro-stress backtests, local labor-market drift monitoring, and segment-level calibration can veto use of a model even when average response or spend prediction is strong.

3.3 Will the Newly Issued Card Be Used?

3.3.1 Adoption is not usage

The second requested problem is the probability that a customer who receives a new card will use that card. The payment-choice literature warns against collapsing this into a single response probability. Koulayev et al. [53] estimate a structural model of adoption and use of payment instruments by U.S. consumers. Their central relevance here is that possession of a payment instrument and use of that instrument are distinct choices. A customer may open a card, activate it, and still use another issuer’s card, debit card, cash, ACH, or a digital wallet at the point of sale.

Mathematically, the useful structure is a two-stage adoption/use problem. A stylized version is:

$$A_{im}^* = Z_i' \alpha_m + \nu_{im}, \quad A_{im} = \mathbf{1}\{A_{im}^* > 0\}, \quad (26)$$

$$U_{ijm} = X_{ijm}' \beta_m + \phi_m A_{im} + \varepsilon_{ijm}, \quad m_{ij}^* = \arg \max_{m \in \mathcal{M}_i} U_{ijm}. \quad (27)$$

Here A_{im} denotes whether consumer i has adopted payment instrument m , U_{ijm} is the utility of using instrument m for transaction context j , and $\mathcal{M}_i$ is the set of adopted instruments available to consumer i . The exact specification in Koulayev et al. [53] is richer, but Equations (26)–(27) capture the modeling lesson for us: “has the card” gates the choice set, while “uses the card” is a conditional payment-choice decision. Our card model should therefore estimate activation and usage conditional on the customer’s competing payment portfolio rather than treating issuance as usage.

For a credit-card campaign, the operational funnel should therefore separate:

1. offer exposure and response;
2. application and approval;
3. card opening and physical or digital activation;
4. first purchase;
5. repeated purchase;
6. share-of-wallet capture;
7. durable active state, dormancy, closure, or charge-off.

Each transition has different covariates and different economic meaning. Approval is partly a risk and underwriting outcome. Activation is partly a friction and onboarding outcome. First purchase is an early usage outcome. Share-of-wallet is a competitive payment-routing outcome. Durable activity is a lifetime-value outcome.

3.3.2 Rewards, reactivation, and early spend

Rewards can alter usage, but the bank must distinguish profitable use from costly promotional pull-forward. Agarwal et al. [2] study cash-back rewards using administrative data from a large U.S. financial institution. The important point for our later model is that the paper is not just a qualitative statement that “rewards matter.” It suggests three account-level response objects:

$$A_{it}^{\text{react}} = \mathbf{1}\{S_{i,t-3:t-1} = 0, S_{it} \geq s_0\}, \quad (28)$$

$$S_{it} = \text{monthly purchase spend on the card}, \quad (29)$$

$$B_{it} = \text{monthly card debt or revolving balance}. \quad (30)$$

Here A_{it}^{react} is a reactivation indicator among accounts that were inactive before the reward, S_{it} is the usage outcome, and B_{it} is the balance outcome. These are closely related booked-account reward- intervention objects: did an existing account become active again, how much spend did it generate, and did it also create debt exposure? They are useful analogues for new-card valuation, but they are not direct acquisition-stage activation evidence. Newly issued card activation and early use still require issuer-specific campaign or onboarding evidence.

A reduced-form implementation for a randomized or quasi-randomized reward offer can be written as

$$Y_{it} = \alpha_i + \lambda_t + \tau \text{Reward}_{it} + \eta' X_{it} + \epsilon_{it}, \quad (31)$$

where Y_{it} may be purchase spend, revolving debt, or an indicator for reactivation among previously inactive cardholders. The estimand is

$$\tau_Y = \mathbb{E}[Y_{it}(r = 1) - Y_{it}(r = 0)], \quad (32)$$

with r denoting the reward treatment. In a bank implementation this should be estimated heterogeneously,

$$\tau_Y(x, r) = \mathbb{E}[Y_i(r) - Y_i(0) \mid X_i = x], \quad (33)$$

where X_i includes prior inactivity, existing card holdings, pre-period merchant mix, line utilization, relationship depth, channel, credit quality, and macro state. Equation (33) is the project bridge from Agarwal et al. [2] to the later model: it proposes a segment-level reward response surface rather than importing a single portfolio average. The bridge is valid for production only after the bank validates the response in the relevant acquisition, cross-sell, or booked-account population.

The output of this block should feed the NPV simulator in four places:

$$\Pr(Z_{i,t+1} = \text{active} \mid Z_{it}, X_{it}, a) \leftarrow \text{activation/reactivation effect}, \quad (34)$$

$$\mathbb{E}[S_{i,t+1} \mid Z_{i,t+1}, X_{it}, a] \leftarrow \text{spend effect}, \quad (35)$$

$$\mathbb{E}[B_{i,t+1} \mid Z_{i,t+1}, X_{it}, a] \leftarrow \text{debt/balance effect}, \quad (36)$$

$$C_{i,t+1}^{\text{reward}}(a) = q_a S_{i,t+1} + F_a + R_a^{\text{signup}}, \quad (37)$$

where q_a is the earn rate or reward cost per dollar, F_a is fixed program cost, and R_a^{signup} is signup or statement-credit cost. The promotion is attractive only if the incremental contribution implied by (34)–(37) is positive after cannibalization and losses. This is the concrete way to avoid importing a headline reward effect as if it were a universal activation parameter.

Behavioral evidence also suggests that payment instruments can affect purchase behavior. Prelec and Simester [67] find in experimental settings that willingness to pay can be higher under credit-card payment than cash payment. For a bank NPV model, this is mechanism evidence: payment

method may change the purchase decision, not simply route a fixed purchase through a different rail. The result is not a portfolio-level NPV estimate and should not be used as a direct calibration without issuer data.

The conservative model use is a mechanism prior for the ticket-size component. Let N_{ijt} be transaction count in category j and $\bar{s}_{ijt}$ average ticket size. Then

$$S_{ijt} = N_{ijt} \bar{s}_{ijt}, \quad (38)$$

$$\log \bar{s}_{ij,t+1} = \mu_i + \kappa_j + \beta' X_{it} + \theta_1 \text{CreditCard}_{ij,t+1} + \theta_2 \text{RewardSalience}_{ij,t+1} + \epsilon_{ij,t+1}. \quad (39)$$

The Prelec and Simester [67] result supports allowing θ_1 or θ_2 to differ from zero, but the bank must estimate those parameters with transaction data. This block is explanatory unless validated on issuer data; it should not be a hard-coded multiplier in NPV.

3.3.3 Cannibalization and incremental usage

For existing clients, the central challenge is cannibalization. A newly issued card can create genuinely incremental spend, shift spend from another issuer, shift spend from the bank's existing card, move checking/debit activity onto credit, or induce balances whose loss and reward costs exceed revenue. To make this estimable, define the following observed spend aggregates over horizon H :

$$S_{iH}^{\text{new}} = \text{spend on the newly issued card}, \quad (40)$$

$$S_{iH}^{\text{oldbank}} = \text{spend on the bank's pre-existing cards}, \quad (41)$$

$$S_{iH}^{\text{debit}} = \text{bank-observed debit/checking purchase activity}, \quad (42)$$

$$S_{iH}^{\text{bank}} = S_{iH}^{\text{new}} + S_{iH}^{\text{oldbank}}. \quad (43)$$

When outside-card spend is not observed, competitor-card substitution is latent and must be inferred from bureau/payment-network aggregates, wallet estimates, or experimental lift in total bank spend. The observed new-card spend path should therefore be decomposed before it is valued:

$$S_{iH}^{\text{new}}(a) = S_{iH}^{\text{new cons}}(a) + S_{iH}^{\text{comp shift}}(a) + S_{iH}^{\text{oldbank shift}}(a) + S_{iH}^{\text{debit shift}}(a) + S_{iH}^{\text{promo timing}}(a) + e_{iH}^S(a). \quad (44)$$

Equation (44) is an accounting discipline for model design, not a claim that every term is directly observed. The first term is genuinely new consumption. The second is spend shifted from competitor cards. The third is internal bank-card cannibalization. The fourth is debit or deposit-rail substitution. The fifth is intertemporal pull-forward or decay created by promotions. The last term collects measurement error and unobserved routing changes. These terms have different economics. Competitor shift can be valuable to the issuer; old-bank-card shift may merely move interchange and reward cost across products; debit substitution may reduce deposit economics; promotion timing may disappear after the offer; and all terms must still be netted against rewards, fraud, funding, servicing, losses, and capital.

$$W_{iH}^{\text{out}} = S_{iH}^{\text{comp}} + B_{iH}^{\text{comp,rev}} + U_{iH}^{\text{outside}}, \quad W_{iH}^{\text{move}}(a) = m_{iH}(a) W_{iH}^{\text{out}}. \quad (45)$$

Here W_{iH}^{out} is an outside card-wallet state combining competitor-card spend, competitor revolving balances, and other outside payment or borrowing components relevant to the decision; $m_{iH}(a)$ is the offer-specific movable fraction; and $W_{iH}^{\text{move}}(a)$ is the portion the offer can plausibly move to the bank. The model should distinguish this structural movable fraction from an operational reporting ratio such as observed new-card spend divided by estimated outside wallet. A large outside wallet

does not imply large movable wallet, and large movable wallet does not imply positive NPV after rewards, losses, and funding.

$$\begin{aligned}\Delta S_{iH}^{\text{new}}(a) &= \Delta S_{iH}^{\text{new cons}}(a) + \Delta S_{iH}^{\text{comp shift}}(a) + \Delta S_{iH}^{\text{oldbank shift}}(a) \\ &\quad + \Delta S_{iH}^{\text{debit shift}}(a) + \Delta S_{iH}^{\text{promo timing}}(a) + \Delta e_{iH}^S(a).\end{aligned}\tag{46}$$

Equation (46) is the version that belongs in incremental NPV. It says that the treatment effect on new-card spend should be allocated to sources before it is valued. A response model that estimates only ΔS^{new} can rank new-card volume, but it cannot by itself rank incremental NPV. The estimands are:

$$\tau_i^{\text{new card spend}}(a) = \mathbb{E}[S_{i,t:t+H}^{\text{new}}(a) - S_{i,t:t+H}^{\text{new}}(0) \mid \mathcal{I}_{it}],\tag{47}$$

$$\tau_i^{\text{bank spend}}(a) = \mathbb{E}[S_{i,t:t+H}^{\text{all bank cards}}(a) - S_{i,t:t+H}^{\text{all bank cards}}(0) \mid \mathcal{I}_{it}],\tag{48}$$

$$\text{Cannibalization}_i(a) = \tau_i^{\text{new card spend}}(a) - \tau_i^{\text{bank spend}}(a).\tag{49}$$

Equation (47) can be positive while Equation (48) is small or even negative. That is why new-card activation and new-card spend are not sufficient promotion criteria.

In a randomized campaign, these are directly estimable by treatment-control differences:

$$\hat{\tau}^{\text{new}}(x) = \mathbb{E}[S_{iH}^{\text{new}} \mid T_i = 1, X_i = x] - \mathbb{E}[S_{iH}^{\text{new}} \mid T_i = 0, X_i = x],\tag{50}$$

$$\hat{\tau}^{\text{bank}}(x) = \mathbb{E}[S_{iH}^{\text{bank}} \mid T_i = 1, X_i = x] - \mathbb{E}[S_{iH}^{\text{bank}} \mid T_i = 0, X_i = x],\tag{51}$$

$$\hat{\kappa}(x) = 1 - \frac{\hat{\tau}^{\text{bank}}(x)}{\hat{\tau}^{\text{new}}(x)} \quad \text{when } \hat{\tau}^{\text{new}}(x) > 0.\tag{52}$$

Here $\hat{\kappa}(x)$ is the estimated cannibalization rate from the bank's own existing-card book. If $\hat{\kappa}(x)$ is close to one, most new-card spend is shifted from old bank cards; if it is close to zero, new-card spend is mostly incremental to the bank. If it is negative, the new card may also lift old bank card spend, for example through relationship deepening.

The latent competitor-transfer component can be represented as

$$\tau_i^{\text{competitor shift}}(a) = \tau_i^{\text{bank spend}}(a) - \tau_i^{\text{debit shift}}(a) - \tau_i^{\text{new consumption}}(a),\tag{53}$$

The debit-shift term can sometimes be estimated from the bank's checking and debit data. The new-consumption term is harder to identify without network or merchant-panel data. Equation (53) should therefore be treated as a decomposition with an uncertainty interval, not as a precisely observed quantity.

Each component of Equation (44) should carry an evidence grade. For example, old-bank-card shift can often be internally anchored from the bank's own card book; debit shift can be internally anchored when the customer has deposit activity at the bank; competitor shift may be experimentally identified by all-bank-spend lift in a holdout design, externally benchmarked from bureau or network aggregates where available, or model-imputed when no anchor exists; and promotion timing should be validated from post-promo decay. The component should not allow model-imputed outside wallet to be reported as experimentally identified wallet capture.

Randomized campaign holdouts are the cleanest way to separate organic propensity from treatment effect. When randomization is unavailable, the causal-inference and uplift literatures provide tools, but they do not remove the need for credible identification. The propensity-score framework of Rosenbaum and Rubin [69], true-lift marketing models such as Lo [62], causal forests [88], and

double/debiased machine learning [25] are useful methodological references. In a regulated bank environment, they should be used with overlap checks, sensitivity analysis, segment calibration, and careful model-governance review.

For later modeling, the cleanest operational implementation is to estimate a conditional average treatment effect for each downstream outcome,

$$\tau_Y(x) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x], \quad (54)$$

where Y may be activation, first purchase, first-90-day new-card spend, all-bank-card spend, revolving balance, loss, or full NPV. The causal/uplift module should feed the simulation engine by replacing raw predictions $\mathbb{E}[Y_i \mid T_i = 1, X_i]$ with counterfactual contrasts $\mathbb{E}[Y_i(1) - Y_i(0) \mid X_i]$. If randomization is not available, the model must record the identification assumption and cannot be promoted solely on predictive fit.

The handoff to Equation (10) is explicit. Let r_j^{int} be net interchange by merchant category, $q_j(a)$ reward cost by category and offer, m_{it}^{finance} net finance margin on balances, and ℓ_{it} expected loss. Then a horizon- H incremental contribution block is

$$\Delta\Pi_{iH}(a) = \sum_j \{r_j^{\text{int}} - q_j(a)\} \hat{\tau}_{ijH}^{\text{bank}}(a) + \hat{\tau}_{iH}^{\text{finance}}(a) - \hat{\tau}_{iH}^{\text{loss}}(a) - C_i^{\text{campaign}}(a) - C_i^{\text{servicing}}(a), \quad (55)$$

$$\Delta\text{NPV}_i(a) = \sum_{h=0}^H \delta_h \mathbb{E}[\Delta\Pi_{ih}(a) \mid \mathcal{I}_{i0}]. \quad (56)$$

The key design choice is that interchange and reward revenue use $\hat{\tau}^{\text{bank}}$, not raw new-card spend S^{new} . This is how the cannibalization model changes the optimization target.

The usage evidence also translates into modeling obligations.

Koulayev et al. [53]: adoption versus conditional use

- Mathematical object: adoption and conditional use of payment instruments.
- Model implication: separate opened, activated, first-use, repeated-use, and payment-instrument choice states.
- Boundary: adoption of an instrument is not proof that all eligible spend will route to that instrument.

Agarwal et al. [2]: rewards as treatment

- Mathematical object: reward effects on spend, debt, and reactivation-like outcomes.
- Model implication: estimate $\tau_Y(x, r)$ for activation, spend, balances, and reward costs, and feed Equations (34)–(37).
- Boundary: booked-account reward response does not supply a universal new-card activation parameter.

Prelec and Simester [67]: payment medium and ticket size

- Mathematical object: experimental payment-medium effect on willingness to pay.
- Model implication: use it as mechanism support for Equation (39).
- Boundary: estimate issuer-specific payment-medium coefficients before using the mechanism in NPV.

Uplift and CATE methods: causal contrasts

- Mathematical object: $\mathbb{E}[Y(1) - Y(0) \mid X = x]$ for activation, spend, balances, losses, and value.
- Model implication: estimate new-card lift, all-bank-card lift, and cannibalization in Equations (50)–(52), then feed contribution through Equation (55).
- Boundary: predictive fit on observed spend is not a causal estimate of incremental all-bank spend.

3.3.4 Implication for the NPV component

The adoption, rewards, and cannibalization literature implies that the component must model a funnel of states rather than a single probability that “the customer uses the card.” The minimum state sequence is offer exposure, application or acceptance, approval where relevant, booked account, activation, first purchase, repeated use, wallet-share capture, dormancy, attrition, and charge-off. Each transition has its own covariates, counterfactual, and economic meaning. A marketing caller may care about contact-versus-suppress; an underwriting caller may care about approve-versus-decline; an activation caller may care about treatment-versus-no-treatment after the account is booked. The NPV component should value those contrasts separately and should not promote a model simply because it predicts booked-account spend.

For existing clients, the response object should report at least two distinct spend concepts: new-card spend and all-bank incremental spend. The first is a card-usage diagnostic; the second is the input to bank value after cannibalization. Where the bank can observe old-card, debit, deposit, or other relationship behavior, those channels should be measured explicitly. Where outside-card behavior or new consumption is latent, the component should return an uncertainty range or warning rather than treat the decomposition as known. This bridge is what connects the literature evidence to the later request and response contracts: callers must declare the decision context, and the response must label activation, first use, new-card spend, all-bank spend, balances, reward cost, and cannibalization separately.

3.4 Lifetime Expenditure and Customer Lifetime Value

3.4.1 Customer-base models

The third requested problem is how to model lifetime expenditure of a credit card user. Marketing science provides a natural starting point. Schmittlein et al. [72] develop a model for inferring which customers are still active and what they will do next from transaction timing and frequency. This is directly relevant for card accounts because a customer can silently become inactive before formal closure.

Fader et al. [35] propose the BG/NBD model as a tractable alternative to the Pareto/NBD model for predicting repeat-purchase behavior. Fader et al. [36] connect recency, frequency, and monetary value to customer lifetime value. These models are useful baselines for transaction frequency, active/dormant status, and spend per transaction. They are especially relevant for transactors whose value is driven primarily by purchase volume and interchange rather than finance charges.

Relationship duration is another core object. Bolton [20] models the duration of a customer’s relationship with a continuous service provider. Credit cards have similar duration dynamics: fees, service quality, fraud events, line changes, disputes, rewards devaluations, and pricing changes can alter retention, dormancy, and usage.

3.4.2 Why card NPV is broader than plain CLV

Credit cards are not a pure repeat-purchase setting. A bank needs the lifetime path of at least six linked processes:

$$S_{it} = \text{purchase spend and transaction count,} \quad (57)$$

$$B_{it} = \text{statement and revolving balance,} \quad (58)$$

$$P_{it} = \text{payment amount and payment rate,} \quad (59)$$

$$D_{it} = \text{delinquency/default state,} \quad (60)$$

$$Z_{it}^{\text{use}} = \text{active, dormant, closed, or charged-off usage state,} \quad (61)$$

$$Q_{it} = \text{rewards, fraud, disputes, servicing, and other costs.} \quad (62)$$

These processes are jointly determined. High spend can be profitable when it is durable and low loss, but low value when driven by high rewards cost or low-margin categories. Revolving balances can raise interest income and expected loss simultaneously. A durable active customer may be unprofitable if servicing, fraud, or rewards costs dominate revenue.

Thus, a bank should treat CLV models as transaction and duration components of a broader card-NPV engine. Lifetime expenditure is necessary but not sufficient: the model must translate expenditure into interchange, rewards cost, balances, finance charges, losses, capital, and servicing cost.

3.4.3 Implication for the NPV component

The lifetime-value literature should enter the replacement component as a state and duration layer. The component should track whether an account is prospect, booked, activated, active transactor, revolver, promotional borrower, cash-advance user, dormant, delinquent, closed, converted, or charged off. Those states determine which cash flows are possible and how long they are expected to persist. A transactor path mainly converts purchase volume into interchange, rewards, fraud, servicing, and attrition. A revolver path adds finance charges, funding cost, utilization, EAD, and loss. A promo path adds reward liability, teaser-rate economics, funding timing, post-promo decay, and attrition risk.

This state layer is also the practical answer to terminal value. Rather than attach one opaque terminal-value multiple to every account, the component should estimate terminal value conditional on state, vintage, months-on-book, product, relationship depth, risk, macro state, and downstream account management policy. The bank will still need approved terminal-value conventions, but the model should make the behavioral source of terminal value visible. Validation should include vintage holdouts, active/dormant calibration, attrition calibration, and decomposition checks showing whether long-horizon value is driven by durable behavior or by fragile assumptions.

4 Component Architecture

A good proposal should not ask the reader to infer the system from a table. The architecture is therefore described in prose first. The component receives a scored population, a decision context, candidate actions, a baseline action, a scenario and valuation-semantics bundle, and downstream-policy assumptions. It returns incremental NPV with a decomposition into economic drivers and explicit use limits. The modules below are not separate products; they are the internal pieces needed to simulate and value a coherent monthly path.

What this section establishes

- The component boundary is narrow: caller systems choose and execute business actions, while the replacement component computes value for a declared action relative to a declared baseline.
- The component is modular because the economics are modular, but the modules are not independent. Spend, wallet routing, balances, losses, rewards, attrition, and PPNR must be joined before valuation.
- The output is a governed value object, not just a scalar: it carries decomposition, status, lineage, warnings, evidence grade, and allowed use.

4.1 Integrated Design of the Replacement NPV Component

This section states the proposal in build terms before the detailed evidence sections. The replacement is a valuation component that converts a declared decision context, a candidate action, a customer state, product terms, scenario assumptions, and downstream policy assumptions into a decomposed incremental NPV object. The component must be modular because card value is not generated by one behavior. It is generated by a linked path of adoption, spend, payments, balances, losses, rewards, funding, capital, servicing, attrition, and relationship effects.

For every module, the design rule is the same. The public literature or supervisory source identifies a mechanism and a modeling slot. The bank's internal data or experiment estimates the issuer-specific parameter. The component exposes the resulting path or cash-flow driver in the NPV response. Validation determines whether that module may be used for a given decision context. This is the practical meaning of evidence integration in this proposal.

4.1.1 Module 1: spend, macro, and employment state

The spend module forecasts purchase volume, transaction count, ticket size, and merchant-category mix under the declared action and macro scenario. The literature on unemployment, income shocks, liquidity, and macro disruptions supports the mechanism that spending is state dependent. The component therefore should not use a static average spend curve for all campaign conditions. It should condition on customer liquidity, payroll or deposit signals where permitted, utilization, local labor-market variables, rates, inflation, and scenario tags.

The bank must estimate how those variables map into its own card spend, payment rate, balance build, and attrition. Public papers can justify including unemployment or liquidity states; they cannot identify this bank's category spend elasticity or routing share to the new card. The output implication is that NPV responses should expose spend and interchange by category and horizon, plus scenario sensitivity. The validation implication is that macro backtests, time-based holdouts, and segment calibration can veto the spend module even if average first-spend prediction is strong.

4.1.2 Module 2: new-card adoption, activation, use, and cannibalization

The adoption and usage module separates offer response, application or acceptance, approval where relevant, booking, activation, first use, repeated use, wallet-share capture, dormancy, attrition, and charge-off. Payment-choice and rewards evidence supports the distinction between having a payment instrument and using it. It also supports modeling rewards and promotions as treatments that can change activation, spend, debt, and reactivation.

For existing clients, the core output is not raw new-card spend. The component must distinguish new-card spend, all-bank incremental card spend, debit or deposit substitution where observed, old-bank-card cannibalization, and uncertain outside-card or new-consumption components. The bank

must estimate these quantities using randomized holdouts, campaign experiments, or credible quasi-experimental designs. The response object should label activation, first-use probability, repeated-use probability, new-card spend lift, all-bank-spend lift, reward cost, balance lift, loss lift, and cannibalization. A model that predicts new-card spend but cannot identify incremental all-bank value should remain diagnostic for cross-sell decisions.

For prospect acquisition and existing-client cross-sell, the module should also estimate an outside-wallet state and an offer-specific movable-wallet state. These are project valuation objects, not quantities directly supplied by the public literature. Outside wallet is the customer's spend or revolving balance not currently on the bank's card products. Movable wallet is the fraction of that outside wallet that the specific offer can plausibly move to the bank over the declared horizon. Both quantities are latent for many customers and issuer-specific for this bank. They should therefore be treated as directly observed, internally anchored, externally benchmarked, experimentally identified, quasi-experimental, model-imputed, or scenario-only, with the evidence grade exposed in the response. Product terms, rewards, promotional duration, channel, card-on-file friction, relationship depth, segment, macro state, and time since offer are hypotheses and controls for these quantities; they do not by themselves prove movable wallet.

4.1.3 Module 3: lifetime state, attrition, dormancy, and terminal value

The lifetime module turns early behavior into a state path. A card customer can be a transactor, revolver, promotional borrower, cash-advance user, dormant account, retained account, converted account, delinquent account, or charged-off account. CLV, customer-base, and duration models support the use of transaction frequency, recency, monetary value, active-state inference, and relationship duration, but card NPV must extend those models to balances, losses, rewards, funding, and servicing.

The component should estimate transition probabilities among these states and carry state-specific terminal-value assumptions. Terminal value should not be one hidden scalar multiple; it should depend on state, vintage, months on book, product, relationship depth, risk, macro state, and downstream policy. The bank must validate lifetime state forecasts with vintage holdouts, dormancy calibration, attrition calibration, and terminal-value sensitivity. If terminal value dominates rankings, the response should expose that fact as a sensitivity or warning.

4.1.4 Module 4: balances, payments, lines, promotions, and cash advances

The balance module enforces stock-flow consistency. Purchases, cash advances, balance transfers, fees, interest, payments, credits, charge-offs, and adjustments should reconcile to statement and principal balances. Credit line, APR, promo duration, balance-transfer terms, rewards, and disclosures are candidate action terms, not passive descriptors. They can change both revenue and risk through utilization, revolve behavior, payment rates, and EAD.

The bank must estimate line and term response from its own account history, experiments, policy changes, and approved quasi-experimental variation. The component output should report line, utilization, purchase balances, cash-advance balances, balance-transfer balances, promotional balances, payments, finance charges, fees, funding cost, and balance-linked EAD. A balance forecast that violates stock-flow accounting or mixes principal and non-principal balances should fail component validation.

4.1.5 Module 5: losses, PD, LGD, EAD, vintage, and balance subledgers

The loss module uses the bank-standard PD/LGD/EAD vocabulary, but it must be connected to the card lifecycle. PD should be a transition or hazard over delinquency and charge-off states. LGD should reflect recovery, collections, bankruptcy, hardship programs, and macro state. EAD should be linked to balances, line, utilization, cash advances, balance transfers, accrued fees, interest, and undrawn draw behavior.

The component should expose expected loss and its drivers by horizon and option. It should also preserve vintage, months-on-book, and principal/non-principal balance dimensions, because newly booked accounts season differently and promotions can shift both timing and composition of loss. Public credit-risk literature supports the structure, but the bank must validate calibration and treatment response using internal vintages, booked account outcomes, line-policy tests, and stress periods. A line or promo option should not be ranked on revenue unless the linked EAD, PD, LGD, capital, and downside risk paths are also valid.

4.1.6 Module 6: PPNR, rewards, funding, capital, tax, cost, and relationship value

The finance module converts behavior into cash flows. It should decompose incremental value into net interest income, non-interest income, non-interest expense, expected loss, capital cost, tax treatment, acquisition cost, terminal value, and approved relationship value. PPNR and stress-testing sources support separating balances, revenue, expense, and losses under scenarios; CLV literature supports discounting future contribution; bank governance requires the decomposition to reconcile to finance and risk definitions.

Rewards, signup bonuses, servicing, fraud, disputes, collections, charge-off sale economics, funding, liquidity, capital, and tax must not be afterthoughts. They are often what turns high spend into low or negative value. Deposit or other-product value should be included only when the card action is credibly causal for relationship change; otherwise it should be excluded or reported separately. The component should expose both the decomposed contribution path and the valuation-semantics identifier that defines discounting, funding, capital, tax, cost allocation, terminal value, and relationship-value rules.

4.1.7 Module 7: sequential policy dependence

The component must tag the downstream policy assumed when valuing an action. Acquisition value depends on future approval, line, repricing, retention, collections, closure, and conversion policies. Underwriting and line choices change the population and state distribution that account-management systems later see. Account-management changes can invalidate upstream acquisition NPV if the old model assumed different future treatment rules.

The practical requirement is a downstream-policy assumption bundle. Each NPV response should state which policy bundle was used, and each production use case should be approved for that bundle. If the bank changes line-management, collections, retention, closure, or conversion policy, upstream NPV should be revalidated. This keeps the component from optimizing a future operating world the bank no longer runs.

4.1.8 How the modules connect to caller systems

Caller systems do not need to know every internal model detail, but they must receive a value object whose meaning is safe for their decision. A marketing caller needs contact-versus-suppress

value, campaign cost, response and activation lift, cannibalization, expected loss, and portfolio constraints. An underwriting caller needs approve-versus-decline or approve-with-terms value under credit policy, with loss, capital, compliance, and adverse-action boundaries handled by the caller. A line-assignment caller needs option-level value, usage, utilization, EAD, loss, capital, and status for each feasible line. An existing-account treatment caller needs forward value under retention, conversion, promo, closure, or no-treatment alternatives. Finance and risk consumers need decompositions for reconciliation, not authority to treat the component as the official ledger, CECL, or stress platform.

This is why the operating contracts later in the document exist. They are not the product. They are the guardrails that keep a sophisticated but fragile NPV estimate from becoming a misleading universal score.

4.2 Recommended Modeling Architecture

A defensible bank design synthesis is a modular state-transition architecture. The literature motivates the components, while the bank must validate the issuer-specific transitions and cash-flow mappings. Let X_{it} collect customer attributes, internal relationship history, bureau variables, card terms, macro variables, and campaign information. Let Z_{it} denote the latent card state:

$$Z_{it} \in \{\text{prospect, opened, activated, first-use, active, dormant, delinquent, closed, charged-off}\}.$$

A practical model can be written as

$$\Pr(Z_{i,t+1} = z' \mid Z_{it} = z, X_{it}, a), \quad (63)$$

$$\mathbb{E}(S_{i,t+1}, B_{i,t+1}, P_{i,t+1}, D_{i,t+1}, Q_{i,t+1} \mid Z_{i,t+1}, X_{it}, a). \quad (64)$$

Equations (63) and (64) can be implemented as a hurdle model, survival model, dynamic panel, machine-learning model, structural dynamic discrete-choice model, or simulation engine. The choice depends on data scale, interpretability, regulatory constraints, and the decision horizon. The dynamic discrete-choice tradition, exemplified by Rust [70], is useful conceptually because account states and bank actions evolve over time, although a full structural model may be unnecessary for many targeting applications. Dynamic panel methods such as Blundell and Bond [14] provide another reference point when lagged outcomes and persistent customer effects are central.

The architecture should contain the following modules:

1. **Eligibility and offer module.** Forecast approval, line, APR, annual fee, reward terms, disclosures, and compliance constraints.
2. **Causal response module.** Estimate incremental open, activation, first-use, and wallet-share effects of the campaign.
3. **Activation and early-usage module.** Forecast activation, digital-wallet provisioning, first purchase, first 90-day spend, and early dormancy.
4. **Spend and category module.** Forecast purchase volume, transaction count, ticket size, merchant-category mix, and seasonality.
5. **Balance and payment module.** Forecast revolving behavior, promotional balance-transfer behavior, payment rate, and interest revenue.
6. **Risk and loss module.** Forecast delinquency, default, exposure at default, loss given default, fraud, disputes, and recovery.

7. **Attrition and dormancy module.** Forecast inactivity, closure, product switching, retention response, and charge-off exit.
8. **Finance module.** Convert simulated paths into discounted contribution after rewards, servicing, funding, capital, tax/accounting treatment, and acquisition cost.

The output should be a distribution for $\Delta\text{NPV}_i(a)$, not only a point estimate. Marketing decisions should consider expected value, uncertainty, downside tail risk, portfolio concentration, and operational constraints. A portfolio optimizer may rank customers by expected incremental NPV subject to risk, fairness, capacity, and regulatory constraints.

This architecture creates a contract obligation. If a caller asks for NPV, the component should not return a scalar whose source is opaque. It should return the state path, behavioral drivers, financial decomposition, uncertainty, scenario, valuation semantics, and status sufficient to tell whether the value is usable for the declared decision. For example, a line-option score should carry usage, utilization, EAD, expected loss, capital, and sensitivity paths; an existing-client cross-sell score should carry new-card spend and all-bank incremental spend separately; a retention score should carry attrition and concession-cost paths. The operating specification in Section C is therefore not an independent API exercise. It is the engineering form of the model architecture.

4.3 Component Decomposition for a Bank-Grade Card NPV Engine

The proposed decomposition into losses, balances, pre-provision net revenue (PPNR), and portfolio-management overlays is directionally right. It resembles how banks already organize credit-card forecasting in risk, finance, and stress testing: expected credit loss is commonly decomposed into probability of default (PD), loss given default (LGD), and exposure at default (EAD); account-level balances and usage drive both revenue and exposure; and PPNR converts balances, rates, fees, expenses, and relationship effects into pre-loss operating profit. The Basel credit-risk framework formalizes the PD/LGD/EAD vocabulary for regulatory capital [9]. U.S. stress-testing practice also separates loan losses from PPNR, while forecasting portfolio balances and revenues under macro scenarios [15, 16]. Accounting standards for current expected credit losses require lifetime expected-loss measurement based on historical experience, current conditions, and reasonable and supportable forecasts [39]. These references do not dictate a marketing NPV model, but they support the decomposition as a bank-native control structure.

This decomposition is also the clearest ownership boundary for the replacement component. The component owns the valuation logic that turns behavioral paths into incremental PPNR, expected loss, capital, tax, relationship value, and terminal value. It owns the decomposition labels, assumptions, model versions, lineage, and reconciliation package needed to explain that valuation. It does not own the general ledger, CECL production process, supervisory stress platform, campaign optimizer, underwriting policy engine, account-servicing workflow, rewards fulfillment platform, or enterprise reporting cube. Those systems consume or reconcile component output; they do not become part of the component build.

The classification needs three amendments before it can serve as an NPV optimization architecture. First, it should separate *state variables* from *cash-flow identities*. Purchases, cash advances, payments, lines, promotional balances, delinquency, and inactivity are state or flow variables; interest income, interchange, rewards cost, fraud, servicing, and credit losses are cash-flow consequences of those states. Second, it should separate *forecasts* from *treatment effects*. A high forecast of spend on the new card is not the same as incremental bank spend after cannibalization, as Section 3.3 emphasized. Every component must therefore produce $Y_i(a) - Y_i(0)$, not only $Y_i(a)$. Third, it should add several missing economic items: interchange and merchant-category economics, rewards

and signup-bonus liability, fraud and dispute losses, servicing and collections expense, funding and liquidity cost, capital cost, tax/accounting treatment, fair-lending and adverse-action constraints, and relationship spillovers to deposits or other products.

4.3.1 Loss Module: PD, LGD, and EAD

The loss block is the most standard part of the proposed taxonomy. Credit scoring and behavioral scoring literatures model the probability that a borrower or account will default or enter delinquency using application, behavioral, bureau, and macro variables [43, 59, 79]. For a card account, a horizon- h expected-loss identity is

$$EL_{i,t,h}(a) = PD_{i,t,h}(a) LGD_{i,t,h}(a) EAD_{i,t,h}(a), \quad (65)$$

with the incremental object

$$\Delta EL_{i,t,h}(a) = \mathbb{E}[EL_{i,t,h}(a) - EL_{i,t,h}(0) \mid \mathcal{I}_{it}]. \quad (66)$$

Equation (65) is an accounting/risk identity, not an independence assumption. In credit cards, PD, LGD, and EAD are linked through utilization, credit line, payment behavior, delinquency state, collections treatment, and macro conditions. A line increase may raise activation and spend, but it can also raise utilization and EAD. A balance-transfer promotion may raise near-term balances and interest income, but it changes both the timing of EAD and the risk mix of accounts.

The PD component should therefore be a transition or hazard model, not a static score detached from the card lifecycle:

$$\Pr(D_{i,t+h} = 1 \mid Z_{it}, X_{it}, B_{it}, P_{it}, L_{it}, M_{gt}, a), \quad (67)$$

where D is default or charge-off, Z is lifecycle state, B balance, P payment behavior, L line, and M macro state. Months-on-book and vintage should enter this equation because newly originated accounts season differently from mature accounts. That point is operationally important for campaign NPV: an apparently low-loss first-six-month window may simply precede the period in which a vintage reaches peak delinquency.

LGD should be modeled as a recovery and charge-off severity process conditional on default:

$$LGD_{i,t,h} = 1 - \frac{\mathbb{E}[\text{Recoveries}_{i,t:t+h} \mid D_{i,t+h} = 1, \mathcal{I}_{it}]}{\mathbb{E}[EAD_{i,t,h} \mid D_{i,t+h} = 1, \mathcal{I}_{it}]}. \quad (68)$$

The LGD literature emphasizes that recovery rates vary with collateral, seniority, macro state, and workout environment [73]; unsecured credit-card LGD must additionally account for collections strategy, bankruptcy, hardship programs, and sale of charged-off debt. For a marketing NPV engine, LGD is usually less sensitive to the offer than EAD and PD, but it is not constant across macro states or customer segments.

EAD must be tied to the balance and line module:

$$EAD_{i,t,h}(a) = B_{i,t+h}^{\text{principal}}(a) + B_{i,t+h}^{\text{nonprincipal}}(a) + \text{Accrued}_{i,t+h}(a) + \text{UndrawnDraw}_{i,t,h}(a). \quad (69)$$

This is where principal versus non-principal balance separation belongs. The model should maintain a balance subledger because purchase principal, cash advances, balance transfers, fees, interest, and promotional balances have different APRs, payoff rules, recoverability, revenue recognition, consumer behavior, and regulatory treatment. Rolling them into one balance can make a promotion look profitable by mixing low-margin teaser balances with high-margin revolving balances or by assigning losses to the wrong balance type.

4.3.2 Balance, Line, and Usage Module

The balance block is the mechanical bridge between usage, revenue, and loss. At monthly frequency, a card account should satisfy a stock-flow identity such as

$$B_{i,t+1} = B_{it} + S_{it}^{\text{purchase}} + S_{it}^{\text{cashadv}} + BT_{it} + Fee_{it} + Int_{it} - Pay_{it} - Credit_{it} - Chargeoff_{it} - OtherAdj_{it}, \quad (70)$$

where purchases, cash advances, balance transfers, fees, interest, payments, credits, charge-offs, and adjustments are separately retained. This identity is not optional in a production NPV model: without it, the model can forecast sales volume, balances, interest income, and losses that are mutually inconsistent.

The behavioral forecasting pieces around the identity are:

$$\Pr(Z_{i,t+1} \mid Z_{it}, X_{it}, a), \quad (71)$$

$$\mathbb{E}[S_{ij,t+1}^{\text{purchase}} \mid Z_{i,t+1}, X_{it}, M_{gt}, a], \quad (72)$$

$$\mathbb{E}[Pay_{i,t+1} \mid B_{it}, S_{it}, Z_{it}, X_{it}, M_{gt}, a], \quad (73)$$

$$\mathbb{E}[L_{i,t+1} \mid L_{it}, X_{it}, B_{it}, D_{it}, a], \quad (74)$$

with utilization $Util_{it} = B_{it}/L_{it}$. The household-finance evidence in Section 3.2 enters (72) and (73); the liquidity and credit-line evidence in Gross and Souleles [42] enters (74); the payment-choice and rewards evidence in Section 3.3 enters (71) and (72).

The proposed list of balance variables is mostly right but should be amended. “Originations” should be split into application approval, booked account, initial line, activation, and first use. “Sales volume” should be category-level purchase volume, not only total spend, because interchange, rewards, disputes, fraud, and macro cyclicalities vary by merchant category. “Lines” should be modeled as a bank action and constraint, not only a descriptive variable. “Cash advance” should be separated from purchases and balance transfers because it has different fees, APR, risk selection, and customer intent. “Interest fees” should be decomposed into finance charges, late fees, annual fees, balance-transfer fees, cash-advance fees, and fee waivers, especially after the CARD Act evidence that regulation and disclosure alter fee economics [3].

4.3.3 PPNR and Contribution Module

PPNR is a useful finance lens because it separates operating revenue and expense from credit losses. In a card NPV engine, however, it should be defined at account-month level before aggregation:

$$PPNR_{it}(a) = NII_{it}(a) + NIR_{it}(a) - NIE_{it}(a), \quad (75)$$

where

$$NII_{it}(a) = Int_{it}(a) - FundCost_{it}(a), \quad (76)$$

$$NIR_{it}(a) = Interchange_{it}(a) + FeeIncome_{it}(a) + OtherIncome_{it}(a), \quad (77)$$

$$NIE_{it}(a) = RewardCost_{it}(a) + ServicingCost_{it}(a) + FraudLoss_{it}(a) + DisputeCost_{it}(a) + CollectionCost_{it}(a) + AcqCost_{it}(a). \quad (78)$$

A bank-level stress and planning view is consistent with this separation [15, 16]; the marketing and CLV literatures support discounting future net contribution [11, 71, 87]. For our purposes, the monthly incremental contribution is

$$\Delta\Pi_{it}(a) = \Delta PPNR_{it}(a) - \Delta EL_{it}(a) - \Delta K_{it}(a) - \Delta Tax_{it}(a), \quad (79)$$

and the customer NPV is the discounted sum of (79), plus any one-time acquisition or conversion costs not already embedded in monthly PPNR.

Deposits should be included, but not as an undifferentiated PPNR line item. For a card-only campaign, deposits are a relationship spillover:

$$\begin{aligned}\Delta RelValue_{it}(a) = & \Delta DepositMargin_{it}(a) + \Delta LoanMargin_{it}^{other}(a) \\ & + \Delta FeeIncome_{it}^{other}(a) - \Delta Servicing_{it}^{other}(a).\end{aligned}\tag{80}$$

This term should be estimated only when the campaign plausibly changes relationship depth, primary-bank status, or deposit behavior. Otherwise, adding deposit PPNR to card NPV risks attributing organic household banking value to a card offer. In the existing-client setting, randomized holdouts are especially important because relationship value is strongly confounded by pre-existing engagement.

4.3.4 Cost-Cutting and Portfolio-Management Overlays

Cost-cutting and portfolio-management topics are not separate objectives from NPV; they are control and segmentation layers around the same component engine. Inactivity/attrition should be a state-transition model,

$$\Pr(Z_{i,t+1} \in \{\text{dormant}, \text{closed}\} \mid Z_{it}, X_{it}, S_{it}, B_{it}, Service_{it}, a),\tag{81}$$

building on duration and customer-base models [20, 35, 72]. Its NPV role is to shorten or lengthen the cash-flow horizon and to trigger retention, line-management, or closure actions. Cost-cutting based on inactivity is only valid if it does not destroy positive future option value or create adverse selection in the remaining portfolio.

Promotional accounts should be represented by explicit promo states:

$$PromoState_{it} = (APR_{it}^{promo}, \text{expiry}_{it}, B_{it}^{promo}, \text{reward liability}_{it}).\tag{82}$$

This is necessary because signup bonuses, teaser APRs, balance transfers, and deferred-interest offers create cash-flow timing that differs from ordinary revolving balances. A promo account can have high early sales volume and negative NPV if reward and funding costs arrive before durable profitable balances or if post-promo attrition is high.

Product conversion should be modeled as a treatment-induced transition among products:

$$\Pr(Product_{i,t+1} = p' \mid Product_{it} = p, Z_{it}, X_{it}, a),\tag{83}$$

with the incremental value measured against organic conversion. This is the same causal problem as new-card issuance: the bank should not credit the conversion campaign for customers who would have converted anyway, and it should subtract cannibalized annual fees, rewards economics, or portfolio balances from the old product.

Months-on-book and vintage should be required dimensions, not optional reporting labels. They enter PD, spend, payment, attrition, and profitability because accounts season after origination, campaigns are launched in different macro states, and underwriting or pricing policies change across cohorts. A production reporting cube should therefore support segmentation by at least product, channel, campaign, offer terms, vintage, months-on-book, FICO or risk band, geography, macro state, line/utilization band, revolver/transactor status, principal/non-principal balance type, promotional status, relationship depth, and protected-class proxy monitoring where legally appropriate. Flexible segmentation is a reporting and governance requirement; it should not imply that every segment receives a separate unconstrained model. Hierarchical or pooled models are often preferable when segment-level data are sparse.

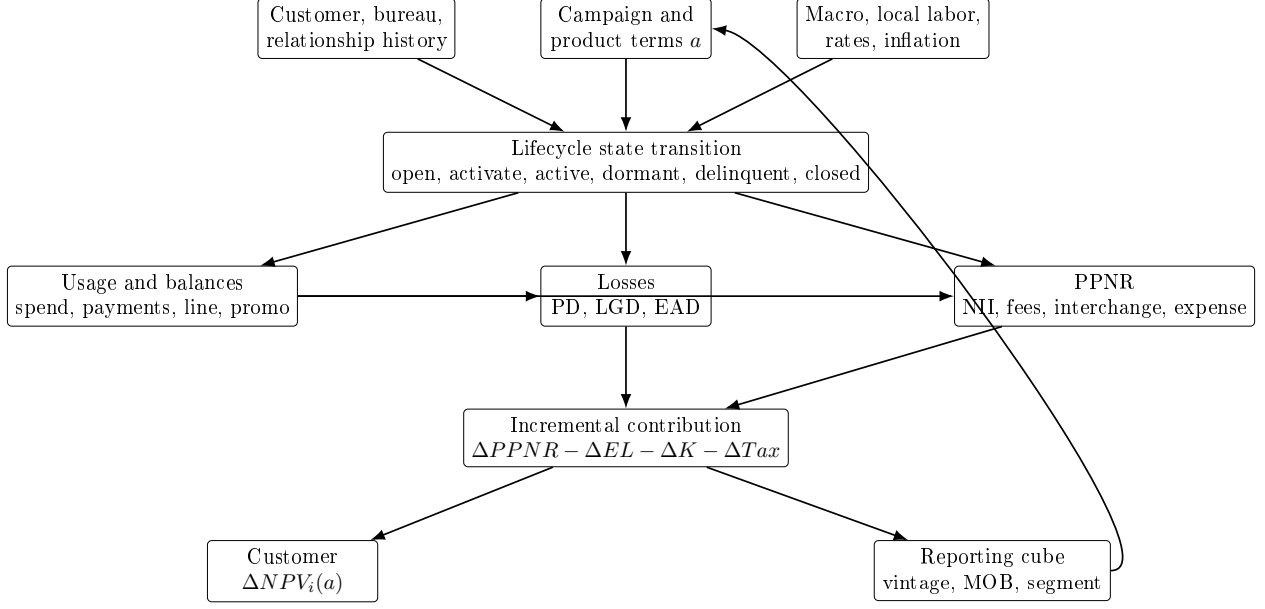


Figure 6: Schematic architecture for a component-based incremental NPV engine.

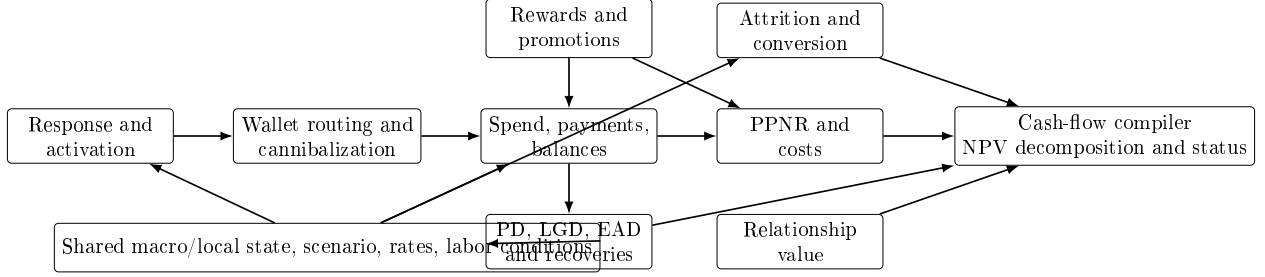


Figure 7: Model dependency graph. The modules are not independent plug-ins: wallet routing affects usage, usage affects balances and losses, rewards affect both spend and cost, macro state affects multiple modules, and the cash-flow compiler is where accounting and valuation semantics are enforced.

4.3.5 Schematic System Design

Figure 6 shows the intended system architecture. The important feature is that customer state, bank action, and macro state feed all behavioral modules; the finance layer then imposes accounting identities and converts simulated paths into incremental cash flows. The reporting layer is downstream of the model, so segmentation can be flexible without changing the definition of NPV.

4.3.6 Connection Back to Incremental NPV

The component architecture is useful only if every module feeds the same counterfactual cash-flow primitive introduced in Equation (9). The final object should be

$$\Delta NPV_i(a; d, s, \pi) = -C_i^{\text{acq}}(a) + \sum_{h=0}^H \delta_h \mathbb{E}[\Delta CF_{i,t+h}(a, \pi; s) \mid \mathcal{I}_{id}] + \delta_H \mathbb{E}[\Delta TV_{i,t+H}(a, \pi; s) \mid \mathcal{I}_{id}], \quad (84)$$

where each Δ is relative to no offer, no conversion, or the relevant business-as-usual action. The module-level PPNR, expected-loss, capital, tax, and relationship-value terms enter through ΔCF ; acquisition cost and terminal value are shown separately because they are often governed by different finance assumptions. The component handoff should be readable as a chain of economic gates rather than as a grid.

Response and activation

- Required outputs: $\Delta \text{Pr}(\text{open})$, $\Delta \text{Pr}(\text{active})$, and $\Delta \text{Pr}(\text{first use})$.
- NPV role: these probabilities gate all later spend, balance, expense, loss, and terminal-value paths.
- Main danger: a response or activation lift is a proxy unless it is reconciled to later cash flows.

Usage and balances

- Required outputs: incremental purchases, cash advances, balance transfers, payments, lines, promotional balances, and utilization.
- NPV role: these objects drive interchange, rewards, finance charges, funding cost, EAD, and capital.
- Main danger: new-card activity can be cannibalized from an old bank product or debit activity.

Losses

- Required outputs: ΔPD , ΔLGD , ΔEAD , and ΔEL under the same simulated path.
- NPV role: expected loss is subtracted from PPNR and also informs capital and downside-risk constraints.
- Main danger: booked-account risk models do not automatically transport to prospects or rejected applicants.

PPNR

- Required outputs: incremental net interest income, non-interest income, and non-interest expense.
- NPV role: PPNR supplies pre-loss contribution before expected loss, capital, tax, and labeled relationship spillovers.
- Main danger: high spend can look profitable if rewards, servicing, fraud, funding, and capital are omitted.

Attrition and conversion

- Required outputs: incremental dormancy, closure, retention, and product-migration paths.
- NPV role: these paths determine cash-flow duration and whether value is new, retained, or cannibalized from another bank product.
- Main danger: a retention or conversion action can pay customers who would have stayed or converted organically.

Relationship module

- Required outputs: incremental deposit margin and other-product contribution where the card action plausibly causes them.
- NPV role: relationship value is added only as a separately labeled term with its own evidence grade.
- Main danger: organic household quality can be misattributed to the card action.

Reporting segmentation

- Required outputs: aggregation by product, vintage, months-on-book, risk, geography, promotion state, and balance type.
- NPV role: segmentation supports governance, monitoring, and cost actions.
- Main danger: a reporting segment is not itself a new optimization target or an unconstrained model.

This framing also clarifies what should be amended in the original classification. Losses, balances, and PPNR are essential, but the production system should explicitly include rewards, interchange, fraud/disputes, servicing and collections, capital and funding, taxes/accounting, relationship spillovers, and causal cannibalization. The model should preserve the principal/non-principal subledger and vintage/months-on-book dimensions because they affect both economics and governance. Finally, cost-cutting decisions should be evaluated as actions inside the same NPV system rather than as external heuristics; an inactive account, a promotional account, or a product conversion is valuable only to the extent that its incremental discounted contribution is positive after losses, costs, and cannibalization.

4.4 Model Implementation Blueprint for the Replacement Component

The preceding sections define the economic object and the module decomposition. This section translates them into an implementation blueprint. It is not a platform build plan. It is the design of the NPV component itself: how the component constructs analytic bases, estimates module paths, connects modules through accounting identities, propagates evidence grades, calibrates to finance and risk, and returns a governed value object to existing bank systems.

4.4.1 Analytic bases by decision context

The component should not have one universal training table. It should have decision-context analytic bases, each built from the information legally and operationally available at the decision time. This is the practical way to avoid post-treatment leakage and funnel-selection bias.

The prospect or prescreen base should contain every eligible record at the campaign feature-cut time, including suppressed controls, contacted nonresponders, responders, applicants, declines, approvals, booked accounts, inactive accounts, and charge-offs. Its target objects include response, application, approval conditional on application, booking, activation, first use, early spend, early balances, campaign cost, and observed-horizon incremental value. The baseline action is suppress or no-offer.

The applicant and underwriting base should contain submitted applications with application-time bureau, internal, fraud, policy, and offer data. It supports approve, decline, refer, line, APR, and terms valuation inside approved credit policy. It should not use post-booking payment, utilization, or delinquency variables as if they were known at application time. Declined and referred

applications should remain in the base for selection analysis, even when long-run card behavior is unobserved.

The booked-account base should begin at booking and preserve activation, digital-wallet provisioning, first use, early transaction behavior, card-on-file events, rewards, statement cycles, balances, and payments. It supports onboarding, activation, usage persistence, early loss, and early dormancy modules. It should be used only for decisions made after booking unless a transportability argument has been approved.

The active-account base should contain monthly or cycle-level transaction, balance, payment, line, delinquency, rewards, servicing, fraud, dispute, and account-management histories. It supports line changes, retention, product conversion, repricing, promotional treatment, closure, and collections-related valuation where permitted. Because active accounts are selected survivors, the base should carry survival and treatment-policy lineage rather than being treated as representative of prospects.

The relationship base should link card accounts to customer, household, deposit, debit, ACH/payroll, other-product, digital-engagement, and service records subject to legal and permitted-use constraints. It supports all-bank incremental value, internal cannibalization, debit substitution, and relationship-value diagnostics. Relationship outcomes should enter causal card NPV only when the action’s effect on those outcomes is identified.

Each base should store as-of date, decision timestamp, feature-cut timestamp, outcome-window start and end, macro-data vintage, policy-bundle identifier, valuation-semantics identifier, caller use case, assignment probability where applicable, and the source lineage of every major feature. Without these fields, the component cannot distinguish a forecast from a causal effect or a valid score from a replay-impossible artifact.

4.4.2 Baseline and challenger model classes

Every material module should begin with an interpretable baseline. Baselines are not a concession to simplicity; they are controls that let model-risk, finance, risk, and engineering reviewers understand whether a more flexible model is adding real value or exploiting leakage and selection.

Funnel transitions should have logistic, multinomial-logit, hazard, or discrete-time survival baselines for response, application, approval, booking, activation, first use, active use, dormancy, closure, delinquency, default, and charge-off. Spend, transaction count, and ticket-size modules should have hurdle or two-part baselines that separate probability of use from conditional amount. Balance and payment modules should have stock-flow baselines that enforce Equation (70). Loss modules should have scorecard, logistic, survival, or competing-risk baselines for PD, separate severity models for LGD, and exposure models tied to line and utilization. Attrition and duration modules should include survival or customer-base baselines consistent with the CLV literature [20, 35, 72]. Finance baselines should be identity-driven: if behavior is held fixed, PPNR, loss, funding, capital, rewards, and cost calculations should reproduce approved finance definitions.

Challenger models should be added only where they answer a specific weakness of the baseline. Gradient boosting, random forests, neural networks, or other machine-learning models may improve prediction of nonlinear spend, payment, loss, fraud, attrition, or cost paths. Causal forests, uplift models, and double/debiased machine-learning estimators may help estimate heterogeneous treatment effects when the identification design supports such claims [25, 88]. Dynamic panel or state-space models may help when lagged utilization, payment behavior, and policy feedback are material [7, 14]. Dynamic treatment-regime or Markov decision-process models may help when downstream account-management policy is central [66, 68].

The component should keep baseline and challenger ledgers. For each module, the ledger should

state the target, decision context, population, feature stage, model class, training window, validation window, allowed use, evidence grade, and veto diagnostics. A challenger can replace a baseline for production only when it improves the primary criterion without failing calibration, leakage, stability, support, causal-validity, or reconciliation vetoes. A better AUC or validation loss is not enough for a valuation module.

4.4.3 Monthly path engine and cash-flow compiler

The component should construct monthly paths rather than one-period summary scores. A path engine makes three things explicit: the state transition, the stock-flow identity, and the downstream policy. At each month h , the engine updates lifecycle state, line, balance subledgers, payment, utilization, rewards, delinquency, losses, servicing, fraud, and attrition under action a , scenario s , and downstream policy π :

$$\hat{S}_{i,t+h+1}(a) \sim \hat{P}(S_{i,t+h+1} \mid S_{i,t+h}, X_{it}^d, a, \pi, M_{g,t+h+1}), \quad (85)$$

$$\hat{B}_{i,t+h+1}(a) = \mathcal{B} \left(\begin{array}{c} \widehat{B}_{i,t+h}, \widehat{Spend}_{i,t+h}, \widehat{CashAdv}_{i,t+h}, \widehat{BT}_{i,t+h}, \\ \widehat{Fee}_{i,t+h}, \widehat{Int}_{i,t+h}, \widehat{Pay}_{i,t+h}, \widehat{Chargeoff}_{i,t+h} \end{array} \right), \quad (86)$$

where $\mathcal{B}$ enforces the balance stock-flow identity. The same engine should run the candidate action and the baseline action so that incremental value is a path difference, not a difference between two unrelated model runs.

The cash-flow compiler converts each monthly state into component cash flows: interchange by merchant category, finance charges by balance type, fees, reward cost, acquisition cost, servicing, fraud, dispute cost, collections cost, funding cost, expected loss, capital charge, tax convention, and approved relationship value. The compiler should be deterministic conditional on the behavioral path and assumption bundle. This separation lets modelers improve behavioral forecasting without silently changing valuation semantics, and it lets finance owners update funding, capital, rewards, or cost conventions without retraining behavioral models.

The path engine should support deterministic expected-value paths and simulation paths. Expected-value paths are useful for high-volume batch scoring and reconciliation. Simulation or bootstrap paths are useful for uncertainty, downside risk, tail loss, and rank-stability analysis. In either mode, the component response should expose the horizon split, terminal value, discount factors, and decomposition residual.

4.4.4 Calibration and reconciliation

Calibration should happen at several levels. Module calibration asks whether activation, spend, balances, payments, attrition, PD, LGD, EAD, rewards cost, fraud, servicing, and losses match observed outcomes by horizon. State calibration asks whether transition rates among prospect, booked, activated, active, dormant, delinquent, closed, and charged-off states match observed paths. Vintage and months-on-book calibration ask whether new accounts season correctly. Segment calibration asks whether the model is reliable by product, channel, offer, risk band, geography, relationship depth, promotion status, revolver/transactor state, and balance type. Macro-regime calibration asks whether the model remains coherent across unemployment, rate, inflation, income, and credit-cycle changes.

Finance reconciliation is a separate gate. The component should tie aggregate and segment-level PPNR, rewards, fees, funding, capital, fraud, servicing, loss, recoveries, and acquisition cost to approved finance and risk definitions within pre-agreed tolerances. Where the model's valuation

semantics differ from finance reporting, the difference should be explained by bridge items: incremental versus total value, pre-tax versus post-tax, marginal versus allocated cost, modeled expected loss versus booked loss, current-policy versus target-policy, and terminal-value convention.

Risk reconciliation should compare PD, LGD, EAD, delinquency migration, charge-off, and recovery paths against credit-risk models and vintage experience. The NPV component does not replace risk models, but it must not produce value paths that imply impossible or unapproved risk behavior. For example, a line-increase option that raises spend while leaving utilization, EAD, PD, and capital unchanged should be treated as suspicious unless the data and policy context justify it.

Migration reconciliation should compare the replacement component to the current component. The comparison should include rank correlation, top-decile overlap, decision movement, value movement, segment aggregate tie-out, component decomposition, no-score and degraded-score rates, and disagreement classification. A change in ranking should be decomposed into behavioral model change, valuation semantics change, data-lineage change, evidence-grade change, or unsupported use-case change. That is the only way to distinguish genuine improvement from silent redefinition.

4.4.5 Evidence-grade propagation

The component’s output label should be inherited from its weakest material evidence dependency. Let each module m have an evidence grade G_m and a materiality weight $w_{im}(a)$ for customer i and action a . A module is material when changing it within its uncertainty range can change the sign, rank, or allowed use of $\Delta\text{NPV}_i(a)$. A simple propagation rule is:

$$G_i^{\text{output}}(a) = \min_{\{m:w_{im}(a) \text{ material}\}} G_m, \quad (87)$$

where the ordering is randomized, quasi-experimental, observationally adjusted, predictive, scenario-only. The ordering is not a mathematical theorem; it is a governance convention that prevents a high-quality response experiment from laundering weak cannibalization, terminal-value, or relationship-spillover assumptions into causal NPV.

The response should therefore include module-level grades and aggregate allowed-use labels. A score can be causal incremental NPV for response and activation while only policy-conditional for losses and scenario-only for relationship value. In that case, the aggregate value should not be reported as fully causal unless the scenario-only term is excluded or immaterial. If a caller requests causal action ranking and a material module is predictive or scenario-only, the component should downgrade the score, return a warning, or no-score according to the use-case contract.

4.4.6 Monitoring and refresh triggers

Monitoring should cover more than predictive drift. The component should monitor data drift, treatment-policy drift, macro drift, calibration drift, overlap and support drift, finance/risk reconciliation drift, and usage drift by caller. Data drift includes missingness, staleness, schema changes, identifier changes, feature distribution changes, and macro-vintage failures. Treatment-policy drift includes changes in campaign eligibility, channel mix, offer terms, underwriting policy, line policy, rewards, retention, collections, or closure rules. Macro drift includes unemployment, rates, inflation, income, credit cycle, and competitive changes outside the training support.

Refresh or reapproval should be triggered when a monitored quantity crosses a predefined threshold, when a major source system changes, when a caller uses a new decision context, when a new offer family or product is introduced, when funding or capital conventions change, when reward

economics change, when a material experiment supersedes prior evidence, or when finance/risk reconciliation fails. The response contract should carry enough lineage to identify which historical scores were produced under the old model, assumption bundle, evidence grade, and policy bundle.

4.4.7 Implementation artifacts

The component build should produce concrete artifacts that reviewers and engineers can inspect. The minimum set is:

- decision-context analytic-base specifications, with feature-cut and outcome-window rules;
- feature lineage and permitted-use inventory for internal, public, and licensed data;
- module model cards covering target, population, model class, training data, validation, evidence grade, limitations, and allowed use;
- valuation-semantics registry covering horizon, discounting, funding, capital, tax, rewards, costs, relationship value, terminal value, scenario, and downstream policy;
- path-engine and cash-flow compiler specifications, including balance identities and decomposition additivity checks;
- experiment and quasi-experiment ledgers connected to module updates;
- calibration and reconciliation reports by segment, vintage, horizon, and finance/risk view;
- request and response schema contracts, including status semantics, degraded-score handling, no-score handling, replay fields, and lineage hashes;
- shadow-score comparison against the current component and a disagreement register before cutover.

These artifacts are the build specification for the NPV component. They keep the scope narrow enough to be implementable while making the valuation object strong enough to plug into the larger bank ecosystem without becoming another opaque score.

5 Dynamic State-Space Model and Policy Simulation

The component has to value actions whose consequences unfold over time. A static model can rank customers under today’s observed features, but it cannot state what happens when a line, offer, retention action, or downstream policy changes future behavior. The state-space assembly gives the proposal a single forward object: under a declared policy, simulate the distribution of future states, observations, cash flows, constraints, and terminal value.

The mathematical material in this section is intentionally explicit. It is the bridge between ordinary NPV scoring and later policy simulation, offline policy evaluation, or constrained reinforcement-learning research. The proposal does not claim that production RL is ready on day one; it claims that the replacement NPV component should be built on a dynamic object that would make such analysis well posed later.

What this section establishes

- A monthly NPV score is meaningful only after the proposal states the forward law: the state being simulated, the observation process, the action, the downstream policy, the reward ledger, and the terminal convention.
- The state-space model is a disciplined approximation, not a claim that the bank observes a customer’s complete economic state. Latent wallet, liquidity, engagement, and risk appetite must be treated as beliefs with evidence grades and downgrade rules.
- Reinforcement learning is framed as a possible later research program. The day-one requirement is fixed-policy valuation and replayable simulation under approved action sets and governance constraints.

5.1 Why a Dynamic State-Space Model Is Needed

The path engine in Section 4.1 already says that the component should simulate monthly state and cash-flow paths under a declared policy. For committee review, however, that is only the starting point. A replacement NPV component must say what state it is simulating, which parts of that state are observed by the bank, which parts are latent, how modules move together, how new observations update beliefs, and what it means to evaluate a future policy. Otherwise the component remains a collection of submodels whose outputs are added together without a coherent forward law.

This section supplies that forward law. It is the most mathematical part of the proposal because it is where ordinary NPV scoring becomes policy-conditional simulation. The proposal does not claim that production reinforcement learning is ready on day one. It claims something narrower and more important: the NPV component should be built around a dynamic object that can simulate a declared policy, expose unsupported extrapolation, and later support offline policy evaluation or constrained policy-improvement research if the evidence warrants it. The control framing is anchored in sequential decision-process and dynamic treatment-regime references, in the credit-card management precedent of Bank One, and in the partial-observability and Bayesian-filtering literatures [30, 32, 51, 66, 68, 75, 76, 81, 89].

Two qualifications should be explicit before the equations begin. First, the state described below is an engineered state abstraction, not the customer’s complete economic history. The literature supports writing a Markov or belief-state model when the state is sufficient for future transition and reward prediction, but it does not prove that any particular bank feature set is sufficient. Markov sufficiency is therefore a validation question: omitted history, data lags, unobserved policy changes, and unmodeled relationship variables can break the approximation. Second, the joint transition kernel is a structured approximation built from conditional blocks. State-space and filtering references justify this modular construction as a tractable way to simulate and update a high-dimensional process [30, 32, 57, 89]; they do not make the chosen factorization exact. The component should therefore validate both the component blocks and the coherence of the simulated joint path.

The section is intentionally explicit about provenance. Some equations are reused directly from the proposal; some are adaptations that preserve the same economic object while changing notation for stochastic-process clarity; and some are new assembly derivations needed to make the entire process well defined. The committee should be able to see which is which.

To avoid a common source of confusion, this section separates three different policy objects. The symbol π^{down} denotes the downstream operating policy bundle assumed in valuation: future line-management, repricing, retention, collections, closure, hardship, and conversion rules. The symbol μ denotes the historical logged behavior policy that generated observed actions. The symbol φ denotes

a candidate or challenger decision rule to be simulated, evaluated, or, much later, optimized. The separate symbol π^{gov} denotes the governance and admissibility bundle that defines hard eligibility, compliance, policy, operational, and allowed-use restrictions. Earlier proposal equations sometimes use π generically for the declared downstream policy; in this assembly, policy comparisons should state explicitly whether they mean π^{down} , μ , φ , or the governance bundle π^{gov} .

5.1.1 Crosswalk to existing proposal sections and equations

The assembly in this section follows the existing proposal as follows.

1. **Directly reused valuation objects.** The global valuation contract follows Section 2.3, especially Equations (2), (3), (4), (5), and (6). Those equations already define the state vector, one-period cash-flow primitive, policy-conditioned NPV, and spend-source accounting objects used here.
2. **Directly reused module equations.** The macro, product-term, usage, wallet, balance, loss, PPNR, attrition, promo, and conversion equations follow the proposal’s own module sections, especially Equations (14), (17), (18), (19), (21), (25), (26), (27), (34)–(37), (44), (45), (46), (65), (67), (68), (69), (70), (72), (73), (74), (75), (76), (77), (78), (79), (80), (81), (82), and (83).
3. **Directly reused simulation semantics.** The monthly path engine, cash-flow compiler, calibration, and evidence-grade logic follow Section 4.4.3 and Equations (86) and (87).
4. **Directly reused policy-value semantics.** The dynamic policy and downstream-policy logic follow Section 5.5.8 and Equations (158) and (159).
5. **New assembly derivations.** The observed/latent state partition, the observation equations, the Bayesian filtering recursion, and the constrained belief-state Bellman recursion are new derivations in this proposal. They are introduced because the earlier sections name the ingredients of a dynamic system, while this chapter assembles them into a full state-space model.
6. **New component-interface implication.** The same assembly also implies callable interfaces for policy simulation, action-grid scoring, offline policy evaluation, and policy registration. Those interfaces are stated below because the bank is replacing an NPV component, not writing an academic control model in isolation.

5.2 State, Belief, and Feasible Actions

The first construction task is to define what the bank knows when it acts. In credit-card valuation, that information is neither a fully observed Markov state nor an unconstrained feature table. It is a time-stamped decision state, augmented by beliefs about quantities that matter economically but are only partially observed: outside wallet, routing frictions, repayment stress, promotion sensitivity, attrition propensity, and relationship spillover.

The reader should interpret the state below as the proposal’s best operational information state. In an exact POMDP the full posterior over latent state is a sufficient statistic for control [51, 75]. In the bank’s implementation, the posterior will be compressed into calibrated summaries. That compression is acceptable only if the component can show that the summaries preserve the behavior and value-relevant information needed for the approved use case.

5.2.1 Timing, information, and filtration

The following timing contract is a new assembly step. It makes explicit the information structure that is only implicit in Sections 3.2, 3.3, 4.3, and 4.4.3. Writing this timing contract explicitly is standard in partially observed sequential decision models [51, 75].

Within month t , the dynamic system evolves as follows.

- T1.** At the opening of month t , the bank has pre-action history H_{it}^- and a posterior belief b_{it}^- over the latent state.
- T2.** The bank observes the decision-time feature state O_{it} constructed from internal ledgers, bureau information, relationship records, approved external data, and all release-vintage macro variables legally available at the decision time.
- T3.** The bank chooses action $a_{it} \in \mathcal{A}_t(O_{it}, b_{it}^-; d, s, \pi^{gov})$, where the feasible set is decision-specific and already governed by underwriting, compliance, pricing-floor, contact, capital, and operations rules.
- T4.** Conditional on the full latent and observed state and on action a_{it} , within-month activation, purchase, routing, balance, payment, delinquency, reward, attrition, collections, and relationship outcomes are realized.
- T5.** Raw signals $Y_{i,t+1}$ are then observed with the source-system lags, partial observability, and measurement noise specific to each module.
- T6.** The posterior is updated to $b_{i,t+1}^-$, and the one-period reward is compiled through the same cash-flow semantics already defined in Equation (3).

The filtration should be written in two steps. The pre-action information set is

$$\mathcal{F}_{it}^- = \sigma(H_{it}^-, O_{it}, b_{it}^-), \quad (88)$$

and the action must be measurable with respect to $\mathcal{F}_{it}^-$:

$$a_{it} = \varphi_t(O_{it}, b_{it}^-, d, s, \pi^{down}) \quad \text{or} \quad a_{it} \sim \varphi_t(\cdot \mid O_{it}, b_{it}^-, d, s, \pi^{down}). \quad (89)$$

After the action and within-month observations are realized, the post-observation information set is

$$\mathcal{F}_{i,t+1}^+ = \sigma(\mathcal{F}_{it}^-, a_{it}, Y_{i,t+1}, O_{i,t+1}), \quad (90)$$

where $O_{i,t+1}$ is the next decision state constructed from raw source-system signals and approved feature rules. This is the state-space version of the proposal's earlier downstream-policy language and uses standard partially observed control logic: the bank acts on information available at decision time, not on the unobserved latent state itself [51, 75].

5.2.2 Observed state, latent state, and full economic state

The observed and latent partition starts from the existing proposal's global economic state vector in Section 2.3, especially Equations (2) and (5)–(6). The partition itself is new because the proposal already relies on quantities such as outside wallet, movable wallet, latent repayment condition, and relationship spillover that are economically material but not fully observed at decision time. The methodological justification for lifting those objects into a belief-state control model comes from the POMDP and state-space literatures [32, 51, 75, 89].

The proposal's economic state vector was

$$S_{it} = (Z_{it}, B_{it}, L_{it}, P_{it}, D_{it}, Rwd_{it}, Rel_{it}, M_{gt}), \quad (91)$$

which already appears in Equation (2). For the full state-space specification, split this into observed and latent blocks.

Let the observed decision state be

$$O_{it} = (G_{it}, B_{it}, L_{it}, P_{it}, D_{it}, Rwd_{it}, Rel_{it}^{obs}, M_{it}^{obs}, C_{it}), \quad (92)$$

with the symbols unpacked as follows.

- G_{it} is the lifecycle regime: prospect, targeted, applicant, approved, booked, activated, active, dormant, delinquent, collections, closed, or charged-off, depending on the use case.
- B_{it} is the vector of observed balance subledgers, retained separately so purchases, cash advances, balance transfers, fees, accrued interest, and charged amounts are not collapsed into one scalar.
- L_{it} is the observed line and term state, including current credit line, APR schedule, fee terms, and promotional term status.
- P_{it} is observed payment behavior, such as payment amount, payment rate, minimum-payment compliance, or payoff behavior.
- D_{it} is the observed delinquency/default/cure state.
- Rwd_{it} is the observed reward and promotion state: earn schedule, signup-bonus status, promo APR state, and reward liability indicators.
- Rel_{it}^{obs} is the observed relationship state: current deposits, loans, product holdings, service interactions, and other directly observed cross-product relationship features.
- M_{it}^{obs} is the observed macro and local-labor signal vector available at decision time, including only information released by the feature-cut date.
- C_{it} is decision-context metadata, such as channel, approved offer family, feature-cut timestamp, analytic-base tag, and eligibility flags.

The point of Equation (92) is that the proposal's existing economic state is being partitioned into an explicit bank-observed vector before any latent-state augmentation is added. Each coordinate belongs in O_{it} because it is either a retained internal ledger state, an approved feature derived from retained source systems, a released external series available by the feature-cut date, or decision-context metadata needed to replay what the bank knew when it acted. If a field is observed only after the action or reconstructed with future information, it is an outcome or audit variable, not a decision-state coordinate.

The balance subledger should remain explicit rather than compressed:

$$B_{it} = (B_{it}^{pur}, B_{it}^{cash}, B_{it}^{BT}, B_{it}^{fee}, B_{it}^{int}, B_{it}^{chg}), \quad (93)$$

so that purchase principal, cash-advance balances, balance-transfer balances, fees, accrued interest, and charged amounts keep the different economics already required by Equation (70).

The latent economic state is a new assembly derivation, but every block is tied back to proposal objects that are already present in the literature and module sections:

$$U_{it} = (M_{it}^{lat}, W_{it}^{out}, \Theta_{it}^{route}, \Theta_{it}^{repay}, \Theta_{it}^{attr}, \Theta_{it}^{promo}, Rel_{it}^{lat}). \quad (94)$$

The symbols in Equation (94) are deliberately modular.

- M_{it}^{lat} is a latent macro or local-economic regime. It is useful when the bank wants to smooth noisy releases, nowcast local conditions, or distinguish the underlying economic state from the release-vintage variables actually observed at decision time.
- W_{it}^{out} is outside wallet or outside balance opportunity, i.e. competitor-card or nonbank payment/borrowing opportunity not currently on the bank's card products.
- Θ_{it}^{route} is the latent routing and card-on-file friction state: issuer preference, stored-card position, and other unobserved frictions affecting how cardable spend is allocated across payment rails.
- Θ_{it}^{repay} is the latent repayment, liquidity, and distress state not fully summarized by observed balances, payments, or bureau variables.
- Θ_{it}^{attr} is the latent attrition, dormancy, and retention propensity state.
- Θ_{it}^{promo} is the latent promotion-response and promotion-decay state, governing how teaser APRs, bonuses, and rewards affect future behavior beyond what is directly observed.
- Rel_{it}^{lat} is the latent relationship-spillover state, capturing cross-product depth or future relationship value not fully represented by current observed deposits, loans, or service variables.

The purpose of Equation (94) is not to claim that these coordinates are already identified. It is to make explicit where the proposal's unobserved economic drivers live in the stochastic process. The coordinates fall into two classes. Θ_{it}^{repay} , Θ_{it}^{attr} , and Θ_{it}^{promo} are latent versions of concepts already used by the proposal's payment, attrition, and promotion modules. W_{it}^{out} , Θ_{it}^{route} , and Rel_{it}^{lat} are more clearly closure variables: they are introduced because wallet capture, routing, and relationship spillover affect NPV but cannot be fully represented with observed bank-card ledgers alone. Their inclusion should therefore carry stronger evidence-grade and scenario-boundary labels.

The full economic state is

$$\tilde{S}_{it} = (O_{it}, U_{it}). \quad (95)$$

Equation (95) simply says that the full state is the union of the bank-observed state vector and the latent economic state. Nothing mysterious is intended here: if a quantity is observed at decision time it belongs in O_{it} , and if it is economically material but not directly observed it belongs in U_{it} .

The control object is not a point estimate of U_{it} but the posterior belief state

$$b_{it}^-(u) = p(U_{it} = u \mid H_{it}^-, s, \pi^{gov}). \quad (96)$$

Here u is a generic point in the latent-state space. Equation (96) means that before the action is chosen, the bank does not know the true latent state; it only has a posterior distribution over plausible latent states given all observed history and the declared scenario/governance bundle. This is the object the policy should depend on in the mathematically correct formulation. That claim follows directly from the belief-state control literature: under partial observability, the posterior distribution is the sufficient control object, while point estimates are only approximations [51, 75]. The bank may implement a finite-dimensional approximation to Equation (96), but the conceptual sufficient statistic is the posterior distribution itself. For readability, later equations usually write b_{it} or simply b for this pre-action belief unless the distinction between pre-action and post-observation beliefs is material.

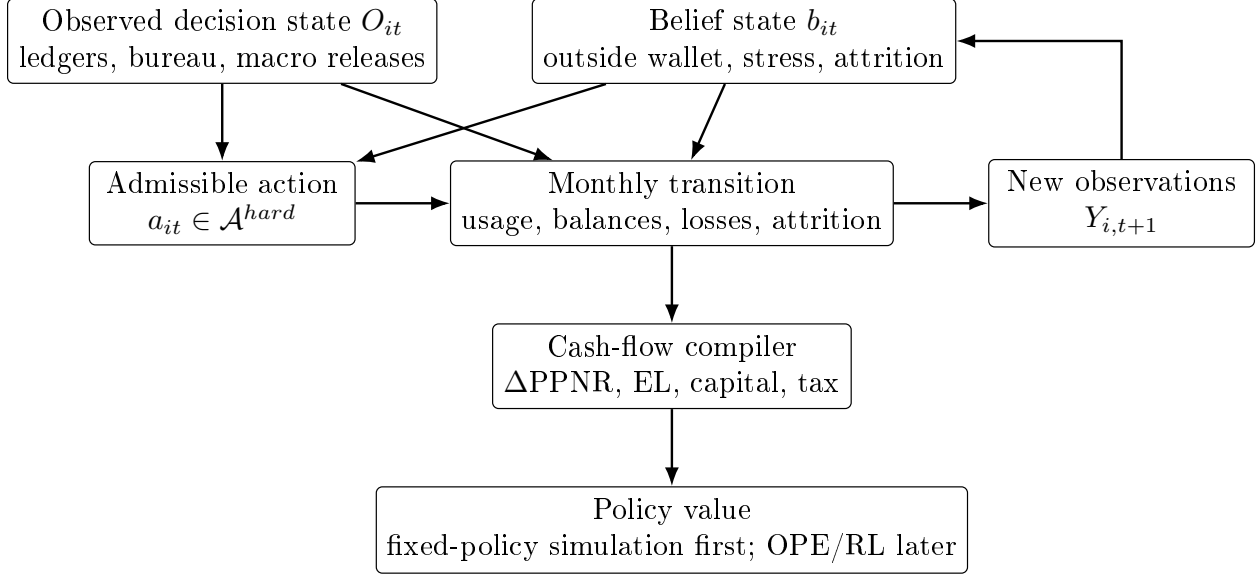


Figure 8: Belief-state valuation loop: the component values admissible actions using observed state, calibrated latent beliefs, controlled transitions, observations, and a cash-flow compiler. The loop supports fixed-policy simulation before any optimization language is promoted.

Operational belief approximation. The first production version should not promise exact high-dimensional POMDP filtering. It should expose a finite-dimensional belief summary whose elements are interpretable, calibrated, and tagged by evidence grade. A useful initial belief vector is

$$\hat{b}_{it} = (\hat{p}_{it}^{primary}, \hat{W}_{it}^{out}, \hat{p}_{it}^{promo}, \hat{p}_{it}^{revolver}, \hat{p}_{it}^{stress}, \hat{p}_{it}^{attr}, \hat{M}_{it}^{lat}, \hat{Rel}_{it}^{lat}), \quad (97)$$

where the entries denote posterior or calibrated estimates of primary-card adoption, outside wallet, promotion-only behavior, revolver state, liquidity or payment stress, attrition/dormancy, latent macro regime, and relationship spillover. Equation (97) is not a claim that those coordinates are uniquely correct. It is the bridge between the belief-state theory and a replacement component that must be built, validated, and explained. The response should also carry uncertainty summaries, such as posterior variance, bootstrap intervals, scenario ranges, or evidence-grade flags, for belief entries that materially move NPV.

This operational belief vector is therefore not a theorem-level sufficient statistic. It is a production approximation. Its adequacy should be checked by calibration of the belief coordinates, path-simulation backtests, sensitivity of NPV to alternative belief summaries, and stability of action rankings under reasonable compression choices. If a use case is sensitive to an unobserved quantity whose belief coordinate is weakly calibrated, the response should be degraded, scenario-only, or no-score rather than labeled causal incremental NPV. The approximation errors that matter are not abstract posterior-distance errors. They are errors that change valuation or decisions: materially miscalibrated belief coordinates, unstable NPV decomposition, reversal of action rankings, understated tail loss, or materially different constraint-cost estimates under reasonable alternative belief summaries. The bank does not need an exact POMDP posterior to report diagnostics, but it does need an adequately calibrated information state before using a latent block for action ranking.

5.2.3 Constrained action space by lifecycle regime

The feasible-action set follows the decision-specific objective logic of Equation (155) and the governed product-term action vector in Equation (14). What is new is the placement of that action menu inside a belief-state transition system. The literature support here is for sequential control language and policy classes, not for any issuer-specific action menu [66, 68, 81].

Let the feasible action correspondence be

$$a_{it} \in \mathcal{A}_t(O_{it}, b_{it}^-; d, s, \pi^{gov}), \quad (98)$$

where the dependence on d and π^{gov} enforces the current proposal’s claim that targeting, underwriting, line assignment, account management, retention, collections, and conversion are different decision problems with different feasible action menus.

The product-term action vector still follows directly from Equation (14):

$$a_{it} = (\ell_{it}, r_{it}^{pur}, r_{it}^{cash}, r_{it}^{BT}, r_{it}^{promo}, T_{it}^{promo}, f_{it}, q_{it}, b_{it}^{signup}, d_{it}^{disc}). \quad (99)$$

No new economics is introduced here; the point is to embed the proposal’s own action semantics into the state-space model.

5.3 Transition Blocks: How the State Evolves

Once the state and feasible actions are defined, the component needs a joint transition system. The following blocks are not independent silos. They are a readable decomposition of one monthly transition: macro conditions influence spend and payment capacity; usage and routing create balances; balances, payments, line, and macro state drive delinquency and exposure; rewards and promotions affect usage and costs; attrition and conversion determine whether future cash flows continue.

Formally, the full transition kernel could be written as

$$p(\tilde{S}_{i,t+1}, Y_{i,t+1} \mid \tilde{S}_{it}, O_{it}, a_{it}, s, \pi^{down}),$$

but estimating that object without structure is not realistic. The proposal’s module blocks are a factorization of this kernel into economically interpretable conditional laws. The factorization must preserve stock-flow identities, timing, and shared shocks. It should be stress-tested by asking whether simulated purchase, payment, balance, delinquency, recovery, attrition, rewards, and relationship paths remain jointly plausible, not merely whether each module looks acceptable in isolation.

5.3.1 Macro block

This block follows the proposal’s macro/state-dependence arguments in Sections 3.2 and 3.2.4, especially Equations (11), (12), (13), and (25). The latent/observed macro split and the explicit state-space transition below are new assembly derivations added so that the proposal’s macro-sensitive spend, payment, loss, and attrition language becomes an actual stochastic-process block. The literature support from state-space/Bayesian filtering sources justifies the latent-release structure, while the economic content still comes from the proposal’s own macro equations and their cited empirical papers [8, 32, 41, 42, 50, 89].

The macro state should be common across modules rather than reintroduced as an unrelated control in every equation. A customer in a worsening local labor market may spend less, pay more slowly, draw more on available credit, become more likely to become delinquent, and generate

different funding and capital economics. Treating those effects as unrelated regressors can produce internally inconsistent scenarios. Data-rich macro and factor-model literatures support using lower-dimensional latent common states to summarize broad macro conditions [12, 13, 77]. In this proposal, that literature justifies the common-shock architecture; the bank must still estimate the relevant local and portfolio-specific elasticities.

The intended architecture is shared by geography and time. Customers in the same geography-time cell see the same released macro signals and the same latent local-economic factor, while customer-specificity enters through exposures, lifecycle state, balances, income proxies where permitted, product terms, and interaction terms. This is not merely a smoothing device for noisy releases: it is the common shock that should feed spend, payment, delinquency, attrition, funding-sensitivity, and stress-scenario modules. If the latent macro block does not improve scenario calibration, vintage stability, or regime-transfer backtests relative to an observed-scenario layer, it should be removed or downgraded to an observed macro-control block rather than retained as an unnecessary latent process.

A convenient baseline macro specification is

$$M_{i,t+1}^{lat} = A_M M_{it}^{lat} + \eta_{t+1}^M, \quad \eta_{t+1}^M \sim \mathcal{N}(0, Q_M), \quad (100)$$

$$M_{i,t+1}^{obs} = H_M M_{i,t+1}^{lat} + \nu_{t+1}^M, \quad \nu_{t+1}^M \sim \mathcal{N}(0, R_M), \quad (101)$$

where the observed macro vector may include local labor, rate, inflation, income, and other release-vintage signals available at decision time. Equation (100) is not meant as the unique macro model. It is the minimal state-space representation needed to turn the proposal’s macro sensitivity language into an explicit stochastic process. The general state-space and Bayesian-filtering justification for writing a latent macro block with noisy observed releases comes from the state-space literature [32, 89]; the specific economic drivers still follow the proposal’s own macro equations and citations in Section 3.2. The elasticities linking M^{lat} and M^{obs} to category spend, payment rate, delinquency, attrition, and loss severity remain bank-specific empirical parameters. They must be estimated with geography-time variation, vintages, stress periods, quasi-experiments, and backtests; the state-space notation alone does not identify them.

5.3.2 Usage, category spend, routing, and wallet block

This block follows and adapts the proposal’s usage and wallet sections, especially Equations (25), (26), (27), (5), (6), (44), (45), and (46). The proposal already says that card use, routing, cannibalization, and outside wallet are distinct objects; the new content here is the explicit stochastic routing system and latent wallet process needed to make those accounting objects dynamic. The supporting literature adds weight to the adoption/use distinction and to the need for latent-state or belief-state treatment under partial observability [2, 51, 53, 67, 75].

First, lift the category-spend equation into a controlled state transition for latent cardable spending opportunity:

$$\log(1 + C_{ij,t+1}) = \mu_i^C + \kappa_j + \beta_j^{M^\top} M_{i,t+1}^{lat} + \chi_j^\top O_{it} + \omega_j^\top (M_{i,t+1}^{lat} \otimes O_{it}) + \epsilon_{ij,t+1}^C, \quad (102)$$

which is an adaptation of Equation (25). The adaptation is explicit: the proposal’s category spend $S_{ij,t+1}$ becomes a latent cardable-spend opportunity $C_{ij,t+1}$ so that routing and wallet shares can be modeled separately.

Second, retain the proposal’s adoption/use logic but place it inside the monthly transition sys-

tem:

$$A_{im,t+1}^* = Z_i' \alpha_m + \gamma_m^\top O_{it} + \xi_{im,t+1}, \quad A_{im,t+1} = \mathbf{1}\{A_{im,t+1}^* > 0\}, \quad (103)$$

$$U_{ijm,t+1}^{use} = X_{ijm}' \beta_m + \phi_m A_{im,t+1} + \rho_m^\top (\Theta_{it}^{route}, \Theta_{it}^{promo}) + \varepsilon_{ijm,t+1}, \quad (104)$$

which follows Equations (26) and (27) but explicitly inserts latent routing and promo state. The usage block should be read as staged, not as one structural demand equation. There are four distinct objects: account opening or possession, activation and first use, repeated-use intensity conditional on having an account, and transaction-level routing across rails. Equation (102) models category-level opportunity before routing. Equation (103) models possession or enabled-account status for instrument m in the relevant decision context. Equation (104) models conditional use intensity or payment-rail utility after possession and activation states are defined. These equations are a reduced-form and semistructural shell for staging the objects; they are not yet a fully identified structural demand system.

Third, convert the proposal's spend-source accounting into a stochastic routing system. Let $r \in \{\text{newbank, oldbank, outside, debit, cashother}\}$ index payment rails. Define latent routing utilities

$$V_{ij,t+1}^r(a) = \alpha_{jr} + \Gamma_{jr}^\top O_{it} + \Lambda_{jr}^\top U_{it} + \Psi_{jr}^\top a_{it}, \quad (105)$$

and corresponding shares

$$q_{ij,t+1}^r(a) = \frac{\exp(V_{ij,t+1}^r(a))}{\sum_{r'} \exp(V_{ij,t+1}^{r'}(a))}. \quad (106)$$

Then rail-specific spending is

$$S_{ij,t+1}^r(a) = q_{ij,t+1}^r(a) C_{ij,t+1}. \quad (107)$$

Equations (106)–(107) are new assembly derivations. They are not stated in the proposal; they are introduced because Equations (6) and (44) already require a routing allocation but do not make it stochastic.

Observed wallet anchors versus latent wallet states. The wallet block is economically central and empirically fragile, so the component should keep six objects separate. First, new-bank-card spend is a ledger observation once the account is open. Second, old-bank-card spend is also ledger observed for existing bank-card customers and is the main internal cannibalization anchor. Third, bank debit, deposit flows, balance transfers, payoff events, card-on-file changes, digital-wallet provisioning, and servicing signals may anchor internal substitution where data-use rules permit. Fourth, external wallet proxies such as bureau aggregates, network aggregates, account-aggregation panels, and competitor product benchmarks are intermittent or licensed anchors, not full observations. Fifth, competitor-card wallet and movable-wallet fractions remain latent unless directly identified by an experiment or approved external measurement design. Sixth, any residual outside wallet needed to close the accounting identity is model-imputed or scenario only. The component should expose these labels rather than allowing all of them to appear as one “wallet” number.

The outside-wallet state evolves as

$$W_{i,t+1}^{out} = h_W(W_{it}^{out}, M_{i,t+1}^{lat}, G_{it}, \Theta_{it}^{route}, a_{it}) + \epsilon_{i,t+1}^W, \quad (108)$$

and the proposal's movable-wallet object becomes the derived functional

$$W_{it}^{mov}(a) = m(U_{it}, O_{it}, a, d, s, \pi), \quad (109)$$

which is the state-space version of Equation (45). The proposal’s outside-wallet and movable-wallet language is therefore being recast into explicit latent-state notation; the need for such a latent or belief- state treatment is supported by the partial-observability literature, while the actual wallet equations remain proposal-specific assembly derivations [51, 75].

This is a deliberately conservative treatment of wallet. Payment-choice and reward studies support the distinction between adoption, use, routing, and behavioral response [2, 53, 67]; uplift and causal-effect methods support treatment-effect framing; state-space methods support latent-state representation. None of those sources observes this issuer’s competitor-card wallet or identifies the offer-specific movable fraction. The component should therefore report the evidence grade for each wallet term: directly observed bank activity, anchored outside-wallet proxy, bounded estimate, model-imputed value, or scenario assumption. Model-imputed outside wallet may support sensitivity analysis; it should not be labeled captured wallet or causal NPV without an experiment, quasi-experiment, or approved anchor. Old-bank-card shift and some debit substitution can often be anchored internally; competitor shift, outside-card opportunity, and movable-wallet fractions usually remain latent. If these components are weakly anchored and material to the decision, the wallet block should be excluded from causal action ranking or reported as scenario NPV with a sensitivity range.

5.3.3 Lifecycle, activation, and regime-transition block

This block follows the proposal’s funnel-state logic, the lifecycle architecture in Sections 3.3, 3.4, and 4.2, and the general transition language in Equations (63) and (64). The transition kernel below is an explicit regime-level restatement of that earlier proposal logic. The literature weight for viewing this as a sequential state problem comes from customer-base / duration models and dynamic decision-process language, while the actual lifecycle regime set and business interpretation remain proposal-specific [20, 35, 66, 68, 72].

Let the lifecycle regime evolve as

$$\Pr(G_{i,t+1} = g' \mid G_{it} = g, O_{it}, U_{it}, a_{it}, s) = p_G(g' \mid g, O_{it}, U_{it}, a_{it}, s). \quad (110)$$

Equation (110) is the regime-level version of the proposal’s lifecycle transition objects: response, application, approval, booking, activation, first use, repeated use, dormancy, delinquency, closure, and charge-off. It is not meant to collapse these stages into one empirical model. Prospect response, application, underwriting approval, booking, activation, first use, repeat use, dormancy, delinquency, and closure have different analytic bases and different counterfactuals. If the bank cannot separately estimate the material stage transitions for a use case, the lifecycle block should remain a forecasting shell rather than a causal behavioral law for action ranking.

5.3.4 Balance, payment, and line block

This block follows the proposal’s balance stock-flow identity and the behavioral bridge in Equations (70), (72), (73), and (74). The subledger recursions below are not new economic claims; they are an explicit stochastic-process unpacking of the proposal’s already stated accounting identity and line/payment modules. The literature support here comes mainly through the proposal’s own balance and line equations, plus the credit-line/liquidity evidence already cited earlier in the document [42].

$$B_{i,t+1}^{pur} = B_{it}^{pur} + S_{i,t+1}^{newbank,pur} - Pay_{i,t+1}^{pur} - Chg_{i,t+1}^{pur}, \quad (111)$$

$$B_{i,t+1}^{cash} = B_{it}^{cash} + S_{i,t+1}^{cashadv} - Pay_{i,t+1}^{cash} - Chg_{i,t+1}^{cash}, \quad (112)$$

$$B_{i,t+1}^{BT} = B_{it}^{BT} + BT_{i,t+1} - Pay_{i,t+1}^{BT} - Chg_{i,t+1}^{BT}, \quad (113)$$

$$B_{i,t+1}^{fee} = B_{it}^{fee} + Fee_{i,t+1} - Pay_{i,t+1}^{fee}, \quad (114)$$

$$B_{i,t+1}^{int} = B_{it}^{int} + Int_{i,t+1} - Pay_{i,t+1}^{int}, \quad (115)$$

$$B_{i,t+1}^{tot} = B_{i,t+1}^{pur} + B_{i,t+1}^{cash} + B_{i,t+1}^{BT} + B_{i,t+1}^{fee} + B_{i,t+1}^{int}, \quad (116)$$

which is a subledger adaptation of Equation (70). The flows indexed by $t+1$ are realized during the interval $(t, t+1]$ after action a_{it} is chosen; they are not pre-action features. For historical months, purchases, cash advances, balance transfers, fees, interest accruals, payments, and charge-offs are ledger-observed or mechanically computed from ledger records. For simulated future months, they are forecasts constrained by the same subledger identities. Any implementation that violates these identities, or fails material payment/utilization reconciliation, should be vetoed before policy use.

Payment behavior evolves as

$$Pay_{i,t+1} = g_P(B_{it}^{tot}, Util_{it}, G_{it}, D_{it}, O_{it}, \Theta_{it}^{repay}, a_{it}, M_{i,t+1}^{lat}) + \epsilon_{i,t+1}^P, \quad (117)$$

which is an adaptation of Equation (73). Line and term state evolves as

$$L_{i,t+1} = g_L(L_{it}, B_{it}^{tot}, D_{it}, O_{it}, a_{it}) + \epsilon_{i,t+1}^L, \quad (118)$$

which is the stochastic-process version of Equation (74). The functions g_P and g_L are empirical transition objects, not structural repayment or credit-supply laws unless a separate structural model is approved. They should be learned from issuer transaction, statement, line-policy, delinquency, and treatment-history data with the feature-cut discipline used elsewhere in the proposal.

5.3.5 Risk, delinquency, exposure, and recovery block

This block follows the proposal's loss module directly, especially Equations (65), (67), (68), and (69). The controlled risk transition below is the state-space re-expression of that PD/LGD/EAD structure. The literature support is the same as in the proposal's loss section: credit- risk and LGD references justify the decomposition, while the exact transition kernel and recovery dynamics remain issuer-specific empirical objects [43, 59, 73, 79].

Let the delinquency/default/cure process satisfy

$$\Pr(D_{i,t+1} = d' \mid D_{it} = d, G_{it}, B_{it}^{tot}, Pay_{i,t+1}, L_{it}, O_{it}, U_{it}, a_{it}, M_{i,t+1}^{lat}) = p_D(d' \mid \cdot), \quad (119)$$

which is the controlled state-space version of Equation (67). The conditioning on $Pay_{i,t+1}$ means this equation is written after the within-month payment realization; in implementation payment and delinquency may be estimated jointly or sequentially as long as the timing convention is fixed and replayable. Expected exposure at default remains tied to the balance and line system:

$$EAD_{i,t+1} = B_{i,t+1}^{pur} + B_{i,t+1}^{cash} + B_{i,t+1}^{BT} + B_{i,t+1}^{fee} + B_{i,t+1}^{int} + UndrawnDraw_{i,t+1}, \quad (120)$$

which adapts Equation (69). Recovery and LGD evolve as

$$Rec_{i,t+1:t+W}^{work} = g_R(D_{i,t+1}, O_{it}, U_{it}, a_{it}, M_{i,t+1}^{lat}) + \epsilon_{i,t+1}^R, \quad (121)$$

$$LGD_{i,t+1} = 1 - \frac{\delta^{work} Rec_{i,t+1:t+W}^{work}}{\max(EAD_{i,t+1}, \varepsilon_{EAD})}, \quad (122)$$

which is an explicit stochastic-process version of Equation (68). Here $Rec_{i,t+1:t+W}^{work}$ is discounted recovery over the workout window, conditional on a default or charge-off state, and ε_{EAD} only avoids division by zero in notation. In implementation, LGD should be defined only on the default/exposure population and reconciled to the bank’s approved recovery and charge-off conventions. The risk block is anchored in observed delinquency, charge-off, exposure, and recovery histories, but the causal response of delinquency, EAD, recovery, or LGD to line, APR, promotion, servicing, or collections actions is an issuer-specific estimation problem. If action-sensitive risk transitions cannot be validated, the block may support risk forecasting under current policy but should not be promoted to causal action ranking.

5.3.6 Rewards, promotions, attrition, conversion, and relationship block

This block follows Equations (34)–(37), (81), (82), (83), and (80). The new content is not a different economic story; it is the explicit coupling of those proposal blocks into one controlled transition system. The literature weight comes from the same reward, attrition, duration, and dynamic decision references already used elsewhere in the proposal [2, 20, 35, 66, 68, 72].

Promo state evolves as

$$Rwd_{i,t+1} = g_{Rwd}(Rwd_{it}, S_{i,t+1}^{newbank}, a_{it}, \Theta_{it}^{promo}) + \epsilon_{i,t+1}^{Rwd}, \quad (123)$$

$$\Theta_{i,t+1}^{promo} = A_{\Theta,p} \Theta_{it}^{promo} + \Gamma_p a_{it} + \eta_{i,t+1}^p, \quad (124)$$

so that observed promo state and latent promo sensitivity are separated. Observed reward schedules, earn, burn, bonus qualification, and promo clocks are ledger or product-term facts; Θ^{promo} is a latent response and decay state. Promo response can often be anchored with internal campaign and promotion experiments, but teaser-period lift should be downgraded unless post-promo persistence is measured or explicitly bridged.

Attrition and dormancy evolve as a controlled hazard,

$$\Pr(G_{i,t+1} \in \{\text{dormant}, \text{closed}\} \mid G_{it}, O_{it}, U_{it}, S_{it}, B_{it}, a_{it}) = p_A(\cdot \mid \cdot), \quad (125)$$

which is the state-space version of Equation (81). Dormancy, closure, inactivity, and product nonuse are observed lifecycle outcomes, while Θ^{attr} is the latent propensity that summarizes unobserved retention risk. The hazard can support forecasting from account histories, but causal retention value requires approved treatment variation or a credible quasi-experiment. Product conversion follows

$$\Pr(Product_{i,t+1} = p' \mid Product_{it} = p, G_{it}, O_{it}, U_{it}, a_{it}) = p_{conv}(p' \mid p, \cdot), \quad (126)$$

which follows Equation (83). Relationship spillover is represented as a latent and observed process block:

$$Rel_{i,t+1}^{lat} = g_{Rel}(Rel_{it}^{lat}, Rel_{it}^{obs}, a_{it}, G_{it}, M_{i,t+1}^{lat}) + \epsilon_{i,t+1}^{Rel}, \quad (127)$$

which is the dynamic counterpart to Equation (80). This representation does not promote relationship value into core card NPV by itself. The reporting hierarchy remains: core card NPV first; core card NPV plus associated relationship value as a separately labeled view; and causal all-bank relationship value only when the card action is shown to cause the relationship outcome. Without direct identification, relationship spillover should remain an associated-value or scenario view because it is especially vulnerable to double counting and pre-existing household-quality confounding.

5.4 Observation, Filtering, and Policy Value

The transition system describes the latent economic process; the bank sees it through source systems. That distinction is crucial for outside wallet, cannibalization, repayment stress, and promotion decay. The observation and filtering equations below state how raw signals become a belief state, and the reward equations state how that belief state becomes policy-conditional NPV.

5.4.1 Observation system

This part is mostly new assembly derivation. The proposal already names many observed bank quantities throughout the usage, balance, loss, and data sections; the equations below turn those named quantities into an explicit observation system. Such equations are needed for a state-space model. The methodological reason for adding them is standard state-space and Bayesian-filtering practice [30, 32, 89]; the actual signal families still come directly from the proposal’s own data and module sections.

Observed new-bank and old-bank spend satisfy

$$Y_{ij,t+1}^{new} = S_{ij,t+1}^{newbank} + \nu_{ij,t+1}^{new}, \quad (128)$$

$$Y_{ij,t+1}^{old} = S_{ij,t+1}^{oldbank} + \nu_{ij,t+1}^{old}, \quad (129)$$

with ν representing reconciliation noise or posting mismatch. These are ledger observations, not latent proxies, once the relevant bank card or account exists. The noise term is for posting timing, returns, adjustments, and reconciliation differences, not for unobserved competitor spend. Statement balances and payments satisfy

$$Y_{i,t+1}^B = B_{i,t+1}^{tot} + \nu_{i,t+1}^B, \quad (130)$$

$$Y_{i,t+1}^{Pay} = Pay_{i,t+1} + \nu_{i,t+1}^{Pay}, \quad (131)$$

and observed delinquency / charge-off states satisfy

$$Y_{i,t+1}^D = h_D(D_{i,t+1}) + \nu_{i,t+1}^D. \quad (132)$$

Statement balances, payments, and delinquency states are also ledger or servicing-system observations, although they may arrive with statement-cycle, posting, charge-off, recovery, or operational lags. Those lags should be stored as observation timing, not silently imputed as if the bank knew the value at the decision time.

Bureau refreshes are periodic and noisy observations of broader external credit condition,

$$Y_{i,t+1}^{bureau} = h_{bureau}(U_{i,t+1}, D_{i,t+1}, B_{i,t+1}^{tot}) + \nu_{i,t+1}^{bureau}, \quad (133)$$

while outside-wallet anchors such as balance transfers, payoff events, network aggregates, or account-aggregation samples are partial observations,

$$Y_{i,t+1}^{wallet} = h_{wallet}(W_{i,t+1}^{out}, \Theta_{i,t+1}^{route}, O_{i,t+1}) + \nu_{i,t+1}^{wallet}, \quad (134)$$

where missingness itself is informative. These equations are new, but they are forced by the proposal’s own claim that outside wallet, movable wallet, and competitor routing are only partially observed. The observation system is therefore strongest where the bank has retained ledger-level signals and weakest where signals arrive through sparse anchors such as bureau refreshes, network aggregates, account aggregation, or licensed external feeds. Missing bureau, wallet, or relationship observations should not be assumed ignorable by default; they may reflect refresh cadence, customer permission, product eligibility, opt-out, vendor coverage, or data-use restriction. If a latent block has no repeatable observation anchor, it should be scenario-only or heavily evidence-downgraded.

5.4.2 Bayesian filtering and blockwise posterior update

This filter is a new assembly derivation. The proposal repeatedly says that outside wallet, cannibalization, future response, and some relationship effects are only partially observed. The filter below is the mathematical consequence of those earlier claims. The Bayesian-update structure and blockwise filtering logic are supported by the POMDP and state-space literatures [30, 32, 51, 75, 89]; the specific latent blocks and observation signals remain proposal-specific. Let $O_{i,t+1} = g_O(Y_{i,t+1}, O_{it}, a_{it})$ denote the next approved decision-state construction from raw source-system signals. The posterior over latent state is updated by

$$b_{i,t+1}(u') \propto p(Y_{i,t+1}, O_{i,t+1} \mid u', O_{it}, a_{it}, s, \pi^{down}) \int p(u' \mid u, O_{it}, a_{it}, s, \pi^{down}) b_{it}(u) du. \quad (135)$$

If $O_{i,t+1}$ is a deterministic function of $(Y_{i,t+1}, O_{it}, a_{it})$, the first factor should be read as the likelihood of the raw observations plus the state-construction rule, not as a separate source of randomness.

The bank should not pretend that every block is filtered the same way. A committee-level note should say so explicitly.

- The macro block in Equations (100)–(101) admits linear-Gaussian or dynamic-linear-model filtering approximations [32, 89].
- Lifecycle and delinquency states may use exact finite-state hidden Markov filtering where the transition kernel is discrete; this is a standard finite-state partially observed control construction [51, 75].
- Outside-wallet and routing blocks are nonlinear and sparsely observed; particle or sequential Monte Carlo approximations are the most natural baseline [30].
- Payment, promo-response, and relationship-latent blocks may use either dynamic-linear-model approximations or more general simulation-based filters depending on empirical fit and tractability.

The policy input is therefore the posterior distribution or a structured approximation to it, not merely a point estimate. A posterior mean can be a feature, but it is not in general a sufficient statistic for a constrained control problem under partial observability. For implementation, it is useful to distinguish three layers: the exact posterior in theory, a structured approximate posterior used by the simulator, and low-dimensional reporting summaries returned to caller systems. The reporting summary can be useful even when it is not sufficient for control. If belief compression materially changes policy rankings, tail-risk estimates, or constraint ordering, the compressed proxy should be treated as inadequate for decision use.

5.4.3 Reward mapping and constrained belief-state policy value

The reward mapping follows Equations (3), (4), (86), (158), (159), and (155). The new content is the explicit conversion of those proposal objects into a belief-state control recursion. The literature support here is for sequential control, belief-state policy evaluation, and offline/safe challenger evaluation; the stage reward itself still follows the proposal’s own NPV cash-flow semantics [46, 49, 51, 52, 56, 66, 68, 75].

How the baseline path enters the dynamic recursion. The dynamic recursion inherits the proposal’s incremental-NPV semantics only when candidate and baseline paths are coupled under the same scenario, valuation convention, terminal-value rule, and downstream-policy assumption. In simulation, this means running the candidate action or policy and the baseline action or policy under common macro paths and common exogenous shocks where appropriate, then differencing the compiled cash flows. One-time acquisition costs, setup costs, product-conversion costs, incentives, and retention concessions enter in the period in which the action creates them; terminal value enters through the same ΔTV convention used in Equation (138). If this coupling cannot be reconciled to the proposal’s original incremental-NPV semantics, the dynamic recursion should remain a scenario-analysis object rather than a governed optimization layer.

Conditional on full state, action, next state, and realized monthly signals, the stage reward is

$$\begin{aligned} r_{i,t+1}(\tilde{S}_{it}, a_{it}, \tilde{S}_{i,t+1}, Y_{i,t+1}; s, \pi^{down}) &= \Delta PPNR_{i,t+1}(\tilde{S}_{it}, a_{it}, \tilde{S}_{i,t+1}; s, \pi^{down}) \\ &\quad - \Delta EL_{i,t+1}(\tilde{S}_{it}, a_{it}, \tilde{S}_{i,t+1}; s, \pi^{down}) \\ &\quad - \Delta Kchg_{i,t+1}(\tilde{S}_{it}, a_{it}, \tilde{S}_{i,t+1}; s, \pi^{down}) \\ &\quad - \Delta Tax_{i,t+1}(\tilde{S}_{it}, a_{it}, \tilde{S}_{i,t+1}; s, \pi^{down}) \\ &\quad + \Delta RelValue_{i,t+1}(\tilde{S}_{it}, a_{it}, \tilde{S}_{i,t+1}; s, \pi^{down}), \end{aligned} \quad (136)$$

which is a direct state-space restatement of Equation (3). The belief-state reward is then

$$\bar{r}_{it}(b_{it}, O_{it}, a_{it}; s, \pi^{down}) = \mathbb{E}[r_{i,t+1}(\tilde{S}_{it}, a_{it}, \tilde{S}_{i,t+1}, Y_{i,t+1}; s, \pi^{down}) \mid b_{it}, O_{it}, a_{it}, s, \pi^{down}]. \quad (137)$$

The expectation in Equation (137) integrates over the current latent state, the controlled transition to $\tilde{S}_{i,t+1}$, and the observation system. This matters because losses, payments, rewards, and attrition are realized over the month, not known at the pre-action decision time.

Let H denote the terminal valuation month for this section. The fixed-policy value under candidate rule φ and downstream policy bundle π^{down} is

$$\begin{aligned} V_t^{\varphi; \pi^{down}}(b_{it}, O_{it}; s) &= \mathbb{E}^{\varphi, \pi^{down}} \left[\sum_{h=t}^{H-1} \delta^{h-t} \bar{r}_{ih}(b_{ih}, O_{ih}, a_{ih}; s, \pi^{down}) \right. \\ &\quad \left. + \delta^{H-t} \Delta TV_{iH}(b_{iH}, O_{iH}; s, \pi^{down}) \mid b_{it}, O_{it} \right]. \end{aligned} \quad (138)$$

which is an adaptation of Equation (159) from fully observed state notation to belief-state notation.

The constrained Bellman recursion is the new assembly derivation needed to make Equation (155) operational in the partially observed setting:

$$\begin{aligned} V_t^*(b, O; s) &= \max_{a \in \mathcal{A}_t(O, b; d, s, \pi^{gov})} \left\{ \bar{r}_t(b, O, a; s, \pi^{down}) \right. \\ &\quad \left. + \delta \mathbb{E} \left[V_{t+1}^*(b', O'; s) \mid b, O, a, s, \pi^{down} \right] \right\} \end{aligned} \quad (139)$$

The hard feasibility constraints are pointwise:

$$a \in \mathcal{A}_t^{hard}(O, b, d, s, \pi^{gov}), \quad (140)$$

where the set encodes compliance, credit-policy, contact-permission, operational, product-availability, and current-account-status rules. Other constraints are usually cumulative or

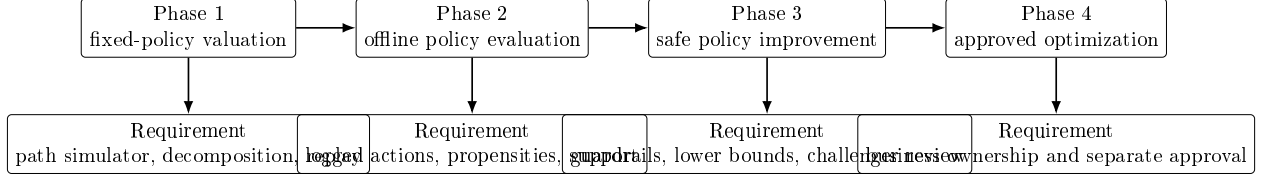


Figure 9: Optimization roadmap. The replacement component should first support fixed-policy valuation. Offline evaluation, safe improvement, and optimization are later governed extensions, not day-one production claims.

portfolio-level. For a fixed candidate policy φ , write the constraint-cost vector as

$$J_{c,k}^{\varphi}(b, O; s) = \mathbb{E}^{\varphi, \pi^{\text{down}}} \left[\sum_{h=t}^{H-1} \delta^{h-t} c_{k,h}(b_{ih}, O_{ih}, a_{ih}; s) \mid b_{it} = b, O_{it} = O \right], \quad (141)$$

with constraints such as

$$J_{c,k}^{\varphi}(b, O; s) \leq \kappa_k \quad k \in \{\text{loss, capital, liquidity, ops, complaints, fairness}\}. \quad (142)$$

The functions $c_{k,h}$ and thresholds κ_k are governance and risk appetite constructs calibrated from bank policy, regulatory/compliance review, finance/risk limits, operating capacity, complaint guardrails, and fair-lending monitoring where applicable. They are not learned freely to make the optimizer look attractive. In computation, these cumulative constraints can be handled by augmenting the state with remaining budgets, by solving a Lagrangian relaxation

$$\max_{\varphi} \left\{ V^{\varphi; \pi^{\text{down}}}(b, O; s) - \sum_k \lambda_k (J_{c,k}^{\varphi}(b, O; s) - \kappa_k) \right\}, \quad (143)$$

or by using constrained policy-optimization methods when the data and policy class justify that level of complexity [1]. The proposal should not imply that all economically material constraints are one-period inequalities; only hard admissibility is naturally pointwise.

5.5 Policy Simulation, Offline Evaluation, and RL Roadmap

The preceding equations are sufficient for fixed-policy simulation: the caller declares a policy, scenario, valuation convention, and horizon, and the component simulates paths and returns decomposed value. Reinforcement-learning language should enter only after that object exists and after the logged data support offline evaluation. The roadmap below is therefore deliberately staged.

The boundary matters institutionally. Fixed-policy valuation asks what a cohort or customer is worth under a declared policy bundle. Offline policy evaluation asks whether a challenger policy would have performed better than a logged or baseline policy, subject to overlap and support diagnostics. Policy optimization asks which rule should be chosen. The first belongs inside the replacement NPV component's initial scope. The second can be a governed analytical interface. The third should remain a later, separately approved extension, because it requires behavior-policy logging, support, safety constraints, and business ownership beyond a valuation call. MDP, dynamic-treatment-regime, off-policy evaluation, and safe-improvement literatures support this staged separation [46, 49, 52, 56, 66, 68].

5.5.1 Identification and rollout boundary for the assembled process

This final subsection does not introduce new economics; it states the evidence boundary for the full stochastic assembly. The literature supports the sequential-control framing, partial-observability language, Bayesian filtering, and offline policy-evaluation toolkit [23, 24, 30, 32, 46, 49, 51, 52, 56, 66, 68, 75, 81, 89]. It does not identify this bank’s exact latent-state coordinates, observation-noise model, reward-ledger calibration, behavior-policy propensities, or safe rollout threshold.

Therefore the assembled process should first support *fixed-policy forward simulation* and only later support offline challenger optimization. The current shorter dynamic-policy preview in Section 5.5.8 should be read as a narrative lead-in; this section is the full mathematical specification. The relevant literature supports exactly this sequencing: first define the dynamic control object and its belief-state representation, then evaluate challenger policies offline, and only then discuss safe improvement relative to a logged baseline [46, 49, 52, 56, 66, 68].

5.5.2 Component interfaces implied by the stochastic assembly

The purpose of the preceding mathematics is not to turn the NPV component into the bank’s decision platform. It is to define what the replacement component must be able to compute when a caller asks for value under a declared policy, scenario, and valuation convention. The state-space assembly implies four component interfaces.

simulate_policy. This interface asks: under a declared candidate rule φ , downstream policy bundle π^{down} , scenario s , valuation convention, and horizon H , what distribution of decomposed NPV and constraint costs is produced for a cohort? The input is a cohort snapshot, a candidate policy identifier, a baseline policy identifier, the scenario and valuation bundle, horizon, simulation controls, and model-version identifiers. The output should include expected NPV, incremental NPV relative to the baseline policy, posterior or simulation intervals where approved, cash-flow decomposition, constraint-cost decomposition, path diagnostics, support or extrapolation warnings, and replay identifiers. This is the first production use of the MDP/POMDP layer.

score_action_grid. This interface is for decision contexts with a finite feasible menu: line amount, APR/reward cell, contact/suppress, retention offer, product conversion, or approved account-management option. For each action $a \in \mathcal{A}_t^{hard}(O, b, d, s, \pi^{gov})$, the component returns Δ NPV, expected loss, EAD, capital charge, funding sensitivity, reward cost, attrition effect, and use-limit diagnostics under the declared downstream policy. The interface should never rank an action that is infeasible under the hard action set, and it should label actions that are feasible but outside the historical logged support of μ .

evaluate_policy_offline. This interface compares a candidate policy φ with the historical logged behavior policy μ or another declared baseline using observed trajectories and the model-based simulator. A trajectory record should include

$$(O_{i0}, \hat{b}_{i0}, a_{i0}, \mu(a_{i0} \mid H_{i0}), r_{i1}, c_{i1}, O_{i1}, \hat{b}_{i1}, \dots),$$

where the behavior-policy probability is logged when known and estimated with uncertainty when not known. The interface should report direct-method, importance-sampling, weighted importance-sampling, and doubly robust estimates where feasible; it should also report effective sample size, weight instability, overlap, support, and model-versus-OPE disagreement diagnostics [31, 46, 49, 80]. Over long card horizons, rare losses and deterministic historical policies can make importance weights unstable; the component should treat that as a veto diagnostic, not a cosmetic warning.

register_policy. This interface records a candidate or baseline policy as a versioned object: policy code or rule definition, action schema, eligibility and feasibility rules, intended decision context, allowed-use restrictions, downstream-policy bundle, required features, and replay metadata. Registration matters because an NPV estimate is not meaningful without the policy that generated the future path. It also prevents the same scalar value from being reused under a changed line, retention, collections, closure, or product-conversion regime.

These interfaces should be documented in the component contract before any RL language is promoted. They are also where the proposal’s evidence grades become operational: a response can be causal incremental NPV, policy-conditional NPV, diagnostic NPV, or no-score depending on the support, identification, and reconciliation status attached to the request.

5.5.3 Offline policy evaluation and staged RL roadmap

The stochastic assembly makes reinforcement-learning language possible, but it does not make production RL justified. Offline RL learns from data generated by a historical behavior policy and is vulnerable to distribution shift, extrapolation error, and overestimated value for unsupported actions [40, 54, 60]. Dynamic-treatment-regime work similarly emphasizes consistency, sequential randomization, and positivity before a sequential rule can be interpreted causally [55, 66, 74]. The correct roadmap is staged.

Phase 0: stochastic contract and fixed-policy simulator. Deliver the observed-state schema, operational belief schema, feasible-action schema, transition registry, reward and constraint-cost compiler, scenario registry, policy registry, and replayable fixed-policy simulator. No learned policy is needed in this phase.

Phase 1: rule-based policy comparisons. Use `simulate_policy` and `score_action_grid` to compare current policy against declared business alternatives: contact/suppress rules, offer cells, line grids, promo terms, retention treatments, inactivity actions, and macro- stress policies. This phase already adds value because it makes downstream policy assumptions and constraint costs explicit.

Phase 2: offline policy evaluation. Use `evaluate_policy_offline` before any challenger policy is proposed for business use. A candidate rule should not be promoted if support is weak, OPE weights explode, direct-method and doubly robust estimates disagree materially, or the policy moves substantial mass into state-action regions not covered by the logged policy μ . The output can still be useful as experiment design, but it should not be sold as decision-grade causal NPV.

Phase 3: narrow offline policy improvement. Only after Phases 0–2 pass should the team consider fitted Q iteration, Q-learning/A-learning style dynamic-treatment-regime methods, conservative offline RL, or batch-constrained policy improvement on a narrow discrete action grid [34, 40, 54, 56, 74]. A direct card-domain precedent exists for credit-limit management as an MDP, but such examples support the naturalness of the formulation, not broad production generalization [4, 81].

Phase 4: constrained pilot, not autonomous deployment. Any learned or optimized rule should first run as a controlled pilot with explicit holdouts, loss/capital/fairness/complaint/servicing veto metrics, and a pre-specified promotion rule. The NPV component should remain a valuation

and policy-simulation service. The owning marketing, underwriting, line-management, retention, or account-management system remains responsible for the final decision workflow.

Several practical concerns about NPV are legitimate. They are not objections to using NPV; they are reasons the NPV system must be explicitly staged, causal, and uncertainty-aware. The marketing and CLV literatures motivate a discounted contribution objective [11, 21, 71, 87], but the card setting adds long-horizon credit, funding, capital, and account management dynamics. The right response is not to replace NPV with response, approval, or sales volume. It is to make clear which NPV is being estimated, for which decision, on which population, and under which downstream policy.

5.5.4 NPV Is a Long-Horizon Latent Quantity

Customer NPV is unintuitive because it is not directly observed. The bank observes partial cash flows over finite calendar time; it does not observe the counterfactual no-offer path, the customer’s full outside-wallet behavior, or the terminal value of the relationship. A more honest notation is therefore

$$\Delta\text{NPV}_i(a; \theta, \pi) = \sum_{h=0}^H \delta_h \mathbb{E}_\theta[\Delta CF_{i,t+h}(a, \pi) \mid \mathcal{I}_{it}] + \delta_H \mathbb{E}_\theta[\Delta TV_{i,t+H}(a, \pi) \mid \mathcal{I}_{it}], \quad (144)$$

where θ collects model parameters and scenario assumptions, π is the future account-management policy, ΔCF is the one-period incremental cash-flow primitive from Equation (9), and ΔTV is terminal value. This notation makes the instability visible: rankings can change when assumptions about spend decay, revolve rate, attrition, default timing, recovery, funding cost, capital cost, or terminal value change.

A bank implementation should therefore report not only $\widehat{\Delta\text{NPV}}_i(a)$, but also a sensitivity or posterior distribution:

$$\left\{ \Delta\text{NPV}_i(a; \theta^{(s)}, \pi^{(s)}) \right\}_{s=1}^S. \quad (145)$$

For campaign selection, the robust decision statistic may be expected incremental NPV, lower-tail NPV, probability of negative NPV, or expected NPV subject to capital and loss constraints. This is consistent with the stress-test view that balance, revenue, and loss paths should be evaluated under macro scenarios [15, 16]. It also prevents proxy metrics such as first-90-day spend from silently becoming promotion criteria for lifetime value.

5.5.5 Card Value Is Behaviorally Heterogeneous

The second issue is that card profitability is not one-dimensional. Two customers with the same first-month spend can have opposite economics if one is a low-risk transactor with high interchange and modest rewards cost, while the other revolves promotional balances, pays slowly, uses cash advances, and later charges off. The appropriate object is a heterogeneous treatment effect over the full cash-flow vector:

$$\tau_V(x, a) = \mathbb{E}[V_i(a) - V_i(0) \mid X_i = x], \quad (146)$$

where V_i is not one outcome but the discounted contribution generated by spend, balances, payments, losses, rewards, fraud, servicing, funding, and capital. Uplift and causal-forest methods are useful for estimating heterogeneous treatment effects [25, 62, 88], and statistical treatment-rule work emphasizes choosing actions by expected welfare or payoff under heterogeneity rather than by average effect alone [65].

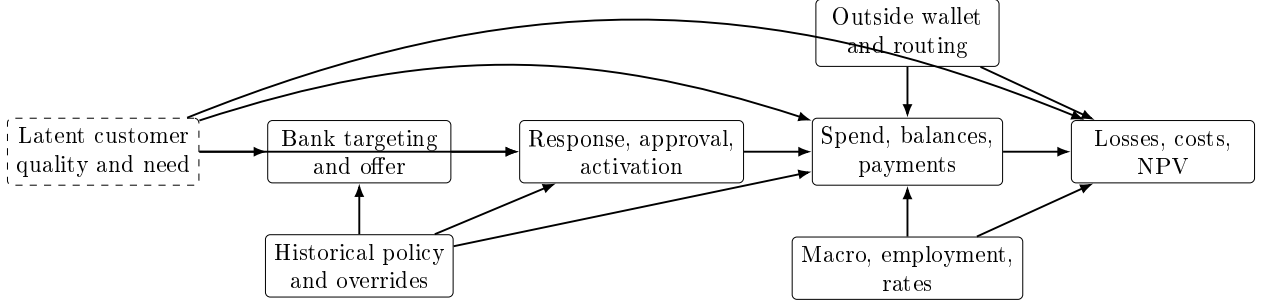


Figure 10: Causal identification map. Observed post-offer value is confounded by latent customer quality, historical targeting policy, macro state, and outside wallet/routing. Experiments or credible quasi-experiments are needed to turn observed outcomes into incremental NPV evidence.

This implies that the model should not rank customers by a single revenue or risk proxy. A useful diagnostic decomposition is

$$\begin{aligned} \tau_V(x, a) = & \tau_{\text{interchange}}(x, a) + \tau_{\text{finance}}(x, a) + \tau_{\text{fees}}(x, a) - \tau_{\text{rewards}}(x, a) \\ & - \tau_{\text{servicing}}(x, a) - \tau_{\text{fraud}}(x, a) - \tau_{\text{loss}}(x, a) - \tau_{\text{funding/capital}}(x, a). \end{aligned} \quad (147)$$

The decomposition need not be estimated by separate independent models; indeed, the components are linked through balances and states. But it should be reported, because a high average NPV segment can hide subsegments whose value comes from fragile assumptions about revolve, funding, or losses.

5.5.6 The Evaluated Population Changes Through the Funnel

The population changes as the customer moves from prospect to applicant to approved account to booked account to active user. This is a selection problem: the covariates observed, the eligible action set, and the conditional value distribution all change after each gate. The sample-selection literature is the classical warning that outcome models estimated only on selected observations need not describe the original population [44]. In the card funnel, the same issue appears in operational form:

$$\mathcal{P}_0 = \{i : \text{marketable prospects}\}, \quad (148)$$

$$\mathcal{P}_1 = \{i \in \mathcal{P}_0 : \text{respond or apply}\}, \quad (149)$$

$$\mathcal{P}_2 = \{i \in \mathcal{P}_1 : \text{approved and line assigned}\}, \quad (150)$$

$$\mathcal{P}_3 = \{i \in \mathcal{P}_2 : \text{booked and activated}\}. \quad (151)$$

The right estimand changes with the conditioning set. For example,

$$\tau^{\text{target}}(x) = \mathbb{E}[\text{NPV}_i(\text{contact}) - \text{NPV}_i(\text{suppress}) \mid i \in \mathcal{P}_0, X_i = x], \quad (152)$$

$$\tau^{\text{approve}}(x) = \mathbb{E}[\text{NPV}_i(\text{approve}) - \text{NPV}_i(\text{decline}) \mid i \in \mathcal{P}_1, X_i = x], \quad (153)$$

$$\tau^{\text{usage}}(x) = \mathbb{E}[\text{NPV}_i(\text{line, terms}) - \text{NPV}_i(\text{baseline}) \mid i \in \mathcal{P}_2, X_i = x]. \quad (154)$$

These are not interchangeable. A targeting model can use prospect and relationship data but not application-only variables. An underwriting model can use application and bureau data that were unavailable at prospect selection. An active-account management model can use payment, utilization,

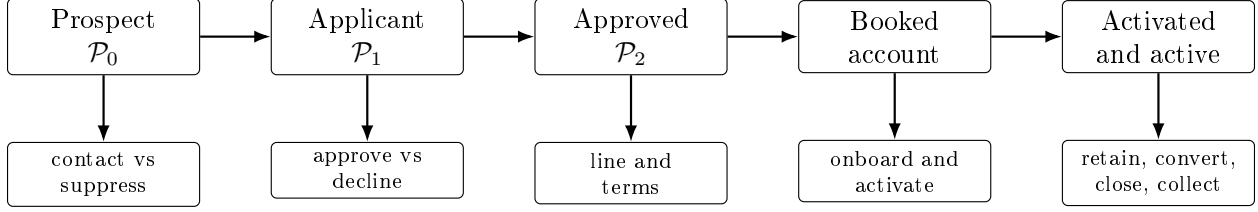


Figure 11: Funnel selection: each stage changes the population, observed data, counterfactual, and valid NPV estimand. A booked-account model is not a prospect model unless a transportability argument is supplied.

transaction, service, and delinquency history that do not exist for non-booked prospects. Therefore, a bank should maintain separate model cards, validation populations, and data-availability timestamps for each funnel stage.

5.5.7 The Counterfactual and Objective Change by Decision

The fourth issue is that “the NPV objective” is not one universal treatment contrast. It is a family of decision-specific contrasts. Direct-marketing selection asks whether contact has positive incremental value net of contact cost [21]. Portfolio budget allocation asks whether card marketing beats redeploying the same budget to another product, channel, or retention program. Underwriting asks whether approval creates value relative to decline, subject to credit policy, fairness, compliance, and risk appetite. Line assignment asks which line maximizes value over a feasible menu while respecting loss and capital constraints. Formally,

$$a_d^*(x) \in \arg \max_{a \in \mathcal{A}_d(x)} \mathbb{E}[U_d\{\Delta \text{NPV}_i(a), \text{Risk}_i(a), \text{Cap}_i(a), \text{Fair}_i(a)\} \mid X_i = x, \mathcal{P}_d], \quad (155)$$

where d indexes the decision stage and $\mathcal{A}_d(x)$ is the feasible action set at that stage. The utility U_d may be expected NPV for low-risk targeting, constrained NPV for underwriting, or portfolio NPV subject to budget, capacity, capital, and fairness constraints.

The operational version is easiest to read as five separate decision contracts. A model validated for one contract should not be reused for another without re-estimating the counterfactual and checking that the conditioning population and action set match.

Targeting

- Population: marketable prospects or existing clients.
- Counterfactual: contact versus suppress.
- Objective: incremental NPV net of campaign cost.

Budget allocation

- Population: eligible campaign universe.
- Counterfactual: card campaign versus redeploying the same budget.
- Objective: portfolio NPV per scarce marketing dollar or capacity unit.

Underwriting

- Population: applicants or responders with application data.
- Counterfactual: approve versus decline, or approve with terms versus decline.
- Objective: risk-adjusted NPV subject to credit policy, compliance, fairness, and risk-appetite constraints.

Line assignment

- Population: approved or preapproved accounts.
- Counterfactual: one feasible line amount versus alternative feasible line amounts.
- Objective: NPV over the line menu, accounting for usage, EAD, losses, funding, and capital.

Account management

- Population: booked accounts.
- Counterfactual: maintain, reprice, retain, convert, line change, or close versus the declared baseline path.
- Objective: forward NPV under updated behavior and risk state.

5.5.8 Decisions Are Interdependent Over Time

Finally, decisions are interdependent. Acquisition changes the population that underwriting sees. Underwriting and line assignment change the balance, utilization, and risk states that account management later sees. Future line, pricing, collections, retention, or conversion policies change the lifetime economics assumed at acquisition. The dynamic treatment-regime and Markov decision-process literatures provide useful language for this problem: the value of an action depends on the future policy that follows it [66, 68, 70].

A compact representation is

$$Z_{i,t+1} \sim p_{\theta}(Z_{i,t+1} \mid Z_{it}, X_{it}, a_t), \quad (156)$$

$$CF_{it} = c_{\theta}(Z_{it}, X_{it}, a_t), \quad (157)$$

$$\pi_t : (Z_{it}, X_{it}) \mapsto a_t, \quad (158)$$

with policy value

$$V_i^{\pi}(z, x) = \mathbb{E}_{\theta}^{\pi} \left[\sum_{h=0}^H \delta_h CF_{i,t+h} + \delta_H TV_{i,t+H} \mid Z_{it} = z, X_{it} = x \right]. \quad (159)$$

The acquisition model should therefore state its assumed downstream policy π : for example, line-increase rules, repricing rules, collections policy, retention offers, dormancy closure, and product-conversion policy. If those policies change, the upstream acquisition NPV should be recalibrated. This is especially important for cost-cutting initiatives: closing inactive accounts, reducing promotional offers, tightening lines, or converting products may improve near-term expense or losses while reducing the value of cohorts that were acquired under older assumptions.

The practical recommendation is to treat card NPV as a policy-evaluation system, not a one-time score. Each major decision should store the population, available variables, action set, counterfactual, downstream policy assumption, cash-flow decomposition, and sensitivity range. That metadata is not bureaucracy; it is what prevents an upstream model from optimizing a downstream world that the bank no longer operates.

6 Identification, Data, and Experimental Evidence

The valuation engine will fail if predictive convenience is confused with causal incremental value. Card data are selected at every stage: who is marketed to, who applies, who is approved, which line is assigned, who activates, who receives a retention treatment, and who survives long enough to be observed. This section collects the endogeneity, selection, data, and experiment material in one place so that the reader can see the evidence ladder rather than a set of scattered caveats.

What this section establishes

- The central statistical risk is not lack of machine-learning capacity; it is confusing observed behavior under bank policy with the counterfactual value of a new action.
- The section separates randomized evidence, credible quasi-experimental evidence, observational adjustment, prediction, and scenario analysis. Those are different evidence grades and should produce different NPV labels.
- Experiments must be admissible before they are useful: population, action set, customer impact, compliance, privacy, fair-lending, servicing, and reputation constraints can veto otherwise clean statistical evidence.

6.1 Empirical Design and Validation

6.1.1 Data requirements

A bank-grade data set should join campaign exposure, channel, creative, offer terms, application, underwriting, activation, card transactions, balances, payments, delinquency, charge-off, recoveries, fraud, disputes, servicing, relationship products, deposits, payroll indicators, bureau refreshes, and local macro series. The data should preserve timing. Variables measured after campaign exposure must not be used as if they were available at targeting time.

For the existing-client new-card problem, the data must also identify cannibalization. At minimum, the bank should measure whether new-card spend comes from:

- a. genuinely incremental customer spend;
- b. spend shifted from a competitor card;
- c. spend shifted from the bank's existing card;
- d. debit or deposit activity shifted onto credit;
- e. temporary promotion-induced spend that decays after the offer.

The first two categories are the cleanest direct sources of new issuing-bank card value, holding other relationship effects fixed. Debit or deposit substitution, internal old-card shift, and promotion timing can be positive, neutral, or negative depending on interchange, rewards, deposit economics, funding, losses, servicing, and durability.

The data set should also identify the evidence grade for outside-wallet and spend-source estimates. Directly observed or internally anchored quantities include old-bank-card spend, new-card spend, bank-observed debit activity, balance transfers into the new card, payoffs from identified accounts, card-on-file changes visible to the bank, and post-promo decay on the bank's own book. Externally benchmarked quantities may use bureau tradeline attributes, network or merchant aggregates where available, public macro and credit series, and licensed market data. Model-imputed or

scenario-only outside-wallet quantities should be retained for sensitivity analysis, but they should not be reported as captured competitor wallet or used as proof of incremental NPV.

6.1.2 Promotion criteria

The primary promotion criterion should be randomized or credibly quasi-experimental lift in incremental NPV. Response AUC, approval AUC, activation lift, first-purchase lift, and first-90-day spend are useful diagnostics, but they are proxy metrics. They should nominate models for further testing rather than establish lifetime NPV validity.

Recommended validation splits include:

- time-based holdouts for economic drift;
- campaign randomized holdouts for treatment effects;
- vintage holdouts for lifetime trajectories;
- macro-stress backtests around unemployment, rate, and inflation shocks;
- segment calibration across score bands, income proxies, geography, tenure, relationship depth, channel, and product;
- counterfactual cannibalization analysis for existing-client offers;
- validation anchors for outside-wallet and movable-wallet estimates, including balance-transfer or payoff records, old-card spend decay, debit/deposit substitution, card-on-file migration, bureau aggregates where available, account-aggregation samples, and post-promotion decay.

These validation designs should map directly to component gates. A spend model that fails macro-drift validation should not be used for scenario-sensitive campaign ranking. A causal lift model that lacks overlap or holdout evidence should not be used to claim incremental NPV, even if it predicts responders well. A booked-account model that uses post-approval payment history should not be used for prospect targeting. A decomposition that does not reconcile to finance and risk views should not be used for production ranking until the assumption difference is explained and approved. An outside-wallet model that has no validation anchor should not be used for precise dollar capture claims. A target-population diagnostic should compare prospects, contacted customers, responders, applicants, approved accounts, booked accounts, activated accounts, and active incumbents before a model trained on one population is used for another. These are veto conditions for component use, not academic footnotes.

6.2 Identification, Endogeneity, and Data Strategy

The proposal so far decomposes card value into modules: spend, usage, balances, payments, losses, PPNR, costs, capital, and relationship effects. That decomposition is necessary, but it creates a serious identification problem. Many of the variables that predict NPV are also variables that the bank controls, targets, or observes only after a selected customer passes through the funnel. A model can predict observed portfolio outcomes under the current policy and still be wrong for the decision question, “what is the incremental NPV if we take action a instead of baseline action a_0 ?”

The causal-inference literature provides the language for this distinction. Let $Y_i(a)$ denote the future cash-flow path, or one component of it, that would occur if customer or account i received action a . The component needs objects such as

$$\mathbb{E}[Y_i(a) - Y_i(a_0) \mid X_i, d, s, \pi], \quad (160)$$

not merely $\mathbb{E}[Y_i \mid T_i = a, X_i]$. The latter is an observed-outcome prediction under the historical treatment rule. It equals the causal object only under an identification argument: randomized assignment, conditional exchangeability with overlap, a valid instrument or encouragement design, a valid discontinuity, a credible difference-in-differences design, or another defensible quasi-experiment [5, 45, 47, 48, 69]. For a production NPV component, this is not a philosophical distinction. It determines whether the output may be used to rank actions or only to monitor expected outcomes.

6.2.1 Where endogeneity enters the card NPV modules

Endogeneity first appears at marketing selection. Campaigns are usually not sent randomly. Customers with better scores, deeper relationships, attractive deposit signals, prior engagement, lower expected loss, or lower contact cost may be more likely to receive an offer. If the model compares contacted customers with uncontacted customers without accounting for this assignment rule, it will credit the campaign for value that was already present. A response model has the same problem in a different form: responders are not a random subset of prospects, and their post-response value does not identify the value of contacting the marginal prospect.

The second problem is funnel selection. The evaluated population changes from prospect to applicant, applicant to approved account, approved account to booked account, booked account to activated account, and activated account to seasoned account. At each stage, variables, available outcomes, and economic meaning change. A model trained on active accounts conditions on response, approval, booking, and activation. It may be useful for servicing an active portfolio, but it does not by itself estimate prospect NPV. The sample-selection problem described by Heckman [44] is therefore not a side issue; it is central to using the component across the funnel.

The third problem is endogenous product terms. Credit line, APR, promotional duration, balance-transfer terms, rewards, signup bonus, fee waivers, and retention offers are assigned by policies that already use risk, expected profitability, capacity, customer tenure, or channel. Those terms then change usage, balances, payments, losses, rewards cost, funding cost, and attrition. Naively regressing NPV on line or APR will often recover a mixture of the customer selection rule and the causal effect of the term. In particular, high-line accounts can look safer because the bank gave high lines to safer customers, while a line increase for a marginal account can still raise EAD, utilization, and loss.

The fourth problem is simultaneity among balances, payments, utilization, and losses. Payments reduce balances, balances generate interest and funding cost, line affects utilization, utilization affects risk scores, delinquency affects collections treatment, and collections treatment affects recovery and attrition. Conditioning on variables that lie after the action, such as early utilization or first-payment behavior, can be appropriate for an account-management decision made later, but it is leakage for a prospect-targeting or initial-line decision. Dynamic panel methods such as Arellano and Bond [7] and Blundell and Bond [14] are useful warnings here: lagged outcomes and unobserved heterogeneity cannot be treated as ordinary controls without thinking through the data-generating process.

The fifth problem is endogenous attrition, dormancy, and account-management policy. An account can become inactive because the customer lost interest, the bank reduced rewards, the line was cut, service quality deteriorated, fraud or dispute experience changed the relationship, another card became more attractive, or the customer entered financial distress. Closure, retention, repricing, hardship, and conversion policies are themselves actions. If the component estimates lifetime value only among surviving accounts, it builds in survivorship bias. If it ignores future policy, it values a customer under an operating regime that may not be the one the bank will actually run.

The sixth problem is macro and geography confounding. Market conditions, employment, rates,

inflation, housing, merchant mix, and local shocks affect spend, payments, balances, and credit losses. But campaigns and policy changes also occur in calendar time and may be geographically targeted. A campaign run in a strong labor market can appear profitable because unemployment was falling, not because the offer had high incremental value. A line-tightening policy implemented during a downturn can look harmful or helpful depending on which macro controls and comparison cohorts are used. This is why the macro literature in Section 3.2 should be connected to identification, not used only as a list of predictors.

Finally, relationship value and deposit spillovers are especially vulnerable to confounding. Customers who take a premium card, keep large deposits, use digital channels, and buy other products are often already more engaged. The component should not count deposit growth, brokerage balances, or other-product profit as card-caused NPV unless there is evidence that the card action changed the relationship. Without that evidence, relationship value should be reported as an associated outcome or scenario sensitivity, not added silently to incremental card NPV.

6.2.2 Identification designs for component modules

The strongest design is randomized treatment assignment with a preserved control group. For marketing and cross-sell, this means campaign holdouts, channel holdouts, offer-cell randomization, and, where feasible, randomization over creative, rewards, promo duration, or line invitation. The component should treat those experiments as first-class data assets. It should ingest the randomization unit, assignment probability, treatment arm, eligibility rule, holdout flag, campaign calendar, and offer terms. When a randomized design is valid and sufficiently powered, the treatment effect can be estimated directly or with covariate adjustment, and heterogeneous treatment-effect methods can be used to learn which customers have positive incremental NPV.

Randomization is not always feasible. Encouragement designs can still help when the bank can randomize an invitation, reminder, channel placement, or prequalified offer while the final take-up decision remains voluntary. The instrumental-variables literature, especially the local-average-treatment-effect framework of Angrist et al. [6], is relevant because the effect is identified for customers whose treatment status is changed by the encouragement, not for every customer. The NPV component should therefore label such estimates as complier effects and should not extrapolate them to customers with very different take-up behavior without a separate model-risk approval.

When assignment is observational but rich pre-treatment data are available, propensity-score and doubly robust methods are useful starting points [48, 69]. The required component discipline is pre-treatment feature construction, overlap checking, balance diagnostics, and explicit logging of the estimand. Doubly robust or double/debiased machine-learning estimators can reduce sensitivity to flexible nuisance models [25], and causal forests can estimate heterogeneous effects [88]. These methods do not make selection vanish. They require that the variables sufficient to make treatment as-good-as-random are observed, measured before treatment, and available for both treated and control populations.

Regression-discontinuity designs can be useful around score cutoffs, credit policy thresholds, line-assignment bands, prescreen eligibility thresholds, or pricing tiers [58]. The strength of the design is local comparison near a threshold; the weakness is that the effect may not generalize far from that threshold. If the component uses such evidence, the response should state the threshold, bandwidth, population, and extrapolation status. For line assignment, this is often a better design than comparing all high-line and low-line accounts.

Difference-in-differences and event-study designs can be useful for policy changes, rewards changes, pricing changes, market shocks, or state-level implementation differences. They require a credible comparison group and pre-trend evidence. Modern difference-in-differences work shows

that staggered treatment timing and heterogeneous effects can make naive two-way fixed-effect event studies misleading [22, 78]. The component should therefore record the cohort timing, comparison population, pre-trend diagnostics, and estimator class. A failed pre-trend or placebo test should downgrade the estimate to diagnostic status.

For outcomes observed only after selection, the component should separate the selection equation from the outcome equation. A prospect-level model may need application probability, approval probability, booking probability, activation probability, and conditional lifetime value as separate components. Selection corrections in the spirit of Heckman [44] can be useful when there is a defensible exclusion restriction. More often, the practical remedy is data preservation: keep suppressed prospects, nonresponders, applicants, declines, approvals, bookings, inactive accounts, closures, and charge-offs in the analytic base, with the decision-time information available at each stage.

For sequential account-management decisions, static treatment effects are not enough. A card acquired today may later receive line increases, repricing, retention offers, hardship treatment, collections treatment, or product conversion. Dynamic treatment-regime and Markov decision-process literature [66, 68] supports the requirement that the component tag the downstream policy bundle used in valuation. If the bank changes the downstream policy, the acquisition NPV should not be assumed unchanged.

The result is a hierarchy of evidence for NPV use. Randomized experiments with preserved controls should support production treatment ranking when finance and risk reconciliation also pass. Credible quasi-experiments can support bounded production use within the identified population. Observational adjustment can support challenger models, monitoring, and carefully governed use when overlap and balance diagnostics are strong. Pure prediction should be used for expected outcome forecasting under current policy, not for causal claims about alternative actions.

6.2.3 Public and licensed external data

External data are valuable, but they should be used for the right job. They support scenario construction, macro controls, benchmarking, vintage context, competitive context, and prior distributions. They generally do not identify the customer-level causal effect of this bank’s card offer, line amount, APR, reward, or retention action. The proposal should state that boundary clearly.

Federal Reserve Enhanced Financial Accounts and Distributional Financial Accounts are useful for household balance-sheet context, distributional wealth and debt aggregates, and macro-financial scenario priors [17]. They can help segment a scenario by household balance-sheet vulnerability or validate whether aggregate portfolio behavior is plausible. They do not reveal how a particular customer would have spent on this bank’s new card under a randomized offer.

Federal Reserve G.19 consumer-credit data provide aggregate revolving and nonrevolving consumer-credit context [18]. Z.1 Financial Accounts provide broader household and financial-sector balance-sheet context [19]. These series are useful for macro controls, scenario alignment, portfolio benchmarking, and explaining whether a backtest period was credit-expansionary or credit-contractionary.

The New York Fed Household Debt and Credit Report and the underlying Consumer Credit Panel ecosystem provide benchmark information on balances, delinquency, credit-card debt, and consumer-credit transitions at aggregate and segmented levels [37]. The useful role for the NPV component is external benchmarking of delinquency, credit-card balance, and transition assumptions, especially by vintage, age, score band, or geography where public cuts are available. The role is not replacement of issuer-level PD, LGD, EAD, or activation calibration.

Labor-market and regional-economic data should feed the spend, payment, and loss modules as scenario and control variables. BLS Local Area Unemployment Statistics can provide local

unemployment and labor-force measures [85]. BEA regional accounts can provide state and local personal income, GDP, and related regional indicators [84]. Census ACS can provide demographic, income, housing, commuting, and household composition controls at aggregate geographies [86]. These data are useful for macro and geography controls, segmentation, monitoring, and fairness-risk review subject to legal and compliance governance. They should be joined using approved geography and time keys, not used to infer prohibited characteristics or make unsupported individual causal claims.

CFPB public credit-card data, including card agreements, terms surveys, and related public consumer-finance information, can help benchmark product terms, fees, APR structures, and market practices [26]. These sources are useful for competitive and policy context, especially when the component compares offer terms or explains why rewards, fees, and disclosure changes should be explicit. They are not a substitute for the bank’s own reward cost, activation, revolve, loss, and attrition experience.

FRED and ALFRED are useful data-delivery and vintage-control sources [38]. ALFRED-style real-time vintages matter because a backtest that uses revised unemployment, income, inflation, or rate data can overstate the information the bank actually had at campaign or line-decision time. The NPV component should preserve both the event date and the macro-data vintage used in training and scoring.

Because the bank has a Refinitiv/LSEG license, the component team should also inventory entitled LSEG Workspace and economic-data feeds [63, 64]. Depending on entitlement, useful categories may include rates and yield curves, inflation and macro series, market-implied recession or credit-risk indicators, consumer-sector equity and earnings data, issuer and competitor financials, credit spreads, asset-backed securities market indicators, and merchant or sector indicators. These feeds can improve funding-cost curves, discounting, stress scenarios, competitive context, and early-warning monitoring. The proposal should not assume that any particular proprietary field is licensed or approved for model use until data governance confirms entitlement, permitted use, lineage, retention, and redistribution limits.

6.2.4 External data operational inventory

The preceding discussion identifies useful public and licensed sources. For a production replacement component, that is not enough. Each external source must be operationalized as a governed feature, scenario input, benchmark, or prior. The data object should carry a source identifier, release date, vintage date, observation date, frequency, geography, transformation, join key, permitted-use classification, and model-use class. The model-use class should be one of: scenario driver, pre-treatment control, portfolio benchmark, prior or regularizer, monitoring indicator, or causal evidence. For the sources below, the default classification is never customer-level causal evidence unless a separate experimental or quasi-experimental design supplies identification.

Federal Reserve Enhanced Financial Accounts and Distributional Financial Accounts should enter as household balance-sheet and distributional macro-financial context [17]. Example variables include household debt, liquid assets, wealth distribution, and related balance-sheet aggregates. The natural frequency is quarterly or release-specific, and the natural geography is national or aggregate distributional cell rather than customer-level geography. The operational use is to construct scenario priors, stress overlays, and plausibility checks for aggregate portfolio behavior. The join path is observation month or quarter and scenario version. These data should not be used to infer an individual customer’s outside wallet or offer response.

Federal Reserve G.19 should be used for monthly consumer-credit cycle context [18]. Revolving and nonrevolving consumer credit, growth rates, and selected rate information can help label

whether a training window was credit-expansionary, credit-contractionary, or unusual. The join path is calendar month and, where available, rate series vintage. Its proper role is a macro control, scenario input, and benchmark for aggregate revolving-balance growth. It is not an issuer PD, EAD, utilization, or revolve-rate estimate.

Federal Reserve Z.1 Financial Accounts should enter as a quarterly balance-sheet and flow-of-funds benchmark [19]. Household-sector liabilities, assets, debt-service context, and financial-sector aggregates can be used to validate broad scenario consistency. The join path is quarter and release vintage. Z.1 is most useful for aggregate plausibility and stress story coherence; it is too aggregated to identify customer-level treatment response.

The New York Fed Household Debt and Credit data should support external credit-cycle benchmarking [37]. Public cuts of balances, delinquency, utilization, credit limits, originations, and transition rates, where available by score band, age, state, or product, can benchmark the bank's portfolio mix and vintage behavior. The join path is time, product, and approved segment cut, not individual customer. These benchmarks are especially useful for asking whether the bank's modeled delinquency, balance, or credit limit paths are plausible in the broader market. They do not replace internal PD/LGD/EAD calibration.

BLS LAUS and related labor-market data should feed local employment and income-stress controls [85]. Example variables include unemployment rate, labor-force participation, employment level, and local unemployment changes. The operational grain can be county, metropolitan area, state, or national, subject to legal and compliance review of geography usage. The join path should be customer or branch geography as of the feature cut and calendar month. Revised labor data can create look-ahead bias, so backtests should use available-at-decision vintages where feasible or an explicit vintage approximation when exact vintages are unavailable.

BEA regional data should provide local income and economic-activity context [84]. Personal income, wages and salaries, GDP, and regional industry composition can help distinguish local labor shocks from customer selection. The frequency is often quarterly or annual depending on the series. The join path is geography and time, with transformations such as trailing growth rates or deviations from regional trend computed only from information available at the feature cut. These features are controls and scenario drivers, not direct measures of individual spend capacity.

Census ACS data should be treated as slowly moving geographic context [86]. Variables such as income distribution, housing tenure, commuting patterns, household composition, and local demographic structure can help describe market context and fairness-risk monitoring. Their use must be reviewed for permitted-use, fair-lending, privacy, and model-risk constraints. Because ACS variables are aggregate and often lagged or multi-year estimates, they should normally be used for segmentation, monitoring, missing-context diagnostics, or fairness review rather than high-frequency customer scoring.

CFPB credit-card agreement and product-term data should support product and competitive context [26]. Agreements, APR structures, fees, promotional terms, and public product terms can help validate whether offer assumptions and reward economics are economically plausible. The join path is product family, issuer or competitor category where permitted, and effective date. These data should not be treated as observed competitor wallet, competitor authorization behavior, or customer-level treatment response.

FRED and ALFRED should be used as data-delivery and real-time-vintage controls [38]. The main operational requirement is not the specific series; it is the ability to reconstruct what a model could have known on the decision date. For rates, inflation, unemployment, income, and credit series, the component should store the release vintage used in training, scoring, and backtesting. A model that backtests with revised macro data can overstate both forecast skill and causal evidence quality.

LSEG/Refinitiv data should be inventoried only after entitlement review [63, 64]. Candidate feeds include Treasury and swap curves, credit spreads, bank funding indicators, consumer ABS market indicators, macro nowcasts or forecasts, issuer and competitor financials, merchant or sector signals, and equity-market indicators for consumer sectors. The join path may be daily valuation date, month-end scenario date, issuer, sector, geography, or product category, depending on the licensed feed. The proper NPV roles are funding and discounting curves, stress and early-warning signals, competitive context, scenario construction, and benchmark validation. Before use, the team must confirm entitlement, redistribution limits, retention rules, permitted modeling use, lineage, and whether values can be persisted in model artifacts or only referenced through controlled snapshots.

The operational implication is that external data should be stored in a separate, replayable feature family from internal customer behavior. A production score should be able to answer: which external source was used, which release and vintage were used, how it was joined, whether it entered as a scenario, control, benchmark, prior, or monitor, and whether the caller is allowed to use the output for the requested decision. If that lineage cannot be reconstructed, the score should be degraded or no-scored for decision-grade use.

6.2.5 Internal data and selection-safe construction

The bank’s internal data are the only plausible source for issuer-specific activation, usage, cannibalization, loss response, reward cost, servicing cost, and relationship spillover. The component should therefore require an analytic base table that preserves the funnel rather than only the booked portfolio. At minimum, it should include campaign eligibility, suppression rules, contact assignment, randomization and holdout flags, channel, creative, offer terms, prescreen variables, application data, approval, decline, refer, override, booking, activation, transaction, balance, payment, delinquency, charge-off, recovery, fraud, dispute, servicing, complaints, rewards, account-management treatments, deposits, payroll indicators, relationship products, bureau refreshes, and local macro joins.

The most important engineering requirement is time. Each row used for training or scoring should have an as-of timestamp, decision timestamp, feature-cut timestamp, outcome window, treatment timestamp, macro-data vintage, and policy-bundle identifier. Variables created after treatment should be available only for later decisions. For example, first-purchase behavior can help value a retention offer after activation, but it cannot be used to decide whether to target the prospect before the card is opened. This feature-cut rule is the practical defense against post-treatment leakage.

For existing-client cross-sell, the internal data should preserve all-bank spend, old-card spend, new-card spend, debit activity, deposit flows, merchant category, digital-wallet provisioning, autopay, promotional balances, balance-transfer activity, cash advances, and reward-redemption behavior. The component should estimate not only new-card spend but all-bank incremental value. If the bank cannot observe competitor-card spend, the proposal should state the missing counterfactual and use experiments, bureau aggregates where available, or sensitivity bounds rather than pretending the outside-card counterfactual is observed.

For losses, internal data must connect underwriting, line assignment, utilization, delinquency, collections, charge-off, recovery, hardship, bankruptcy, and account closure. PD, LGD, and EAD should be estimated on decision-appropriate populations. A booked-account PD model can support booked portfolio forecasting; prospect-level expected loss additionally needs application, approval, booking, and activation selection. Principal and non-principal balances, promotional balances, fees, accrued interest, cash advances, and balance transfers should be kept separate because they have different revenue, loss, payment, and accounting behavior.

For PPNR and costs, internal finance data should provide rewards expense, interchange, fees,

interest income, charge reversals, fraud expense, collections cost, servicing cost, dispute cost, acquisition cost, funding transfer pricing, capital charge, tax convention, and allocated overhead. The component should reconcile behavioral forecasts to finance definitions, but it should not become the official ledger. Differences between model cash flows and finance reporting should be explained through valuation-semantics identifiers and reconciliation reports.

The component should also preserve decision logs from current systems. The historical policy that generated treatment assignment is itself data. Marketing scores, eligibility rules, prescreen criteria, underwriting score cutoffs, manual overrides, line caps, APR matrices, channel capacity, budget constraints, collection queues, retention rules, and closure rules help explain why a customer received an action. Without those logs, observational adjustment is often weak because the most important confounder is the bank's own prior view of the customer.

6.2.6 Internal data feasibility inventory

The preceding paragraphs state the timing principle. A panel will also ask whether the required data exist at a usable grain. The answer should not be a generic assurance that the bank has rich data. The proposal should specify the classes of internal records that must be inventoried, the grain at which they are needed, and the main bias or lineage risk. The inventory below is written as a feasibility checklist for the component team; it does not assume that every source is already clean, joined, or approved for modeling.

Campaign and offer records are needed at the customer, household, channel, and campaign-cell grain. Required fields include eligibility, exclusion rules, prescreen score or selection score, exposure, suppression or holdout flag, assignment probability, channel, creative, offer cell, product, reward, signup bonus, annual fee, APR, promotional APR, balance-transfer terms, contact timestamp, and budget or capacity constraint. These fields identify the contact-versus-suppress and offer-cell counterfactuals. The main risks are lost holdout flags, re-contact contamination, overwritten offer terms, and selection scores that are not retained as of the decision date.

Application and underwriting records are needed at the application and customer grain. Required fields include application timestamp, application channel, bureau and internal attributes available at application time, approval, decline, refer, fraud screen, policy rule, score band, line assigned, APR assigned, fees, adverse-action process where applicable, and manual override. These fields define the approve-versus-decline and approve-with-terms counterfactuals. The main risks are survivorship bias from keeping only approvals, using post-booking variables in application models, and losing policy-rule or override lineage.

Account and transaction records are needed at account-day or account-month grain, with transaction detail where merchant mix and wallet movement matter. Required fields include booking, physical activation, digital-wallet provisioning, first use, repeated use, merchant category, ticket size, card-present/card-not-present, card-on-file indicators where available, cash advance, balance transfer, authorization decline, dispute, fraud flag, and transaction reversal. These fields support adoption, usage, spend source, and fraud modules. The main risks are merchant-category drift, delayed transaction posting, missing card-on-file signals, and conflating authorization activity with settled spend.

Balance and payment subledgers are needed at monthly grain and, for some validations, cycle grain. Required fields include purchase principal, promotional balance, balance-transfer balance, cash-advance balance, fee balance, accrued interest, payments, credits, charge-offs, recoveries, statement balance, minimum due, utilization, line, APR, payment due date, autopay status, and payment channel. These fields enforce stock-flow identities. The main risks are mixing principal and non-principal balances, losing promotional subledger status, treating payment behavior after an action

as if it were available before the action, and inconsistent charge-off timing.

Risk and loss records are needed at account-month, vintage, and collections event grain. Required fields include delinquency bucket, behavioral score, PD, LGD, EAD, utilization, line change, hardship treatment, collections treatment, bankruptcy, charge-off, recovery, sale proceeds, fraud loss, and closure reason. These fields support expected loss, downside risk, and capital charges. The main risks are estimating prospect losses from booked survivors, not separating EAD response from PD response, and losing the collections or hardship policy that changed LGD.

Rewards, cost, and servicing records are needed at account, transaction, campaign, and cost-center grain. Required fields include earn, burn, breakage assumption, signup bonus, statement credit, reward liability, servicing contact, complaint, dispute handling, fraud operations, acquisition cost, mailing cost, digital impression cost, underwriting cost, collections cost, charge-off sale cost, marginal cost, and allocated overhead. These fields convert activity into contribution. The main risks are using allocated cost as if it were marginal, ignoring reward liability timing, and omitting servicing or fraud costs from high-spend segments.

Relationship and deposit records are needed at customer, household, and relationship grain, subject to legal, privacy, and permitted-use constraints. Required fields may include checking balances, savings balances, debit spend, ACH or payroll indicators where permitted, direct deposit continuity, deposit attrition, mortgage or loan relationships, brokerage or wealth relationships, digital engagement, household linkage, and relationship tenure. These fields should not automatically enter card-caused NPV. They are used to measure associated value, to control for pre-existing engagement, and, only with a credible design, to estimate causal relationship spillovers. The main risk is crediting the card action for organic relationship value.

The component team should classify each field as directly observed, internally anchored, externally benchmarked, model-imputed, or scenario-only. That classification should be stored with the feature and propagated to the NPV response where it materially affects value. This is especially important for outside-wallet, movable-wallet, relationship-value, terminal-value, funding, and capital fields, where a precise-looking number can hide a weak measurement basis.

The practical data inventory is straightforward in prose. Campaign records must preserve who was eligible, who was contacted, who was suppressed, what offer terms were shown, and which experiment or policy rule created the offer. Application and underwriting records must preserve the decision-time snapshot, the approved terms, and overrides. Transaction, balance, payment, rewards, servicing, risk, loss, recovery, relationship, and deposit records must be available at the grain needed to reconstruct monthly state and cash-flow paths. Most importantly, each record must keep the timestamp and policy lineage needed to avoid post-treatment leakage and selection bias. Appendix Section B gives the implementation checklist in prose form.

6.2.7 Estimation ledgers and evidence grades

The replacement component should maintain estimation ledgers that match the objects it claims to estimate. These ledgers are not separate governance paperwork; they are the way the component prevents outside wallet, movable wallet, causal response, and NPV from being collapsed into one opaque score.

The selection and funnel ledger should compare suppressed prospects, contacted prospects, responders, applicants, declined applications, approvals, booked accounts, activated accounts, dormant accounts, and active incumbents. It should record target-population weights, overlap diagnostics, and segment calibration. If the model is trained on booked or active accounts but used on prospects, the response should expose that transportability status.

The offer and terms ledger should record contact channel, creative, reward, signup bonus, annual fee, APR, balance-transfer terms, promotional duration, line, fee waivers, repricing, retention treatment, eligibility rule, and any manual or policy override. It should also record whether each term was randomized, policy-driven, negotiated, manually overridden, or imputed. This ledger is essential because line, APR, rewards, and promotions are bank actions, not passive covariates.

The outside-wallet and cannibalization ledger should record the source of each outside spend, outside balance, movable-wallet, and spend-source estimate: directly observed, internally anchored, externally benchmarked, experimentally identified, quasi-experimental, model-imputed, or scenario-only. It should also record the uncertainty interval and any cap or reconciliation constraint, such as estimated moved spend not exceeding plausible outside eligible spend.

The dynamic behavior ledger should decompose observed changes into total spend or borrowing growth, share movement to the bank, internal cannibalization, debit or deposit substitution, promotion timing, balance build, payment response, runoff, and attrition. This ledger should be validated out of time, by vintage, and by stress or macro regime.

The causal evidence ledger should record the experiment, holdout, encouragement, cutoff, policy change, or observational adjustment supporting each treatment-response estimate. It should include overlap, balance, pre-trend, placebo, effective-sample-size, and sensitivity diagnostics. If the evidence is only associational, the output should be labeled as predictive or diagnostic, not causal incremental NPV.

The downstream-policy ledger should record the assumed future line, repricing, retention, collections, closure, hardship, and product-conversion policies. A customer acquired under one assumed future policy can have a different value under another. The NPV output should therefore carry the policy bundle identifier and should be revalidated when that bundle changes.

6.2.8 Production implications for the NPV component

The identification strategy should be visible in the component contract. Each NPV response should carry an identification design identifier, population stage, decision context, treatment definition, baseline definition, feature-cut timestamp, macro-data vintage, downstream-policy bundle, experiment or holdout identifier when available, overlap or support score, causal-evidence level, outside-wallet evidence grade, movable-wallet evidence grade, cannibalization evidence grade, target-population diagnostic, and allowed-use status. This metadata is not decoration. It tells the caller whether the number is an incremental value estimate, an expected-outcome forecast, a scenario diagnostic, or an extrapolation requiring approval.

The component should implement veto diagnostics. It should warn or refuse production action-ranking use when treatment-control overlap is poor, balance diagnostics fail, randomization is broken, pre-trends fail, placebo effects are large, post-treatment leakage is detected, the model is trained on booked or active accounts but called for prospects, macro variables use revised rather than as-of vintages, the caller asks for a decision outside the approved population, CATE estimates are unstable, or finance/risk reconciliation fails. The component should also warn when estimated moved wallet exceeds plausible outside wallet, when temporary promotion lift is being used as durable wallet share, or when a model-imputed outside-wallet value is being used as if it were directly observed or experimentally identified. These gates are how the literature becomes an operating control.

The component should also distinguish three output classes. First, *causal incremental NPV* is supported by randomized or credible quasi-experimental evidence and may be used for governed action ranking within the approved population. Second, *policy-conditional expected NPV* predicts outcomes under the historical or declared policy and may be used for planning, monitoring, and portfolio forecasting, but not as proof that a different action would create value. Third, *diagnostic*

or *benchmark NPV* uses external data, weak observational evidence, or scenario assumptions and should be labeled for sensitivity analysis or management review. A single scalar without this class label is not an acceptable replacement component.

These requirements should be part of the replacement component package. Each module should have an identification note, a data-lineage specification, an experiment and holdout inventory, a decision-stage analytic-base design, a leakage-control checklist, a public and licensed data inventory, and a validation gate that separates predictive accuracy from causal incremental value. These artifacts determine which submodel outputs can enter Equation (1) for each caller and which outputs must remain warnings, diagnostics, or assumptions.

6.3 Practical Experimental Design Program

The identification section explains why the NPV component needs causal evidence. This section specifies how that evidence should be produced and consumed. It is deliberately more formal than an experimentation roadmap because a weak experimental layer would make the entire replacement component look more precise than it is.

The literature supports mechanisms and method classes. Household-finance, payment-choice, rewards, and duration studies justify why unemployment, liquidity, rewards, credit terms, payment instruments, and customer duration belong in the model [2, 20, 35, 41, 42, 50, 53, 67, 72]. The causal inference literature gives the language for treatment assignment, instruments, regression discontinuity, difference-in-differences, heterogeneous effects, sample selection, and dynamic regimes [6, 22, 25, 44, 47, 58, 66, 68, 69, 78, 88]. None of that literature supplies this bank’s activation lift, cannibalization rate, movable-wallet share, reward elasticity, line elasticity, loss response, attrition response, or relationship spillover for a specific product, channel, segment, and policy bundle. Those parameters must be learned from internal randomized experiments, credible quasi-experiments, and explicitly downgraded observational evidence.

6.3.1 Regulatory, compliance, and reputation admissibility

For this bank, statistical identification is necessary but not sufficient. A randomized experiment can be unbiased, well powered, and still unusable if the population, contact rule, offer, credit term, disclosure path, data use, or customer-impact profile is not admissible for the decision context. The NPV component should therefore treat regulatory, compliance, and reputation constraints as part of the experimental estimand, not as a signoff step after the estimate is computed.

Let $\mathcal{A}_e^{\text{raw}}$ be the actions the business could imagine testing in experiment e : contact or suppress, one of several offers, a line amount, an APR, a reward structure, a retention action, a product-conversion path, or a servicing treatment. The experiment should not optimize over $\mathcal{A}_e^{\text{raw}}$. It should optimize or estimate only over the admissible action set

$$\mathcal{A}_e^{\text{adm}}(i, d) = \{a \in \mathcal{A}_e^{\text{raw}} : C_e^{\text{credit}}(i, a, d) \leq 0, \quad C_e^{\text{disclosure}}(i, a, d) \leq 0, \\ C_e^{\text{privacy}}(i, a, d) \leq 0, \quad C_e^{\text{fair}}(i, a, d) \leq 0, \quad C_e^{\text{reputation}}(i, a, d) \leq 0\}, \quad (161)$$

where the constraint functions are approved policy and control objects, not free modeling parameters. The exact legal implementation belongs to the bank’s legal, compliance, fair-lending, privacy, servicing, and business-control owners. The component requirement is more modest but important: every experiment whose evidence feeds NPV must record the eligible population, admissible actions, prohibited actions, customer-impact limits, disclosure and contact treatment, complaint-monitoring path, stop rules, and allowed use.

Several U.S. consumer-credit boundaries are especially relevant for the experiments contemplated here. Regulation B and the Equal Credit Opportunity Act framework are central whenever the action affects credit availability, terms, discouragement, or adverse-action processes [28]. Regulation Z and the credit-card provisions of the Truth in Lending framework matter when the experiment changes disclosures, pricing terms, fees, rate changes, payment allocation, or card issuance [29]. Prescreen and prospect-selection designs that use consumer-report information must be designed within the permissible-purpose, firm-offer, and opt-out boundary of the Fair Credit Reporting Act [82]. Consumer-harm and conduct risk are not exhausted by formal credit approval rules: unfair, deceptive, or abusive acts or practices provide a separate boundary for contact, promotion, servicing, fee, and retention experiments [83]. Complaint volume, complaint themes, dispute patterns, servicing escalations, and social or branch-channel feedback are therefore not merely operational metrics; they are veto or pause signals for customer-impacting experiments. The CFPB complaint database is a public external benchmark for complaint categories and themes, while the bank’s own complaint and servicing records are the primary measurement source for its experiments [27].

This changes the design logic in three ways. First, the experimental population is not simply the population with overlap in a propensity model. It is the population eligible under the approved business rule, credit policy, data-use policy, contact policy, and fair-lending review for that channel and date. A statistically valid estimate outside that population is not production evidence for the component. Second, the baseline cannot be chosen solely for econometric convenience. A no-contact, no-offer, no-increase, or delayed-action control must be approved as an admissible counterfactual; controls cannot be used to withhold legally required notices, benefits, servicing, or protections. Third, the analysis must separate financial value from admissible use. An experimental cell may have positive observed NPV and still be disallowed, paused, or restricted because it worsens complaint rates, dispute rates, protected-class monitoring metrics where legally appropriate, hardship impact, servicing burden, disclosure comprehension, or reputational risk.

For the NPV component, the practical rule is simple: admissibility is a veto diagnostic. If an experiment fails regulatory, compliance, privacy, fair-lending, conduct, servicing, or reputation admissibility, its numerical lift should not update causal production NPV. It may remain in a learning ledger, root-cause analysis, or scenario note, but the output label should say that the evidence is inadmissible for production valuation. This rule is especially important for later optimization or reinforcement-learning work. The feasible policy class is not all policies with high predicted value; it is the constrained class

$$\Phi^{\text{adm}} = \{\varphi : \varphi(X_i, d, e) \in \mathcal{A}_e^{\text{adm}}(i, d) \text{ for every scored unit, experiment, and decision context}\}. \quad (162)$$

Offline policy evaluation, constrained policy optimization, or dynamic programming should be discussed only within this class. A high-value policy outside Φ^{adm} is not a challenger policy; it is a rejected candidate.

6.3.2 Experimental objects and notation

For an experiment or quasi-experiment e , let $\mathcal{P}_e^{\text{raw}}$ denote the analytically reachable population at the feature-cut time and $\mathcal{P}_e^{\text{adm}} \subseteq \mathcal{P}_e^{\text{raw}}$ the population approved for the experiment after business, credit-policy, data-use, fair-lending, compliance, servicing, and reputation exclusions. Let Z_{ie} be the randomized or quasi-randomized assignment, D_{ie} the treatment actually received, a_e the treatment action, a_{0e} the baseline action, d_e the decision context, s_e the scenario and valuation-semantics bundle, and π_e the downstream policy bundle. The primary object should be expressed in the same

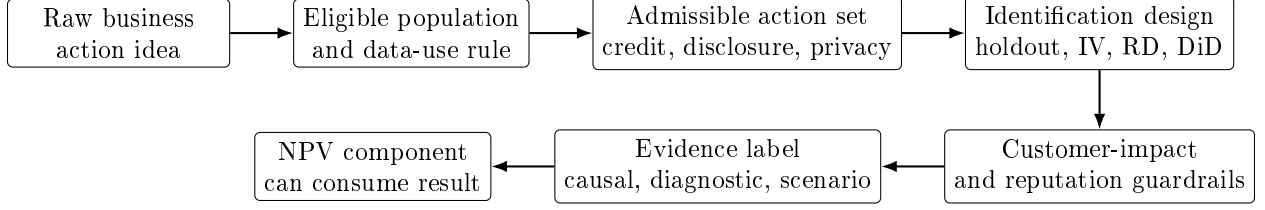


Figure 12: Experimental admissibility flow. Statistical identification starts only after the population, data use, action set, contact/disclosure path, and customer-impact guardrails are admissible for the decision context.

notation as the NPV component:

$$\tau_{\text{NPV}}^e = \mathbb{E} \left[\text{NPV}_i(a_e; d_e, s_e, \pi_e) - \text{NPV}_i(a_{0e}; d_e, s_e, \pi_e) \mid i \in \mathcal{P}_e^{\text{adm}} \right]. \quad (163)$$

This is the experiment-level analog of Equation (1). It is not the same as a response lift, activation lift, new-card spend lift, or model validation statistic. Those quantities can be module estimands, diagnostics, or bridges into NPV, but they are not the final valuation object unless the evidence contract states how they enter the NPV functional and how the remaining cash-flow terms are estimated.

For any module outcome Y such as activation, all-bank spend, revolving balance, EAD, loss, attrition, servicing cost, or deposit spillover, the corresponding intent-to-treat estimand is

$$\tau_{Y, \text{ITT}}^e = \mathbb{E}[Y_i(Z_{ie} = 1) - Y_i(Z_{ie} = 0) \mid i \in \mathcal{P}_e]. \quad (164)$$

If treatment receipt differs from assignment, the experiment may also estimate a complier or local-average-treatment effect:

$$\tau_{Y, \text{LATE}}^e = \frac{\mathbb{E}[Y_i \mid Z_{ie} = 1] - \mathbb{E}[Y_i \mid Z_{ie} = 0]}{\mathbb{E}[D_{ie} \mid Z_{ie} = 1] - \mathbb{E}[D_{ie} \mid Z_{ie} = 0]}, \quad (165)$$

under the instrumental-variables conditions of relevance, exclusion, independence, and monotonicity [6]. Equation (165) is useful, but it is not an average effect for every eligible customer. The component should label it as a complier effect and require a separate transport argument before using it for non-complier populations.

In a clean randomized design, the empirical contrast may be a simple difference in means,

$$\hat{\tau}_{Y, \text{ITT}}^e = \bar{Y}_{Z=1, e} - \bar{Y}_{Z=0, e}, \quad (166)$$

or a pre-specified covariate-adjusted version. In observational or quasi-experimental settings, the estimator changes, but the discipline should not: the proposal must still state the population, counterfactual, assignment mechanism, feature cut, horizon, identifying assumptions, and allowed use.

6.3.3 Design families and estimands

The component does not run experiments. Marketing, product, risk, underwriting, account-management, and relationship systems do. The NPV component must nevertheless specify what experiment evidence it can consume, because otherwise it will inherit proxy metrics and selection bias from caller systems.

Acquisition and prescreen contact/suppress holdouts. The first production-aligned design is a randomized contact-versus-suppress holdout over the controlled prospect or prescreen population. The estimand is

$$\tau_{\text{NPV}}^{\text{acq}} = \mathbb{E}[\text{NPV}_i(\text{contact}) - \text{NPV}_i(\text{suppress}) \mid i \in \mathcal{P}_{\text{acq}}], \quad (167)$$

net of campaign cost, rewards, expected loss, funding, capital, servicing, fraud, and the approved terminal-value convention. The assignment unit should be the unit at which interference is plausibly controlled: customer, household, relationship, address cluster, channel cell, or geography, not automatically account. Response, application, approval, booking, activation, first use, 90-day spend, and 180-day spend are diagnostic unless the evidence contract pre-specifies them as module estimands and reconciles them to Equation (163). A design powered only for response should not be promoted as evidence of positive NPV.

The contact and suppress cells must also be admissible for the prospect population. If the population is selected using consumer-report information, the experiment has to respect the approved permissible-purpose, firm-offer, and opt-out design; if the baseline is suppression, the bank must document that suppressing contact is an approved counterfactual for that population and channel. The holdout cannot be a way to deny required notices, to avoid required servicing, or to create unexplained unequal access to credit. The experiment ledger should therefore store pre-screen source, eligibility rule, firm-offer or solicitation treatment where applicable, opt-out handling, contact-frequency limits, creative/disclosure version, and fair-lending review status in addition to assignment probability and outcome data.

Existing-client new-card cannibalization holdouts. For an existing client, positive new-card spend is not the target. The target is all-bank incremental value after internal cannibalization and any observable debit, deposit, or balance-transfer substitution. The experiment should randomize eligible clients into no-offer and offer cells and estimate

$$\tau_{S,H}^{\text{new}} = \mathbb{E}[S_{iH}^{\text{new}}(\text{offer}) - S_{iH}^{\text{new}}(\text{no offer})], \quad (168)$$

$$\tau_{S,H}^{\text{bank}} = \mathbb{E}[S_{iH}^{\text{all bank}}(\text{offer}) - S_{iH}^{\text{all bank}}(\text{no offer})], \quad (169)$$

$$\kappa_H = 1 - \frac{\tau_{S,H}^{\text{bank}}}{\tau_{S,H}^{\text{new}}} \quad \text{when } \tau_{S,H}^{\text{new}} > 0. \quad (170)$$

These are population-level versions of Equations (50)–(52). The primary NPV bridge is

$$\tau_{\text{NPV},H}^{\text{xsell}} = \mathbb{E} \left[\sum_{h=0}^H \delta_h \{ \Delta \text{PPNR}_{ih} - \Delta \text{EL}_{ih} - \Delta C_{ih} - \Delta K_{ih} \} \mid i \in \mathcal{P}_{\text{xsell}} \right], \quad (171)$$

where ΔPPNR uses all-bank incremental spend and balances, not raw new-card spend. A result with high $\tau_{S,H}^{\text{new}}$ and near-zero $\tau_{S,H}^{\text{bank}}$ is evidence of substitution. It may still be strategically useful, but it should not be booked as incremental causal card NPV without a separate retention, competitor-defense, or relationship-value argument.

Reward, promotion, APR, and line experiments. Rewards, signup bonuses, statement credits, promotional APRs, balance-transfer durations, fee waivers, credit lines, and pricing terms are multi-valued treatments. If $a \in \mathcal{A}$ indexes the approved offer cell, the estimand is not a single lift but an incremental value curve:

$$g_e(a, a') = \mathbb{E}[\text{NPV}_i(a; d_e, s_e, \pi_e) - \text{NPV}_i(a'; d_e, s_e, \pi_e) \mid i \in \mathcal{P}_e], \quad (172)$$

for feasible comparisons $a, a' \in \mathcal{A}$. The design must measure both the numerator and the cost of buying it: promotion-period spend, post-promotion decay, rewards liability, interchange, revolve, payment rate, utilization, EAD, PD, LGD, attrition, servicing, funding, and capital. A teaser-period lift should enter lifetime NPV only through a horizon bridge that models post-teaser persistence and uncertainty. For lines and APRs, randomized variation must remain inside approved risk and compliance boundaries. If randomization is not available, regression discontinuity around a score, pricing, line, or prescreen threshold can be informative, but only locally around the threshold and only with bandwidth, manipulation, and continuity diagnostics [58].

This family is where admissibility most often binds. A line or APR experiment cannot randomly assign terms outside approved underwriting, pricing, ability-to-pay, disclosure, credit-policy, and fair-lending constraints merely to make an elasticity curve easier to estimate. A reward or promotion experiment should likewise pre-approve the advertised terms, eligibility, duration, expiration, fee treatment, disclosure language, complaint-monitoring plan, and remediation path if a cell proves confusing or harmful. The component may still estimate the value curve in Equation (172), but only on the approved support $a, a' \in \mathcal{A}_e^{\text{adm}}$.

Activation and onboarding experiments. For booked accounts, the relevant baseline changes. The experiment no longer asks whether the bank should acquire the customer; it asks whether an onboarding action changes durable use. Let A_{it} be an active-use state and R_{it} a repeated-use state. A first-use nudge should estimate

$$\tau_{R,H}^{\text{onboard}} = \mathbb{E}[R_{i,H}(\text{nudge}) - R_{i,H}(\text{no nudge}) \mid i \in \mathcal{P}_{\text{booked}}], \quad (173)$$

not merely first purchase within a few days. Treatments may include digital wallet provisioning, card-on-file prompts, autopay setup prompts, reminders, channel placement, or first-use incentives. The valuation bridge must subtract contact and incentive costs and should test persistence at post-nudge horizons such as 90 and 180 days. Otherwise the experiment measures operational conversion, not durable value.

Retention, dormancy, and reactivation experiments. Retention and dormancy actions are vulnerable to adverse selection because the bank often targets accounts that already show declining activity or high complaint risk. The experiment should randomize eligible accounts into no treatment and approved treatment cells such as fee waiver, reward offer, retention contact, product conversion prompt, or reactivation incentive. If $T_i(a)$ is remaining active tenure and $\Pi_{ih}(a)$ is net contribution, a retention estimand can be written as

$$\tau_{\text{NPV}}^{\text{ret}} = \mathbb{E} \left[\sum_{h \geq 0} \delta_h \{ \Pi_{ih}(a) - \Pi_{ih}(a_0) \} \mathbf{1}\{h < T_i(a)\} \mid i \in \mathcal{P}_{\text{ret}} \right]. \quad (174)$$

Duration models remain useful for the no-treatment path [20, 35, 72], but they do not identify the incremental value of concessions without a counterfactual control group.

These experiments also carry heightened conduct and reputation risk. Dormant, complaining, delinquent, recently hardship-affected, or fee-sensitive customers may be precisely the accounts where a retention action appears financially attractive and precisely the accounts where contact, fee, reward, or conversion treatment can create complaints or UDAAP risk. The admissibility record should identify vulnerable or sensitive subpopulations, required servicing exclusions, complaint-escalation triggers, maximum contact cadence, agent scripting controls, and remediation rules. A

retention cell with positive short-run NPV but elevated complaint, dispute, hardship, or servicing-escalation signals should not be promoted into production causal NPV without explicit approval and a customer-impact disposition.

Relationship-spillover holdouts. Relationship value should enter core causal card NPV only when the card action is shown to cause the relationship outcome. Let V_{iH}^{rel} denote deposit, brokerage, mortgage, digital-engagement, or other-product value that the bank proposes to include. The needed estimand is

$$\tau_H^{\text{rel}} = \mathbb{E}[V_{iH}^{\text{rel}}(a) - V_{iH}^{\text{rel}}(a_0) \mid i \in \mathcal{P}_{\text{rel}}], \quad (175)$$

measured in both treatment and control groups. Without this evidence, the relationship term should be labeled associated value or scenario value. This is especially important because high-card-value customers may already be high-engagement households; Heckman [44] is directly relevant to this selected population.

Relationship-spillover experiments also need a data-use boundary. Deposit, debit, mortgage, brokerage, servicing, digital-engagement, complaint, and household-linkage data may be analytically relevant, but relevance is not the same as permitted use. The experiment contract should state which relationship data are permitted for experimental analysis, which may be used only in aggregate monitoring, which may not be used for scoring or treatment assignment, and how customer-level outputs are separated from finance or scenario analytics where required.

Macro, geography, and policy-change quasi-experiments. Randomization is often infeasible for policy rollouts, local labor shocks, reward-program changes, rate changes, and regional operating changes. Difference-in-differences and event-study designs can then support module validation if the comparison population is credible. A schematic estimand is

$$\tau_Y^{\text{did}} = \mathbb{E}[(Y_{i,\text{post}} - Y_{i,\text{pre}}) \mid G_i = 1] - \mathbb{E}[(Y_{i,\text{post}} - Y_{i,\text{pre}}) \mid G_i = 0], \quad (176)$$

with cohort-specific or interaction-weighted estimators when treatment timing is staggered and treatment effects are heterogeneous [22, 78]. A failed pre-trend, placebo, or composition diagnostic should downgrade the evidence to scenario or diagnostic status. These designs are valuable for stress and macro-sensitivity calibration; they should not be sold as universal customer-level treatment effects.

6.3.4 Analysis specifications, not just experiment names

A panel will not accept an experiment inventory as evidence unless the analysis specification is also fixed. The NPV component therefore needs the caller experiment to produce an analysis object, not only a campaign result. For each assigned unit, the experiment should construct an observed-horizon value outcome

$$\tilde{V}_{i,H_{\text{obs}}}^e = \sum_{h=0}^{H_{\text{obs}}} \delta_h \{ \widetilde{\text{PPNR}}_{ih} - \widetilde{\text{EL}}_{ih} - \tilde{C}_{ih} - \widetilde{Kchg}_{ih} - \widetilde{Tax}_{ih} + \widetilde{\text{RelValue}}_{ih} \}, \quad (177)$$

using the approved experimental valuation semantics. The tildes matter: some terms are realized during the experiment window, some are accrued accounting quantities, and some, such as still-immature losses, may be expected values from a separately validated risk model. The contract must label which terms are directly observed, model-estimated, or excluded. The analytic base should

include all eligible assigned units, including nonresponders, nonapplicants, declines, unbooked accounts, inactive accounts, and charge-offs. Conditioning only on responders or booked accounts would change the estimand and re-create the sample-selection problem [44].

For a randomized two-cell design, the pre-specified primary analysis can be an ANCOVA or stratified regression:

$$\tilde{V}_{i,H}^e = \alpha + \tau_{\text{NPV,ITT}}^e Z_{ie} + X'_{ie}\beta + \lambda_{\text{stratum}(i)} + \varepsilon_i, \quad (178)$$

where X_{ie} contains only pre-assignment variables and $\lambda_{\text{stratum}(i)}$ represents pre-specified strata or fixed effects. The standard errors should be computed at the randomization or interference unit. With unequal assignment probabilities p_{ie} , a design-based contrast is

$$\hat{\tau}_{Y,\text{IPW}}^e = \frac{1}{|\mathcal{P}_e|} \sum_{i \in \mathcal{P}_e} \left(\frac{Z_{ie} Y_i}{p_{ie}} - \frac{(1 - Z_{ie}) Y_i}{1 - p_{ie}} \right), \quad (179)$$

with the assignment probabilities taken from the experiment ledger rather than re-estimated after the fact. Equation (178) is often more efficient; Equation (179) makes the design logic transparent. Both estimate the ITT contrast, not a selected responder value.

For multi-arm offer, line, APR, or rewards experiments, the primary regression should estimate all approved contrasts against the baseline cell:

$$\tilde{V}_{i,H}^e = \alpha + \sum_{a \in \mathcal{A}_e \setminus a_0} \tau_{a,a_0}^e \mathbf{1}\{Z_{ie} = a\} + X'_{ie}\beta + \lambda_{\text{stratum}(i)} + \varepsilon_i. \quad (180)$$

The experiment should specify which contrasts are confirmatory, which are exploratory, and how multiplicity is handled. A cell that wins on new-card spend but loses on all-bank NPV is not a positive NPV treatment. A cell that wins during the teaser period but loses after reward cost and funding should not seed the lifetime model as a durable uplift.

For encouragement designs, the analysis should report the first stage, reduced form, and ratio estimand. Let D_{ie} be actual treatment receipt. The first stage $\hat{\pi}_1$ in

$$D_{ie} = \pi_0 + \pi_1 Z_{ie} + X'_{ie}\pi_X + u_i$$

establishes relevance, while the reduced form estimates the effect of assignment on Y . The ratio in Equation (165) is a LATE for compliers under the assumptions in Angrist et al. [6]. The component should not silently convert that LATE into an average effect for never-takers, always-takers, or customers outside the take-up support.

For regression discontinuity, the analysis should be local and threshold specific. If R_i is the running variable and c the cutoff, the local linear specification is

$$\tilde{V}_{i,H} = \alpha + \tau^{\text{RD}} \mathbf{1}\{R_i \geq c\} + f_-(R_i - c) \mathbf{1}\{R_i < c\} + f_+(R_i - c) \mathbf{1}\{R_i \geq c\} + \varepsilon_i, \quad (181)$$

estimated inside a declared bandwidth with manipulation and covariate-continuity diagnostics [58]. The result should update the component only for the local threshold population unless the proposal provides a separate transport argument.

For policy rollouts and macro/geographic shocks, an event-study analysis should make the pre-trend test visible:

$$Y_{gt} = \alpha_g + \lambda_t + \sum_{k \neq -1} \beta_k \mathbf{1}\{t - T_g = k\} + \varepsilon_{gt}, \quad (182)$$

or use a modern cohort-specific estimator when treatment timing is staggered and effects are heterogeneous [22, 78]. The pre-treatment coefficients β_k should be reported as diagnostics, not hidden

behind a single average treatment effect. If pre-trends fail, the evidence can still inform scenario discussion, but it should not upgrade a module to causal decision-grade status.

Finally, if the experiment is used to train or validate a customer-level targeting rule, the object is a policy value, not just a treatment effect. Let $\varphi(X_i)$ map pre-treatment features to an action in the approved action set. Under randomized assignment with known propensity $p_i(a) = \Pr(Z_i = a \mid X_i)$, a design-based value estimator is

$$\hat{Q}(\varphi) = \frac{1}{|\mathcal{P}_e|} \sum_{i \in \mathcal{P}_e} \frac{\mathbf{1}\{Z_i = \varphi(X_i)\}}{p_i(\varphi(X_i))} \tilde{V}_{i,H}^e. \quad (183)$$

This is the experimental counterpart of using the NPV component for targeting. It is the right object for comparing scoring policies because it evaluates the action the policy would have chosen, not merely the average treatment effect. Heterogeneous-effect methods such as causal forests and double/debiased machine learning can propose φ or estimate CATEs [25, 88], while statistical treatment-rule work clarifies that the value of the rule is the decision criterion [65]. To avoid overfitting, the rule-learning sample, policy-evaluation sample, and final production population should be separated or evaluated with honest sample splitting. A high CATE validation correlation is not enough if the induced policy value fails against the baseline.

6.3.5 Identification assumptions and veto diagnostics

Each experiment family should have a pre-run identification statement. For a randomized holdout, the required condition is that assignment is independent of potential outcomes within the eligible population:

$$Z_{ie} \perp\!\!\!\perp \{Y_i(a_e), Y_i(a_{0e}), \text{NPV}_i(a_e), \text{NPV}_i(a_{0e})\} \mid i \in \mathcal{P}_e. \quad (184)$$

This condition is violated operationally if the holdout is overwritten, re-contacted, manually suppressed for value-related reasons, merged with another campaign, or lost in data lineage. It is also weakened if the randomization unit is too narrow and treatment spills over across accounts, households, branches, channels, or card-on-file arrangements.

For observational adjustment, the assumption is conditional exchangeability and overlap using pre-treatment variables:

$$\{Y_i(a_e), Y_i(a_{0e})\} \perp\!\!\!\perp D_{ie} \mid X_{ie}, \quad (185)$$

$$0 < \Pr(D_{ie} = 1 \mid X_{ie} = x) < 1 \quad \text{on the target support.} \quad (186)$$

Propensity-score, doubly robust, double/debiased machine-learning, and causal forest methods are useful only after these assumptions are made plausible [25, 48, 69, 88]. They can reduce modeling bias conditional on the design; they cannot repair missing controls, post-treatment leakage, or a no-overlap population.

For regression discontinuity, the component should record the running variable, cutoff, bandwidth, manipulation test, continuity diagnostics, local population, and extrapolation flag. For difference-in-differences, it should record treated and comparison cohorts, timing, pre-trends, placebo outcomes, composition diagnostics, and estimator class. For selection corrections, it should record the selection equation, outcome equation, exclusion restriction if used, and target population [44]. These are not appendix niceties. They determine whether the evidence upgrades a module or remains a diagnostic.

6.3.6 Power, horizon, and measurement discipline

The section should not prescribe a universal sample size. It should prescribe the calculation that every material experiment must contain. For a scalar outcome Y with treatment and control sizes n_1 and n_0 , a simple two-sided minimum detectable effect is approximately

$$\text{MDE}_Y \approx (z_{1-\alpha/2} + z_{1-\beta}) \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0}}, \quad (187)$$

before cluster, stratification, or covariate-adjustment corrections. If the assignment unit is a cluster with average size $\bar{m}$ and intra-cluster correlation ρ , the effective variance should be inflated by the design effect

$$\text{DE} = 1 + (\bar{m} - 1)\rho. \quad (188)$$

This matters because card experiments are often powered for response or spend but not for NPV, credit losses, cannibalization, rare fraud events, or relationship spillovers. If losses are too rare for the experiment horizon, the contract should say whether loss is primary, secondary, monitored, or borrowed from a separately validated risk model. It should not silently treat a non-significant short-horizon loss difference as proof of no loss effect.

Horizon discipline is equally important. Let H_{obs} be the experimentally observed window and H_{val} the valuation horizon. The experiment identifies only the observed portion directly:

$$\tau_{\text{NPV}, H_{\text{obs}}}^e = \mathbb{E} \left[\sum_{h=0}^{H_{\text{obs}}} \delta_h \{ \Pi_{ih}(a_e) - \Pi_{ih}(a_{0e}) \} \mid i \in \mathcal{P}_e \right]. \quad (189)$$

The remaining value,

$$\tau_{\text{NPV}, H_{\text{val}}}^e = \tau_{\text{NPV}, H_{\text{obs}}}^e + \mathbb{E} \left[\sum_{h=H_{\text{obs}}+1}^{H_{\text{val}}} \delta_h \{ \Pi_{ih}(a_e) - \Pi_{ih}(a_{0e}) \} \mid i \in \mathcal{P}_e \right], \quad (190)$$

is a forecast or scenario bridge unless there is longer-run experimental follow-up. The component should expose this split. A 90-day activation win should not be presented as five-year causal NPV unless the tail bridge, uncertainty, and validation evidence are explicit.

6.3.7 How experimental evidence updates the NPV component

The component should consume experimental evidence through a governed update rule, not through ad hoc analyst interpretation. Each module estimate should carry an evidence tuple

$$\mathcal{E}_m = (\text{design}, \mathcal{P}, a, a_0, H, \hat{\tau}_m, \hat{\Sigma}_m, \text{diagnostics}, \text{admissibility}, \text{allowed use}), \quad (191)$$

where m indexes a module such as activation, all-bank spend, balances, loss, attrition, cost, or relationship spillover. The tuple should update the NPV component only within the tested population, action contrast, horizon, and policy bundle. If a production call is outside that support, the component should either downgrade the evidence, require an extrapolation flag, or return a diagnostic status.

The promotion rule should be mechanical. A module can receive randomized causal evidence only when randomization integrity, feature-cut integrity, holdout preservation, contamination checks, admissible population and action set, approved contact/disclosure path, customer-impact guardrails,

complaint and servicing escalation checks, power/MDE documentation, finance and risk reconciliation, and horizon labeling pass. A quasi-experimental module can receive quasi-experimental evidence only when its design-specific diagnostics pass: pre-trends for difference-in-differences, manipulation and continuity checks for regression discontinuity, relevance and exclusion arguments for instruments, and overlap/balance for observational adjustment. If a material module remains predictive or scenario-only, the aggregate component may still return useful NPV, but it should be labeled policy-conditional, diagnostic, or scenario NPV rather than causal incremental NPV.

Finally, the component response should expose enough metadata for caller systems and reviewers to know what they are using: experiment identifiers, holdout identifiers, assignment unit, assignment probability, treatment contrast, target population, outcome horizon, primary estimand, observed horizon versus extrapolated tail, evidence grade, admissibility label, allowed-use label, veto diagnostics, explanatory diagnostics, complaint and customer-impact monitoring results, and unresolved confounders. The component remains a replacement valuation component, not an experimentation platform, but it must be a disciplined consumer of experimental evidence.

6.4 Module-Level Identification Plan

The preceding section gives the general identification strategy. A panel-grade proposal also needs the module-level version: what exactly is being estimated, where endogeneity enters, what evidence is preferred, what fallback evidence is acceptable, and how the output may be used. This section is written as a technical operating plan for the component team. It does not assert that the bank already has every experiment or every variable. It specifies what the component must know before a module output can be promoted from diagnostic to decision-grade NPV.

Response, Approval, Booking, Activation, and First Use

The response/activation module estimates the funnel transition vector

$$\Delta q_i(a) = (\Delta \Pr(\text{respond}), \Delta \Pr(\text{apply}), \Delta \Pr(\text{approve}), \Delta \Pr(\text{book}), \Delta \Pr(\text{activate}), \Delta \Pr(\text{first use})). \quad (192)$$

The main endogeneity problem is targeted exposure: the bank contacts customers because it already expects them to respond, qualify, or be profitable. The preferred evidence is randomized contact or suppression, randomized channel placement, randomized creative, or randomized offer-cell assignment among eligible customers. A weaker but useful design is randomized encouragement, where assignment changes the probability of application or activation but does not force take-up; such estimates should be labeled as complier effects in the instrumental-variables sense [6]. Observational fallbacks require pre-treatment features, overlap diagnostics, balance checks, and leakage tests. A response model without credible assignment evidence may be used for capacity planning or funnel monitoring, but not by itself to claim incremental NPV.

Usage, Spend Source, and Cannibalization

The usage module estimates all-bank incremental spend, not only new-card spend. Its estimands are the components in Equation (6): new consumption, outside-card shift, old-bank-card cannibalization, debit/deposit shift, and promotion timing. The central endogeneity problem is that observed new-card spend is post-treatment and does not reveal its source. For existing-client offers, randomized new-card offer holdouts are the preferred design because they allow the bank to compare old-card spend, debit/deposit activity, total bank spend, and new-card spend under offer and

no-offer paths. Where competitor-card spend is not directly observed, the component should use bounded or evidence-graded estimates based on balance transfers, payoff records, bureau tradeline aggregates where available, card-on-file changes, merchant/network aggregates, account-aggregation samples, and post-promotion decay. A model-imputed outside wallet estimate should not be labeled captured wallet unless it is anchored by an experiment or credible quasi-experiment.

Balances, Payments, Line, APR, and Promotional Terms

The balance module estimates the transition from spend and payments to balances, utilization, interest income, EAD, and attrition. Its main estimands are

$$\Delta B_{i,t+h}(a), \quad \Delta Pay_{i,t+h}(a), \quad \Delta Util_{i,t+h}(a), \quad \Delta EAD_{i,t+h}(a).$$

The main endogeneity problem is policy assignment: lines, APRs, balance transfer terms, rewards, and fee waivers are assigned using risk and value signals. Comparing high-line accounts with low-line accounts therefore mixes customer selection with line effects. Preferred designs include randomized line invitations, randomized promotional terms among eligible accounts, regression discontinuities around line or pricing thresholds, and policy changes with credible comparison cohorts. Observational models are useful for forecasting under historical policy, but line or APR elasticity estimates should be used for action ranking only when overlap and policy-assignment diagnostics pass. A line-assignment NPV should expose the full option grid, not merely the selected line.

Losses: PD, LGD, and EAD

The loss module estimates incremental expected loss,

$$\Delta EL_{i,t+h}(a) = \Delta PD_{i,t+h}(a) LGD_{i,t+h}(a) EAD_{i,t+h}(a).$$

This expression is shorthand, not an independence claim; dependence should be handled by the joint path model. PD, LGD, and EAD are linked through balances, utilization, line, delinquency, macro state, and collections policy. The central endogeneity problem is that risk policies assign line, APR, approval, and collections treatment. A booked-account PD model cannot simply be transported to prospect targeting because prospect NPV also depends on response, application, approval, booking, and activation selection. Preferred evidence includes randomized or quasi-randomized line changes, threshold designs around underwriting or line policies, vintage backtests, and stress-period validation. The module should separately validate principal, non-principal, promotional, cash-advance, and balance-transfer subledgers because they create different revenue and loss paths.

Rewards, Promotions, and Post-Promotion Decay

Rewards and promotions are treatment variables, not merely cost fields. The estimands include incremental activation, incremental spend, reward liability, balance build, post-promotion runoff, attrition, and loss:

$$\Delta(Z, S, B, Pay, RewardCost, Attrition, EL)_{i,t+h}(a^{\text{promo}}).$$

The endogeneity problem is offer targeting: generous rewards and promotions are often offered to customers expected to respond or to balances the bank wants to acquire. Preferred evidence is randomized reward or promotion-cell assignment, with follow-up beyond the teaser or bonus window. If only early spend is validated, the output should be restricted to short-horizon causal

lift and should not be annualized into durable NPV. Post-promotion decay is a veto diagnostic: high early spend that disappears after the bonus can be a negative-NPV acquisition once rewards, funding, and servicing are included.

Attrition, Dormancy, Product Conversion, and Retention

The attrition and conversion module estimates transitions among active, dormant, closed, retained, converted, charged-off, and recovered states. The main endogeneity problem is survivorship and policy response: customers who survive to receive a retention offer are already selected, and the bank may offer retention concessions to customers predicted to leave or to customers it especially wants to keep. Preferred evidence includes randomized retention offers, randomized fee-waiver or reward treatments, policy threshold designs, and event studies around product conversion changes with pre-trend checks. Pure duration models are useful for forecasting the organic path [20, 35, 72], but the incremental value of a retention action requires a counterfactual no-treatment path. The output should separate retained value from concession cost and should flag cases where the concession is given to customers likely to stay without it.

Relationship Spillover and Deposit Value

Relationship value is often the least identified part of card NPV. The estimand is

$$\Delta RelValue_{i,t+h}(a) = RelValue_{i,t+h}(a) - RelValue_{i,t+h}(a_0),$$

not the level of deposits, brokerage balances, mortgage balances, or digital engagement among customers who take the card. The endogeneity problem is severe: customers who accept cards, keep deposits, and buy other products are often already more engaged. Preferred evidence is randomized cross-sell holdout with relationship outcomes measured for both treatment and control, or credible quasi-experimental variation in card access or terms. Without such evidence, relationship value should be reported as associated value or scenario sensitivity, not included silently in causal incremental NPV.

Module Promotion Rule

Each module should carry an evidence grade:

$$G_m \in \{\text{randomized, quasi-experimental, observational adjusted, predictive, scenario only}\}.$$

The NPV output may be labeled causal incremental NPV only for the parts of the value path whose modules have randomized or credible quasi-experimental support in the relevant population and decision context, plus finance/risk reconciliation. If one major module is predictive or scenario-only, the component can still return a useful value, but it should label the result as policy-conditional expected NPV, diagnostic NPV, or scenario NPV. This is how the component prevents a single scalar from hiding mixed evidence quality.

7 Validation, Uncertainty, and Model-Risk Controls

The component should be trusted only to the extent that it reconciles with bank ledgers, survives holdout and experiment checks, respects support limits, and makes uncertainty visible. The main proposal therefore treats validation as a business and statistical argument, not as a checklist. The detailed ledgers and status taxonomies remain available in the appendices.

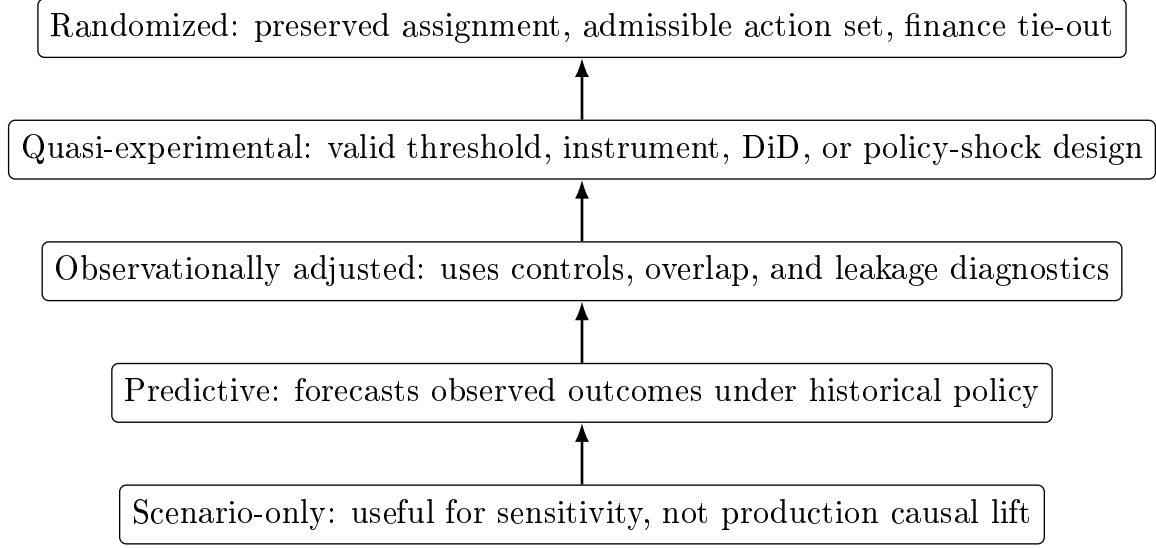


Figure 13: Evidence-grade ladder. The aggregate NPV label cannot be stronger than the material modules that drive the value path, unless weak modules are excluded or immaterial.

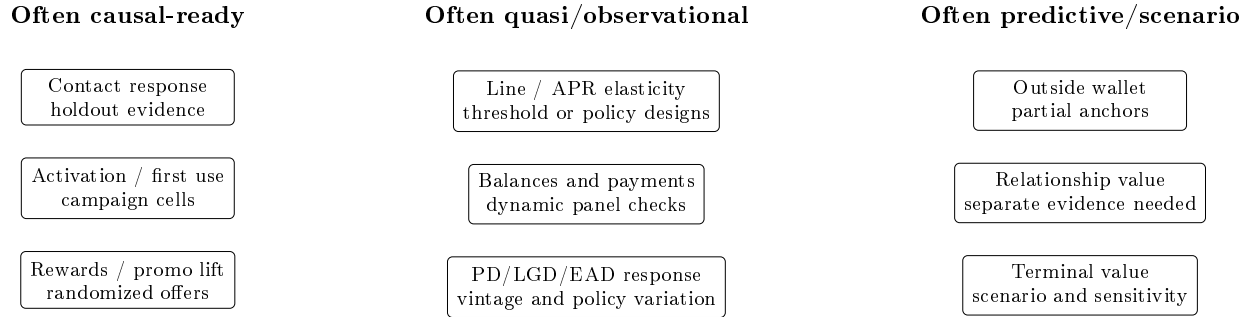


Figure 14: Module evidence heatmap. The labels are illustrative starting points, not permanent grades. The production component should carry the actual evidence grade for each module, population, action set, and decision context.

What this section establishes

- NPV is a functional of estimated paths and assumptions, so the component must report uncertainty, sensitivity, and rank stability rather than only one expected value.
- Validation has veto power. Ledger reconciliation, holdout failure, support violations, stale inputs, implausible terminal value, or evidence-grade mismatch should downgrade or block production use.
- The same value estimate can be appropriate for one decision and inappropriate for another. Model-risk control therefore attaches use limits to the output, not only to the model document.

7.1 Uncertainty Propagation and Model-Risk Control

NPV is not observed; it is a functional of many estimated paths and valuation assumptions. A panel-grade design must therefore propagate uncertainty rather than attach a false sense of precision to one expected value. Let θ collect submodel parameters, η collect scenario and valuation assumptions,

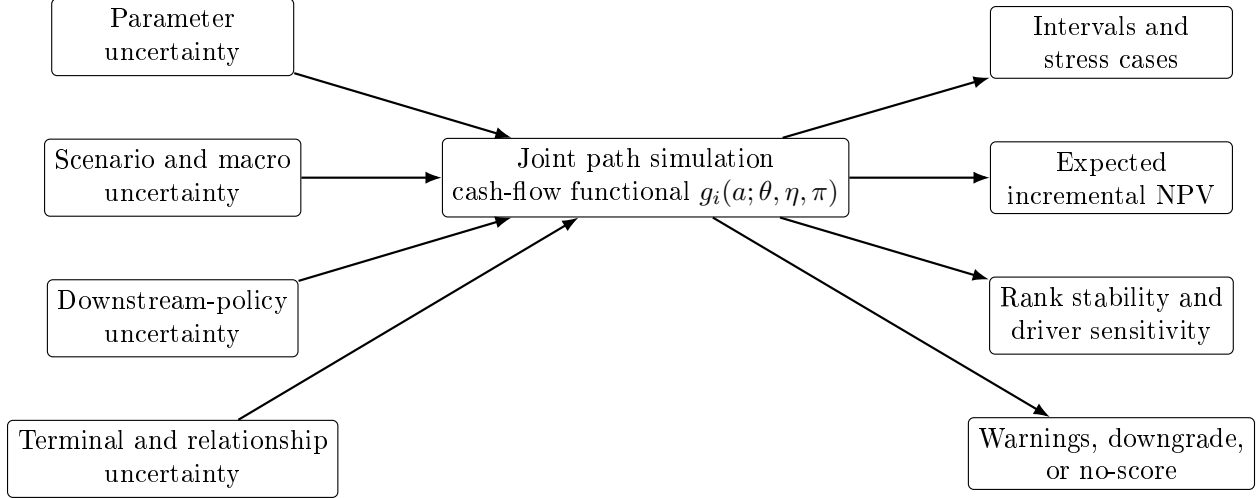


Figure 15: Uncertainty reporting stack. The component should convert uncertainty in parameters, scenarios, policy assumptions, terminal value, and relationship value into reviewer-visible intervals, stress cases, rank-stability diagnostics, and allowed-use limits.

and π denote downstream policy. The component should treat customer value as

$$\Delta\text{NPV}_i(a) = g_i(a; \theta, \eta, \pi), \quad (193)$$

where g_i includes state transitions, cash-flow identities, expected loss, capital, funding, tax, terminal value, and relationship value. The response should report at least an expected value and approved uncertainty or sensitivity summaries:

$$\left\{ g_i(a; \theta^{(r)}, \eta^{(r)}, \pi^{(r)}) \right\}_{r=1}^R. \quad (194)$$

The draws need not imply a fully Bayesian production model. They can be posterior draws, bootstrap draws, scenario grid draws, stress overlays, or approved sensitivity cases. What matters is that the component exposes how rankings change when uncertain drivers move.

The important uncertainty sources are not independent. Spend, utilization, payment behavior, delinquency, attrition, and rewards response are linked. For example, a line increase can raise spend, raise EAD, lower utilization for some accounts, raise utilization for others through borrowing, affect PD, and change attrition. A promotion can increase early spend, create reward liability, alter payment timing, and increase post-promotion runoff. Treating modules as independent plug-ins can understate tail risk and overstate rank stability. The component should therefore preserve covariance where possible: joint path simulation is preferable to separately simulating spend, loss, and attrition from independent marginal models.

The component should decompose uncertainty by driver. A useful response for management and model-risk review is

$$\text{Var}(\Delta\text{NPV}) \approx V_{\text{behavior}} + V_{\text{loss}} + V_{\text{funding/capital}} + V_{\text{terminal}} + V_{\text{relationship}} + 2 \sum_{r < s} \text{Cov}_r^s,$$

where the covariance terms should be retained when they are material. This decomposition tells reviewers whether a high-NPV segment is robust or is mostly a terminal-value, funding, or loss assumption.

Optimization creates an additional model-risk problem: decision engines tend to exploit model error. If the component returns an unconstrained scalar, a campaign optimizer may select customers where the model is most optimistic because of sparse data, poor overlap, missing cannibalization anchors, or unstable terminal value. The component should therefore expose support scores, evidence grades, uncertainty, and downside metrics. Suggested robust decision statistics include expected NPV, lower-confidence or lower-posterior NPV, probability of negative NPV, expected shortfall, and NPV subject to loss, capital, fairness, and operational constraints. High expected NPV with weak support or high downside risk should be eligible for review or experimentation, not automatic production exploitation.

Uncertainty also governs refresh policy. A model should be refreshed or reapproved when the population, offer terms, macro regime, funding curve, capital rule, reward economics, policy bundle, data lineage, or downstream caller changes enough to invalidate the previous evidence. The component should therefore monitor not only prediction drift but also evidence drift: loss of overlap, change in treatment assignment, decay of holdout relevance, breakage of finance reconciliation, and movement in terminal-value sensitivity.

7.2 Formal Validation Gates

The validation section above states the principle: proxy metrics do not prove incremental lifetime NPV. This section gives the more operational version. A production candidate should pass four ledgers: predictive validity, causal validity, valuation/reconciliation validity, and operational validity. Failure in any veto ledger should prevent production action-ranking use, even if a secondary metric looks strong.

Predictive validity asks whether forecasted components match observed paths in the population and feature stage where the model is used. For a component Y such as spend, payment, balance, attrition, PD, LGD, EAD, rewards cost, or servicing cost, calibration can be summarized by segment and horizon:

$$CalErr_{g,h}(Y) = \frac{\sum_{i \in g} w_i \{\hat{Y}_{i,t+h} - Y_{i,t+h}\}}{\sum_{i \in g} w_i |Y_{i,t+h}| + \epsilon}. \quad (195)$$

The proposal should not hard-code universal tolerances, because acceptable error depends on use case and materiality. It should require that tolerances be set ex ante by model owner, finance/risk owners, and model risk, with tighter thresholds for high-stakes or high-volume decisions. Calibration should be checked by time, vintage, months-on-book, risk band, geography, channel, product, relationship depth, promotional state, and balance type.

Causal validity asks whether treatment and control comparisons support the decision contrast. For observational or quasi-experimental designs, the component should report overlap and balance diagnostics. A standard weighted mean-balance diagnostic is

$$SMD_k = \frac{\bar{X}_{k,T=1}^w - \bar{X}_{k,T=0}^w}{\sqrt{(s_{k,T=1}^{2,w} + s_{k,T=0}^{2,w})/2}}, \quad (196)$$

with prespecified thresholds and special scrutiny for variables that drive risk, response, line, and relationship value. Overlap can be summarized by the fraction of scored customers whose estimated treatment propensity lies inside the training support:

$$OverlapRate = \frac{1}{N} \sum_i \mathbf{1}\{e_{\min} \leq \hat{e}(X_i) \leq e_{\max}\}. \quad (197)$$

Poor overlap, failed randomization checks, failed pre-trends, large placebo effects, or evidence of post-treatment leakage should veto causal action-ranking use. They can still support diagnostics or experiment design.

Valuation and reconciliation validity asks whether the component’s cash flows add up and tie to approved finance and risk views. Let the component-level NPV be the sum of discounted driver contributions:

$$\widehat{\Delta\text{NPV}}_i = \sum_{c \in \mathcal{C}} \widehat{\Delta\text{NPV}}_{ic} + \widehat{\Delta\text{TV}}_i - \widehat{C}_i^{\text{acq}}. \quad (198)$$

The decomposition residual

$$\text{Resid}_i = \widehat{\Delta\text{NPV}}_i - \left(\sum_{c \in \mathcal{C}} \widehat{\Delta\text{NPV}}_{ic} + \widehat{\Delta\text{TV}}_i - \widehat{C}_i^{\text{acq}} \right)$$

should be near zero by construction. Segment-level PPNR, loss, funding, capital, rewards, and cost aggregates should tie to finance/risk tolerances or explain the valuation convention difference. A model that predicts behavior well but fails cash-flow reconciliation is not ready for production NPV.

Operational validity asks whether the component can be safely called. It includes schema validation, feature freshness, no-score and degraded-score rates, deterministic replay, lineage hashes, model-version control, assumption bundle versions, and caller-use approval. A batch can pass technically while failing operationally if a material segment no-scores, if missing features are silently imputed, if the caller uses the score outside the approved decision context, or if the output cannot be replayed. Operational validation should therefore be a veto ledger, not an engineering afterthought.

The promotion rule is:

$$\text{Promote} = \mathbf{1}\{V_{\text{pred}} = 1, V_{\text{causal}} = 1, V_{\text{valuation}} = 1, V_{\text{operational}} = 1\},$$

where each V is use-case specific and approved before evaluation. Secondary metrics such as AUC, rank correlation, top-decile overlap, short-horizon spend lift, and validation loss can explain performance and nominate challengers, but they do not override veto failures.

7.3 Endogeneity, Confounding, and Remaining Reviewer Risks

The open risks in this proposal are not cosmetic. They are identification problems that change the estimand if they are handled poorly. A score trained on booked or active accounts does not automatically apply to prospects. A line or APR effect is not exogenous if the bank assigned it using risk and value signals. Observed new-card spend is not the same as incremental bank value. Macro and employment shocks move spend, delinquency, and utilization over time. The literature on sample selection, propensity scores, IV, RD, difference-in-differences, event studies, and dynamic panels exists precisely because these are structural problems, not tuning problems [6, 7, 14, 22, 44, 45, 58, 69, 78].

The module-level identification plan in Section 6.4 states what evidence is required for each submodel. This section explains why that evidence is needed and how the literature suggests handling the main reviewer objections.

7.3.1 Funnel Selection And Transportability

Risk. The customer moves through a funnel: prospect, applicant, approved, booked, activated, active, retained, dormant, closed, charged off. Each stage changes the population, the available data, and the relevant counterfactual. A model estimated on booked accounts can be badly miscalibrated

when applied to prospects because the later stage has already been selected on response, policy, and survival. This is the classical sample-selection problem [44]. It is also why propensity balancing alone cannot make a booked-account model representative for a prospect population [48, 69].

Mitigation. The component should carry stage-specific outputs and allowed-use labels. Prospect targeting should use prospect-stage models and counterfactuals; application and approval should use application-stage and underwriting-stage models; booked and active-account value should use activation, usage, balance, loss, and retention modules. Where the bank wants to transport information across stages, it should do so with explicit overlap checks, inverse-probability weighting or related selection adjustments, and separate validation on the target population. If the target population has little common support, the correct answer is not to force transport; it is to mark the output as out-of-domain or no-score.

Residual gap. No statistical correction removes the need for a decision-specific counterfactual. The same account can have different expected value under contact versus suppress, approve versus decline, line up versus line down, or retain versus close. The proposal therefore treats transportability as a governed exception, not a default assumption.

7.3.2 Policy Variables, Counterfactuals, And Causal Lift

Risk. Line, APR, rewards, fee waivers, promotional length, and retention concessions are not ordinary covariates. They are bank policy variables that are often assigned using latent risk and value information. If they are treated as exogenous predictors, the model learns policy, not customer economics. This is the wrong counterfactual for action ranking. It is also why the literature leans on randomized encouragement, instrumental variables, regression discontinuity, and difference-in-differences designs when policy itself is the treatment [6, 22, 58, 78]. Flexible adjustment methods such as causal forests and double/debiased machine learning can help with high-dimensional nuisance adjustment, but they do not create identification by themselves [25, 88].

Mitigation. The component should separate policy assignment from customer response. Treatment modules should estimate contrasts such as line-up versus line-down, promo versus no-promo, reward-cell A versus reward-cell B, or encourage versus suppress. The output should expose which effects are policy-driven, which are customer-driven, and which are only predictive under historical policy. In the dynamic setting, lagged outcomes and future policy reactions must be handled as part of the state process rather than buried in a static feature set [7, 14, 70].

Residual gap. If the bank cannot randomize or credibly quasi-randomize a policy change, the best it can do is a qualified observational estimate with explicit assumptions. That estimate may still be useful for forecasting or monitoring, but it should not be promoted to causal incremental NPV without the missing design evidence.

7.3.3 Cannibalization, Wallet Decomposition, And Promotion Decay

Risk. Observed new-card spend is not the same as incremental bank value. A new card can shift spend from a competitor card, from an existing bank card, from debit, or from future purchases that would have happened anyway. The direct marketing literature already warns that response probability is not the optimization target by itself [21]. Payment-choice and rewards studies show

that instrument choice and incentives change spending behavior, but they do not magically reveal the source of the shifted wallet for a specific issuer [2, 42, 53, 67].

Mitigation. The component should value all-bank incremental spend, not raw new-card spend. That is why the wallet equations in Section 2.3 split new consumption, competitor-card shift, old-bank-card cannibalization, debit shift, and promotion timing. A randomized holdout or strong quasi-experiment can identify all-bank lift; internal card, deposit, and transaction data can anchor old-bank-card and debit shifts; network, bureau, or merchant-panel data can help bound competitor shift when available. Promotion lift must also be followed beyond the bonus window. If the early lift disappears after the promotion ends, it is not durable wallet-share capture.

Residual gap. Competitor-wallet share and the movable-wallet fraction are still hard to observe directly. The proposal therefore treats them as evidenced estimates or scenario assumptions, not as observed fact. Model-imputed wallet capture must never be reported as experimentally identified value.

7.3.4 CLV, Credit Risk, And Balance-Sheet Cost

Risk. CLV models are helpful baselines for frequency, duration, and active-state forecasting, but they are not the same as card NPV. A marketing CLV model can fit spend and survival and still miss expected credit loss, funding transfer pricing, rewards liability, servicing cost, fraud, capital, and other balance sheet effects. That is a reviewer risk because a polished CLV curve can look complete while leaving out the parts that matter most to a bank [11, 20, 35, 71, 72, 87].

Mitigation. The NPV component should treat CLV as one module, not the final value. The state and cash-flow equations in Sections 2.3 and 6.4 explicitly add losses, rewards, funding, capital, and relationship-value conventions on top of the activity forecast. That is the correct way to preserve the usefulness of CLV without confusing activity forecasting with bank profitability.

Residual gap. Even when CLV and finance modules are both present, the terminal value can dominate long-horizon ranking if it is not well controlled. The proposal therefore keeps terminal value explicit and subject to sensitivity analysis rather than burying it in a hidden extrapolation.

7.3.5 State Dependence, Macro Conditions, And Dynamic Feedback

Risk. Spend, payments, balances, and losses vary with liquidity, unemployment, and macro conditions. Credit limits and interest rates also affect card borrowing and payment behavior, which in turn affect later risk and account-management policy [8, 41, 42, 50]. This creates a dynamic feedback problem: line affects utilization, utilization affects risk, risk affects later line and pricing policy, and policy affects later value.

Mitigation. The component should put macro and employment state inside the transition kernel rather than treating them as static controls. Public macro and labor data are useful as scenarios, controls, and benchmarking inputs. Dynamic panel methods, event studies, and cohort-based difference-in-differences can help with feedback and timing, but they still require a clear estimand and an as-of data protocol [7, 14, 22, 78]. The practical consequence is that the model should be refreshed by state, vintage, and policy regime, not by a single pooled historical average.

Residual gap. State dependence is never fully eliminated; it is managed. The remaining risk is model drift across macro regimes, which is why the proposal requires scenario-based validation and backtesting rather than a one-time fit metric.

7.3.6 External Data, Relationship Spillovers, And Residual Evidence Gaps

Risk. Public and licensed data are valuable, but they do not identify issuer-specific causal lift. Federal Reserve, BLS, BEA, Census, CFPB, and LSEG data can support scenarios, controls, benchmarks, and priors, but they should not be treated as evidence that a given bank action caused customer-level change. The same caution applies to relationship spillovers. Deposit growth, other product uptake, and digital engagement often reflect pre-existing household quality as much as the card action itself. If that spillover is folded into card NPV without a causal design, the card can receive credit for organic relationship value that was already going to occur [17–19, 26, 37, 38, 63, 64, 84–86].

Mitigation. External data should be used as scenario drivers, time/geo controls, benchmarking anchors, and priors for what the bank might reasonably expect under a given macro path. Relationship value should be excluded from core card NPV unless the card action itself has separate randomized or credible quasi-experimental support. When included, it should appear as a distinct line item with its own evidence grade and allowed-use label.

Residual gap. Some quantities remain genuinely proprietary: issuer-specific activation, competing-wallet capture, cannibalization, promo decay, reward elasticity, funding cost, capital charge, and relationship spillover. These are not gaps the literature can close for the bank; they are quantities the bank must estimate from its own experiments, quasi-experiments, internal data, and governed assumptions.

The practical implication of this section is simple: the component should not promise more identification than the evidence supports. Each module should state whether it is randomized, quasi-experimental, observationally adjusted, predictive, or scenario-only, and the output should inherit that label. That is how the proposal stays defensible in front of a skeptical panel instead of becoming a bundle of plausible but unverified claims.

7.4 Alternatives and Tradeoffs

Several simpler approaches are possible, and each is useful for a limited purpose. The reason for recommending a modular NPV component is not that these alternatives are worthless. It is that none of them is sufficient as the replacement valuation component for a large bank once the decision spans marketing, new-card cross-sell, line assignment, offer design, account management, finance reconciliation, and model-risk review.

A response or propensity model is the natural first alternative. It is fast to build, easy to monitor, and operationally familiar. It can improve campaign execution by estimating who is likely to respond, apply, activate, or make a first purchase. The direct-marketing literature, however, already warns against optimizing response alone: the decision should depend on expected net return, not merely on response probability [21]. In a card portfolio, this limitation is sharper because response does not price rewards, losses, funding, capital, servicing, fraud, attrition, or cannibalized spend. Propensity models should therefore remain funnel modules and diagnostics, not the final value object.

A second alternative is a standard CLV model with a finance overlay. This is closer to the economic target because CLV explicitly discounts future contribution and has mature models for

purchase frequency, duration, and retention [11, 35, 71, 72, 87]. It is a useful baseline for lifetime activity. The tradeoff is that plain CLV does not by itself solve the card problems that matter most for bank use: PD, LGD, EAD, line and APR policy, principal versus non-principal balances, rewards liability, funding transfer pricing, capital charges, and within-bank cannibalization. A finance overlay can patch some of these terms, but if the overlay is added after behavior is forecast, it can miss the fact that product terms and account-management policies change behavior itself.

A third alternative is to rely on randomized holdouts and causal lift studies without a full dynamic NPV engine. This is attractive because experiments are the cleanest evidence for incremental value, and the proposal should treat campaign holdouts and offer-cell randomization as first-class assets. The limitation is horizon and policy dependence. A campaign experiment may identify 90-day activation, spend lift, or early loss differences, but it does not automatically identify five-year value under future line increases, repricing, retention offers, collections policies, macro shocks, and terminal value assumptions. Experiments should anchor the causal layer; they do not remove the need for a governed forecast and valuation layer.

A fourth alternative is a monolithic machine-learning value score trained on historical realized profitability. This can produce a strong offline fit and may be useful as a challenger benchmark. It is risky as the production replacement if it hides the target population, treatment assignment, feature timing, finance definitions, and decomposition. A model trained on booked or active accounts can look accurate while being invalid for prospects. A score that uses post-treatment features can leak future information. A realized profitability target can combine organic customer quality with the effect of the bank action. These are not cosmetic model-risk issues; they change the counterfactual being estimated.

The recommended architecture is therefore a modular NPV component with clear valuation semantics. It uses response, activation, CLV, credit-risk, PPNR, capital, cost, and causal-lift models, but it does not collapse them into an uninterpreted scalar. Its cost is additional data lineage, governance, and reconciliation work. Its benefit is that a reviewer can inspect why a customer is valuable, whether the value is causal or predictive, which population the evidence supports, how losses and capital enter, and what assumptions would change the ranking. The practical rollout should still be phased: start with a controlled batch acquisition or prescreen slice, retain simpler models as baselines and diagnostics, and expand use cases only when the component-level evidence and integration contracts support them.

8 Integration With Bank Decisions and Rollout

The replacement component matters because it will be called by existing bank systems. The integration question is therefore not whether the component should own marketing, underwriting, or account management. It should not. The question is how each caller declares the decision, baseline, feasible action set, scenario, valuation convention, and downstream policy so that the returned NPV number is meaningful for that use.

What this section establishes

- Integration is a contract between the caller and the NPV component. The caller supplies the decision context, baseline, feasible actions, feature cut, scenario, and policy bundle; the component supplies a governed value response.
- The first production slice should be narrow enough to validate end to end: a controlled acquisition or prescreen batch with preserved holdouts, replayable feature snapshots, shadow scoring, and finance/risk reconciliation.

- Expansion to cross-sell, line, retention, conversion, or optimization reuses the same component shape but requires new evidence packages and use-case approvals.

8.1 Concrete Caller Examples

The easiest way to see the component boundary is to walk through the calls the bank would actually make. In every case, the caller owns the business decision and execution workflow. The NPV component owns a narrower object: the value of a declared action relative to a declared baseline, under declared assumptions.

8.1.1 End-to-end worked first-slice example

Consider a controlled acquisition or prescreen campaign for one approved offer family. The caller sends the NPV component a frozen feature snapshot for an eligible prospect at the campaign feature-cut date, the approved contact action and offer cell, the suppress/no-offer baseline, the current downstream policy bundle, and the valuation semantics identifier `CARD_NPV_ACQ_BASE_PRETAX_CURRENT_POLICY_V1`. The component does not decide whether to mail the customer. It values the declared action against the declared baseline.

The first gate is assignment and funnel value. The response/activation module estimates the incremental probability of response, application, approval, booking, activation, first use, and repeated use. The output is not labeled causal unless the prospect belongs to a population with preserved holdout or credible quasi-experimental evidence. Nonresponders, declines, unbooked accounts, inactive accounts, and later charge-offs remain in the analytic base.

The second gate is behavior conditional on the path. The usage and wallet modules project new-card spend, old-bank-card spend, debit substitution, balance-transfer behavior, and outside-wallet capture where evidence exists. New-card spend is reported separately from all-bank incremental spend. A customer who produces high new-card purchases by moving spend from an existing bank card may have positive diagnostic new-card activity and low all-bank incremental value.

The third gate is balances, payments, losses, and contribution. The balance engine turns spend, payments, line, APR, and promotional terms into principal and non-principal subledgers. The loss engine produces PD, LGD, EAD, and expected loss on the same path. The finance layer computes PPNR, rewards, servicing, fraud, funding, capital, tax, and terminal value under the approved assumption bundle.

The final response is a governed value object:

- expected incremental NPV against suppress/no-offer;
- decomposition into PPNR, expected loss, funding, capital, rewards, servicing, fraud, acquisition cost, tax, terminal value, and any separately labeled relationship value;
- status label such as causal incremental NPV, policy-conditional NPV, diagnostic NPV, degraded score, or no-score;
- warnings for weak wallet evidence, missing features, out-of-support actions, terminal-value dominance, or failed reconciliation;
- lineage fields needed for replay: feature-cut timestamp, model versions, scenario version, valuation-semantics id, policy bundle, and source snapshots.

This example is intentionally narrow. It is the right first slice because it tests the component's hardest practical obligations without giving the component ownership of underwriting, line

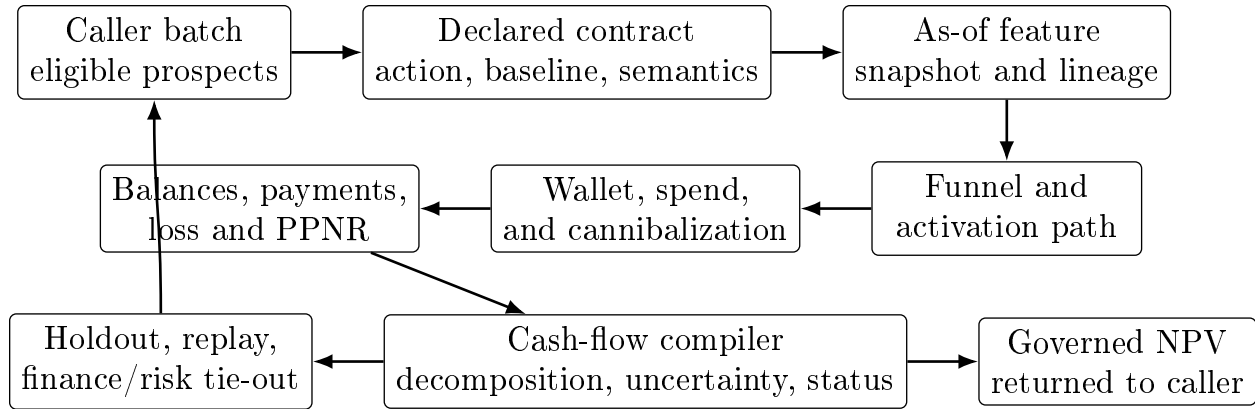


Figure 16: First-slice flow. The proposed initial deployment is a controlled batch acquisition or prescreen use case: the caller declares the decision and baseline, the component values that declared action, and monitoring ties the returned value back to preserved holdouts, replay, and finance/risk ledgers.

assignment, finance reporting, or account management. Expansion to other decisions should reuse the same value object but require its own population, baseline, action set, evidence package, and allowed-use approval.

Acquisition contact versus suppress. A marketing system sends a batch of eligible prospects or prescreen records, the candidate contact channel and offer cell, the suppress/no-offer baseline, the feature-cut timestamp, the current downstream account-management policy, and the requested valuation bundle. The component returns incremental NPV net of contact, acquisition, rewards, servicing, funding, capital, and expected losses, together with activation, spend, balance, loss, and terminal-value decomposition. The result may support campaign ranking inside the approved population and action set. It is not an underwriting approval, and it should not be reused as an adverse-action reason.

Existing-client new-card cross-sell. A CRM or relationship system sends an existing-client population, the proposed new-card offer, the no-new-card baseline, old-card and debit/deposit history where permitted, relationship state, and any relevant current-product terms. The component returns all-bank incremental NPV, not just new-card spend. It therefore separates new-card activation and purchase volume from old-bank-card cannibalization, debit substitution, promotion timing, outside-wallet capture, reward cost, balance build, and loss. The common misuse is to treat high new-card sales volume as value before proving that the value is incremental to the bank.

Initial or ongoing line assignment. A line engine sends an approved or preapproved account, an approved feasible line grid, current risk and relationship state, and the downstream line and account-management policy assumed after assignment. The component scores each feasible line option and returns NPV, expected utilization, EAD, PD, LGD, funding, capital charge, and sensitivity to adverse macro or high-loss scenarios. The component should refuse to rank a line outside the feasible policy set, and it should flag line options whose value relies on unsupported extrapolation beyond historical support.

Retention, inactivity, or product conversion. An account-management or product system sends a booked account, its organic baseline path, and a menu such as no action, retention contact, fee waiver, reward offer, line change, closure, or product conversion. The component returns the incremental value of the action after concession cost, saved attrition, future spend, balance, loss, rewards, servicing, and conversion effects. The key misuse is giving the action credit for customers who would have stayed, spent, or converted without treatment. This is why holdouts, encouragement designs, or credible quasi-experiments are central to this use case.

8.2 Default Valuation Semantics Bundle

The component should not wait until implementation to define valuation semantics. A panel needs a concrete starting convention that finance, treasury, risk, product, tax, rewards, and model-risk owners can debate. The following bundle is a proposed default for the first controlled acquisition or prescreen slice. It is not asserted as final bank policy; it is a disciplined starting point that prevents an untaged number from moving through the bank.

The value basis should be incremental, customer-level, pre-tax or post-tax as declared, and measured against a suppress/no-offer baseline for acquisition or prescreen. The horizon should be long enough to capture activation, early spend, balance build, seasoning, attrition, and loss emergence; the proposal recommends a finite monthly horizon H plus an explicitly modeled terminal value. The terminal value should not be a plug number. It should be a documented continuation-value approximation, decayed lifetime contribution, or closed-form residual value from the state model, with sensitivity analysis that shows whether rankings are terminal-value dominated.

Discounting and funding should be separated. Discounting converts future cash-flow value into present value. Funds-transfer pricing or funding cost charges balance usage and liquidity. The default bundle should specify the curve, observation date, currency, compounding convention, and whether the curve is scenario-specific. A balance-heavy revolver should not be made attractive merely because funding cost was omitted, and a transactor should not be penalized by a funding convention meant for revolving balances.

Capital should be a priced charge, not an undefined subtraction. The bundle should state whether the charge is economic capital, regulatory capital, stress capital, or a business capital allocation. It should define the exposure base, risk-weight or capital methodology, capital price, horizon, and whether capital enters monthly cash flow, terminal value, or a constraint. If capital is used as both a charge and a constraint, those roles should be reported separately.

Rewards, acquisition cost, servicing, fraud, disputes, collections, and other expenses should distinguish marginal and allocated cost. The default decision-grade NPV should use marginal or causally incremental costs for action ranking where available, while allocated overhead can be reported in a reconciliation view. Rewards should be accrued when earned or expected to be redeemed according to the approved rewards finance convention; sign-up bonuses and teaser incentives should be separated from steady-state earn and burn.

Losses should use expected-loss paths consistent with the component's PD, LGD, and EAD definitions, not a single static loss rate. Principal, non-principal, promotional, balance-transfer, cash-advance, fee, and accrued interest subledgers should be preserved where they materially change revenue, loss, or accounting treatment. CECL and stress-testing systems remain outside the component boundary, but the component's expected-loss convention should be reconcilable to finance and risk views.

Relationship value should default to excluded from core card NPV unless approved causal or quasi-experimental evidence supports inclusion for the specific use case. When included, it should be reported both as a separate line item and as an all-bank value view. This protects the card action

from receiving credit for organic deposit or other-product value that would have existed without the card.

The scenario convention should distinguish base, adverse, and sensitivity views. Public and licensed data can supply macro, rate, inflation, credit, and market scenario paths, but they do not by themselves identify customer-level treatment effects. The default bundle should preserve the scenario version, macro-data vintage, and data release vintage so backtests use information available at decision time rather than revised data.

Finally, downstream policy π must be named. Acquisition NPV under one line increase, repricing, retention, collections, hardship, closure, and conversion policy can differ materially from acquisition NPV under another. The first production slice should use a frozen current-policy bundle or an explicitly approved target-policy bundle. A policy change should trigger reassessment of the NPV component or at least a scenario sensitivity.

8.2.1 Concrete first-slice base case

For the first controlled acquisition or prescreen slice, the proposal should define a base case that is specific enough to implement and conservative enough to survive review. A suitable starting identifier is `CARD_NPV_ACQ_BASE_PRETAX_CURRENT_POLICY_V1`. The identifier is not a technical name only; it is a compact contract saying that the score is a card acquisition or prescreen value, incremental to suppress or no-offer, pre-tax, under current downstream account-management policy, and governed by version 1 of the approved assumption bundle.

The base-case value object should be pre-tax incremental card NPV. A post-tax view can be reported in sensitivity or finance reconciliation, but pre-tax is the cleaner first production slice because many campaign, risk, and product decisions are made before final tax allocation conventions are settled. The baseline action is suppress or no-offer; the treatment action is the approved contact and offer cell. The scored unit is the prospect, customer, household, or prescreen record defined by the campaign owner, but the returned value must be customer-level or relationship-level enough to identify internal cannibalization where the customer already has a bank card.

The base-case horizon should be monthly. Let H_{obs} denote the observed experimental or backtest window and H_{val} the valuation horizon. The first slice should report both. A practical default is a three-part path: months 0–6 for response, approval, booking, activation, first use, signup bonus, early reward cost, and campaign cost; months 7–24 for spend persistence, balance build, payment behavior, delinquency emergence, promotion decay, and early attrition; and months 25–60 for seasoned behavior, loss emergence, retention, and runoff. The exact number of months should be approved by finance, risk, and model-risk owners, but the component should not hide the split between observed evidence and extrapolated tail.

The base-case terminal value should be explicit and bounded. A defensible initial convention is a decayed continuation value from the terminal state:

$$\Delta TV_{i,H_{\text{val}}} = \frac{\rho_i \widehat{\Delta CF}_{i,H_{\text{val}}+1}}{r_{\text{disc}} + \lambda_i + q_i}, \quad (199)$$

where ρ_i is an approved persistence factor, λ_i is an attrition or closure hazard, q_i is a conservatism or model-risk decay charge, and r_{disc} is the discount rate expressed on a monthly-compatible basis. This formula is not meant to be a universal truth. It is a transparent placeholder that can be replaced by a state-model residual value. The key review requirement is that the terminal term be reported separately and that rankings be stress-tested with $\Delta TV = 0$, high-decay, and alternative persistence cases. Customer rankings dominated by terminal value should be flagged for review.

Discounting and funding should use different fields. The base-case discount curve should be the approved finance valuation curve as of the valuation date, with currency, compounding convention, observation time, and interpolation rule stored in the assumption bundle. The funding charge should be a balance- and liquidity-sensitive funds-transfer-pricing charge applied to revolving, promotional, balance-transfer, cash-advance, and other funded balance subledgers. A transactor who pays in full should mainly generate interchange, rewards cost, fraud, servicing, and acquisition-cost economics; a revolver also generates interest income, funding cost, capital, and loss economics. The bundle should not let one curve do both jobs.

The base-case capital charge should be a monthly priced charge, not an unlabeled deduction. The first slice can use the bank’s approved business or economic-capital convention if that is the normal decisioning basis, with a reconciliation view to regulatory or stress capital. The response should return the capital basis, exposure base, capital price, and whether capital is used as a charge, a constraint, or both. Basel-style PD/LGD/EAD terminology supports the separation of exposure, probability, and severity components [9]; Federal Reserve stress-methodology documents support the need to keep balances, revenues, losses, and capital-sensitive scenario views distinct [15, 16]. They do not, by themselves, choose this bank’s capital price.

The base-case expected loss path should use monthly PD, LGD, and EAD paths consistent with the state model. It should preserve purchase principal, promotional balance, balance-transfer balance, cash-advance balance, fees, accrued interest, charge-off, and recovery timing where material. The CECL standard supports lifetime expected-credit-loss thinking for financial reporting [39], but the component’s decision-grade loss path should be a valuation convention tied to the action counterfactual and reconcilable to risk and finance. If a loss term is immature during the observed experiment window, the response should label it as modeled rather than realized.

Rewards and incentives should be decomposed. The base-case should report signup bonus, teaser incentive, earn, expected redemption, breakage, statement credits, reward liability timing, and any acquisition incentive separately from steady-state reward cost. The CARD Act and rewards evidence show that fees, disclosures, payment behavior, and incentives can materially affect card economics and behavior [2, 3]; they do not justify treating reward cost as a simple constant percentage for every segment. Promotion decay should therefore be a required sensitivity rather than an afterthought.

Costs should be divided into causally incremental, marginal, and allocated views. For action ranking, the base-case should use incremental campaign cost, acquisition cost, underwriting cost where applicable, marginal servicing, fraud, dispute, collections, and reward costs. Allocated overhead can be reported for finance reconciliation but should not silently change the action ranking unless the approved valuation semantics explicitly require it.

Relationship value should be excluded from core base-case card NPV. It may be reported as a separate associated-value view, and it may enter all-bank NPV only when the specific card action has randomized or credible quasi-experimental evidence for causing deposit, debit, mortgage, brokerage, or other-product value. This convention is intentionally conservative because relationship value is highly confounded: more engaged households are more likely to take cards and more likely to hold other products even without the card action.

Finally, the base case should freeze downstream policy. For acquisition, the policy bundle should name future line-increase rules, repricing rules, retention treatment, collections policy, hardship policy, closure policy, and product-conversion policy. If those policies change, the old NPV estimate is not automatically wrong, but it is no longer the same economic object. The component should then either rescore under the new policy bundle or downgrade the old score’s allowed use.

8.2.2 Sensitivity and challenger bundles

The base case is the starting convention, not the end of valuation discipline. The proposal should require a small set of named sensitivity bundles, each designed to answer a particular reviewer question.

The no-terminal-value bundle sets $\Delta TV = 0$. It answers whether the decision is supported by observed and near-horizon economics or mostly by a long-tail assumption. The high-decay terminal bundle increases attrition, runoff, or conservatism decay in Equation (199). It answers whether high-value segments remain high-value if persistence is weaker than expected.

The adverse macro and labor bundle uses a worse unemployment, income, rate, and credit-loss path. It should be driven by approved public, licensed, or internal scenario sources, with release vintages stored for replay. This bundle answers whether a campaign still creates value when market conditions move against spend, payment capacity, funding, and losses.

The high-funding and high-capital bundle increases funds-transfer-pricing charges and capital price or capital intensity. It answers whether the value comes from genuine customer economics or from an omitted balance-sheet cost. This is especially important for high-line, balance-transfer, cash-advance, and revolver-heavy segments.

The high-rewards-cost and fast-promotion-decay bundle raises redemption, lowers breakage, increases signup-bonus realization, and shortens promotion persistence. It answers whether the component is buying early spend that does not survive after the bonus or teaser period.

The cannibalization-stress bundle raises old-bank-card substitution, debit substitution, and organic-spend substitution. It is mandatory for existing-client new-card cross-sell. It answers whether the value is truly incremental to the bank or mostly a product-transfer accounting movement.

The relationship-excluded bundle is the default for core card NPV; the relationship-included bundle should be allowed only when a separate evidence grade supports causal relationship spillover. Reporting both views prevents a portfolio or relationship team from using the card component to absorb value that belongs to another product or to organic customer quality.

The current-policy and target-policy bundles should be kept distinct. The first asks what the customer is worth under the policy the bank actually runs today. The second asks what the customer would be worth under a proposed future account-management policy. A target-policy view is useful for planning, but it should not be mixed with current-policy production scoring unless the caller explicitly requests that policy and accepts the validation boundary.

Each bundle should generate the same decomposition fields as the base case. The important output is not just a range of NPV values. It is a map of which driver changes the rank: terminal value, macro state, funding, capital, reward cost, cannibalization, relationship value, or downstream policy. That driver map tells the panel whether the component is robust or merely precise under one fragile convention.

8.3 Allowed-Use Taxonomy

The component should not emit a scalar NPV without an allowed-use label. The same numerical estimate can be appropriate for one decision and inappropriate for another. Allowed use depends on decision context, population, feature stage, evidence grade, valuation semantics, scenario, downstream policy, and validation status.

The strongest label is *causal incremental NPV*. This label requires a defined baseline action, decision-stage information set, randomized or credible quasi-experimental evidence for material treatment-response modules, adequate overlap and leakage diagnostics, and finance/risk reconcilia-

tion. It can be used for governed action ranking within the approved population and action set. It does not automatically approve underwriting, adverse action, account closure, collections treatment, CECL, stress testing, or official finance reporting.

The second label is *policy-conditional expected NPV*. This value forecasts expected cash flows under a historical or declared policy and valuation bundle. It can support planning, monitoring, challenger comparison, portfolio forecasting, and scenario analysis. It should not be described as the causal value of changing the action unless an identification design supports that contrast.

The third label is *diagnostic or scenario NPV*. This value uses weak observational evidence, external benchmarks, scenario-only outside-wallet assumptions, or model-imputed quantities. It can be useful for sensitivity analysis, management discussion, experimental design, and prioritizing data collection. It should not drive customer-level production action ranking without additional approval.

The fourth label is *degraded NPV*. The component can score the request, but an approved non-critical feature is missing, stale, imputed, or out of its best-quality range. The response should state the degraded input, affected module, allowed downstream handling, and whether uncertainty or conservative adjustment was applied. A caller should not silently treat degraded values as equivalent to fully validated values.

The fifth label is *no-score*. No-score is not zero NPV. It means the component cannot produce a governed value for the requested context, action, population, feature stage, or valuation semantics. The caller must use an approved fallback, suppress, route to manual review, or request a different approved output class.

The sixth label is *prohibited use*. A score produced for campaign targeting should not be reused for underwriting or adverse-action reasoning. A booked-account value should not be used for prospect targeting unless a transportability argument has been approved. A diagnostic outside-wallet value should not be used as captured-wallet proof. A scenario NPV should not become official finance reporting. These prohibitions should be encoded in the component contract and monitored in caller integration tests.

8.4 Using the Component in Existing Bank Decision Processes

The component should be used only through an explicit decision-context contract. That contract is simple in concept even if the implementation schema is detailed. The caller must say who or what is being scored, which decision is being valued, what action alternatives are feasible, what baseline defines the counterfactual, what information was available at the decision time, what valuation semantics and scenario are being used, and what downstream policy is assumed. The component then returns a decomposed value object, status, warning set, and lineage. The caller remains responsible for action choice and execution.

The most important use cases are different enough that they should not share one generic interpretation of NPV. For new-to-bank targeting, the baseline is usually suppress and the action is contact through a channel and offer. For existing-client new-card cross-sell, the baseline is no new-card offer and the central risk is cannibalizing existing bank-card or debit activity. For underwriting, the baseline is decline or refer and the action must remain inside approved credit policy. For line assignment, the component values a menu of feasible line or term options and must expose usage, EAD, loss, and capital for each option. For retention, inactivity, product conversion, and promotional treatment, the baseline is the organic account path and the action may change attrition, rewards, servicing, balances, and losses.

Valuation semantics must be visible for every use. A value number should not move through the bank unless its horizon, discounting, funding, capital, tax, cost allocation, rewards, relationship

value, cannibalization, terminal value, scenario, and downstream-policy assumptions are tagged. This is not a technical nicety. Without the tag, two callers can compare numbers that are both called NPV but are economically different.

The response should be decomposed enough to diagnose why a customer or option is valuable. A high expected value from interchange is different from a high expected value from revolve. A high value dominated by terminal value is different from one dominated by near-term spend. A positive new-card-spend lift with near-zero all-bank-spend lift is a cannibalization warning, not a success. A high-line option with higher spend but materially higher EAD and capital should not be treated as a pure revenue win. These distinctions are why the component must return paths, decomposition, uncertainty, and status, not only a scalar.

Status semantics protect callers from silent misuse. A no-score because the use case is unapproved is different from a technical timeout. A degraded score because merchant-category history is missing is different from an out-of-domain population. A batch job can succeed while individual rows fail; an option grid can score one line amount while rejecting another. Those states should be explicit in the response and preserved for replay.

8.5 Migration and First Production Slice

The replacement should be introduced as a controlled component migration, not as a one-time model launch. The first production slice should remain narrow: one batch acquisition or prescreen consumer, one offer family, one valuation semantics bundle, frozen feature snapshots, campaign holdouts, and shadow scoring against the current component. This slice is valuable because it exercises the full NPV chain without adding real-time underwriting or account-management risk.

The operational sequence should be explicit.

1. **Data freeze.** Select the campaign or prescreen population, freeze eligibility, assignment, feature-cut, offer, and outcome snapshots, and preserve holdout or randomization flags before any model development outcome refresh.
2. **Feature and lineage replay.** Recreate the features from the frozen as-of timestamps and verify that no post-treatment or revised macro information enters the decision-time state.
3. **Shadow model scoring.** Score the same frozen population with the current component and the replacement component without changing business decisions.
4. **Challenger scoring and decomposition review.** Compare value levels, rank changes, top-decile overlap, no-score and degraded-score rates, and driver decompositions by segment, product, channel, and risk band.
5. **Holdout or experiment evaluation.** Use campaign holdouts, randomized cells, or approved quasi-experimental designs to estimate treatment effects for the modules that claim causal incremental value.
6. **Finance and risk reconciliation.** Tie PPNR, losses, rewards, funding, capital, costs, and terminal-value conventions to approved finance/risk definitions or explain the valuation-basis difference.
7. **Approval gate.** Promote the slice only if prediction, causal-lift, valuation-reconciliation, operational-replay, and allowed-use checks all pass for the named use case.

8. **Limited deployment.** Cut over one approved caller and decision context first, retain rollback to the legacy component, and keep holdouts or monitoring cohorts.
9. **Post-cutover monitoring.** Monitor drift, no-score rates, degradation, rank movement, realized-vs-expected value, finance/risk tie-outs, and use outside the approved decision context.

Migration should begin by inventorying the current component. The bank needs to know every consumer, field, threshold, fallback, owner, latency class, report, and downstream decision that currently depends on the old value. It then needs a valuation-definition comparison: horizon, discounting, funding, capital, tax, cost, reward, terminal value, scenario, and policy assumptions in the old component versus the new one. Without this comparison, a ranking change can look like a model improvement when it is actually an assumption change.

Shadow scoring should compare both values and decisions. The bank should measure rank correlation, top-decile overlap, segment-level aggregate NPV, component decomposition tie-outs, no-score and degraded-score rates, disagreement patterns, and downstream contract behavior. Material disagreements should be put in a register and classified as expected improvements, assumption differences, data issues, unsupported use cases, or blockers. Cutover should occur by controlled consumer and decision context, not globally.

Governance should separate five questions. Does the model predict the behavioral paths it claims to predict? Does it estimate causal lift for actions that change outcomes? Does the decomposition reconcile to finance, treasury, risk, and product definitions? Do the data and integration pipelines preserve as-of timing, schema stability, replay, and status semantics? Is the proposed use case approved by business, model risk, compliance, legal, and technology owners where required? A positive answer for marketing targeting does not approve underwriting, line assignment, account closure, collections, adverse action, CECL, stress testing, or official finance reporting.

Rollback should be designed before launch. Trigger examples include schema breaks, no-score or degraded-score rates above threshold, failed finance or risk tie-out, material unexplained ranking movement, replay failure, feature leakage, out-of-domain drift, or use outside the approved decision context. Rollback may mean returning to the legacy component, suppressing a campaign, using a conservative no-score policy, or routing to manual review, depending on the consumer and use case.

9 Conclusion

The replacement component should estimate incremental, risk-adjusted customer NPV, not response, approval, activation, or new-card spend alone. The literature supports the mechanisms that make this necessary: market conditions, employment, liquidity, credit terms, rewards, adoption, and customer duration affect spend, balances, payments, attrition, and losses. The literature does not supply this bank’s activation rate, cannibalization rate, movable-wallet share, loss elasticity, funding cost, capital charge, or relationship spillover. Those are issuer-specific empirical quantities that must be estimated from the bank’s own experiments, quasi-experiments, internal data, external benchmarks, and finance/risk reconciliations.

The proposal therefore recommends a modular valuation component with explicit decision context, valuation semantics, scenario, downstream-policy assumption, decomposition, status, lineage, and allowed-use metadata. The first slice should be a controlled batch acquisition or prescreen use case, with holdouts, shadow scoring, reconciliation against the current component, and governed cutover. Broader use in underwriting, line assignment, existing-account treatment, finance planning,

or account management should be approved only after the relevant counterfactual, population, data lineage, and validation evidence support that use.

Appendix Reader Guide

The appendices are part of the proposal, not a dumping ground for leftovers. They serve three human purposes. First, they give reviewers a stable vocabulary for the symbols and value objects used in the main text. Second, they preserve the operating contracts that an implementation team would need when replacing the current component. Third, they record source support and unresolved evidence boundaries so that a panel can distinguish a defensible claim from a planned empirical task.

How to use the appendices

- Use Appendix [A](#) when checking equations or comparing the valuation notation across literature, component, and experimental sections.
- Use Appendix [B](#) and Appendix [C](#) when asking whether the proposal is implementable as a replacement component rather than only as a modeling essay.
- Use Appendix [D](#) when checking rollout, monitoring, downgrade, and first-slice obligations.
- Use Appendix [E](#) when pressing whether a claim is supported by literature, supervisory and accounting sources, bank-data evidence, or an explicitly labeled assumption.

A Notation Glossary and Symbol Discipline

This appendix is a reading aid, not a substitute for the main argument. It collects recurring symbols so reviewers can compare equations across the proposal without confusing local notation from different literatures with the stable component vocabulary.

The proposal combines several literatures, and those literatures reuse symbols differently. The main text therefore defines local notation near each equation. The global convention is that a symbol should be interpreted in the smallest section where it is explicitly defined unless it appears in the stable component vocabulary below.

i Scored unit: prospect, customer, household, relationship, application, account, or account-action pair, depending on the decision context.

t and *h*

Calendar time and forecast horizon step. Time is monthly unless stated otherwise. When the distinction matters, H_{obs} is the observed experiment or backtest horizon and H_{val} is the valuation horizon.

d Decision context: targeting, prescreen, new-card cross-sell, underwriting, line assignment, onboarding, retention, product conversion, closure, or portfolio analysis.

a and $a_0(d)$

Candidate action and decision-specific baseline action. Examples are contact versus suppress, approve versus decline, line option versus current line, or offer cell versus no offer.

s Scenario and valuation-assumption bundle. This includes macro path, rate path, funding, capital, tax, rewards, cost, terminal-value, relationship-value, and scenario conventions.

- π Downstream policy bundle: future line-management, repricing, retention, hardship, collections, closure, and conversion policy assumed in valuation.
- X_{it}^d or $\mathcal{I}_{id}$ Information available at the decision-time feature cut for unit i in decision context d . These objects should exclude post-treatment variables.
- S_{it} Economic and behavioral state vector used in the valuation spine. It may include lifecycle state, balances, line and terms, payment behavior, delinquency/default state, rewards and promotion state, relationship state, and macro/local state.
- $B, L, P, D, Rwd, Rel,$ and M**
Local components of the state vector: balance subledger, line/terms, payment behavior, delinquency/default state, reward state, relationship state, and macro/local state. These symbols are local and should not be reused as implementation field names without qualification.
- Z Used in older literature-derived notation for lifecycle state and in the experimental section for randomized assignment. The experimental meaning is local to Section 6.3. In implementation, use distinct names such as `lifecycle_state` and `assignment_flag`.
- D Used locally for delinquency/default state and, in experimental notation, treatment received. In implementation, use distinct names such as `delinquency_state`, `default_flag`, and `treatment_received`.
- Δ Incremental contrast against the decision-specific baseline under the same scenario and downstream policy. A positive new-card value is not necessarily positive all-bank incremental value unless the contrast is explicitly defined that way.
- PPNR, EL, $Kchg$, Tax , $RelValue$, and TV**
Pre-provision net revenue, expected loss, capital charge, tax convention, relationship value, and terminal value. Each should be governed by the valuation-semantics bundle.
- G_m Evidence grade for module m : randomized, quasi-experimental, observationally adjusted, predictive, or scenario-only.

The glossary is deliberately modest. It does not attempt to normalize every symbol inherited from the literature review. Its purpose is to prevent a reviewer or implementer from confusing the stable component objects with local equation notation. A final model specification should use unambiguous implementation names and should maintain a machine-readable data dictionary.

B Detailed Internal Data Feasibility Checklist

The main text states the internal data requirement in prose. This appendix keeps the same implementation checklist, but in a form a reviewer can read without decoding a dense grid. Each domain below states the grain, the modules it supports, and the failure mode that would make the resulting NPV unreliable.

Campaign and offer records

- Required grain: customer or household by campaign cell, channel, offer, and timestamp.

- Modules served: response, offer effects, acquisition cost, experiment assignment, and contact/suppress counterfactuals.
- Main risk: lost holdouts, overwritten terms, re-contact contamination, or missing assignment probabilities. These failures destroy the experiment before any model is fit.

Application and underwriting records

- Required grain: application, customer, product, decision timestamp, policy version, decision, assigned line, APR, terms, and override.
- Modules served: approval, line, APR, fraud screen, expected loss, and approved action set.
- Main risk: approval-only survivorship or missing policy lineage. A booked-account model cannot be transported to applicants unless the selection path is preserved.

Transaction records

- Required grain: transaction and account cycle, with merchant category, channel, posting date, authorization date, returns, disputes, and card-on-file or digital-wallet signals where available.
- Modules served: spend, category mix, activation, first use, usage persistence, rewards, fraud, disputes, and cannibalization diagnostics.
- Main risk: post-treatment leakage, merchant-code drift, or missing routing signals.

Balances and payments

- Required grain: account cycle and account month, with principal, promotional, cash-advance, balance-transfer, fee, interest, and charged-off subledgers kept separate.
- Modules served: revolve, utilization, finance charges, EAD, payment behavior, funding, and capital.
- Main risk: broken stock-flow identity or collapsed principal and non-principal balances.

Risk and loss records

- Required grain: account month, vintage, delinquency state, collections event, charge-off, recovery, and bankruptcy/workout path.
- Modules served: PD, LGD, EAD, expected loss, recovery, stress, and capital.
- Main risk: transporting booked-account risk to prospects without correcting for response, approval, booking, and activation selection.

Rewards, costs, servicing, fraud, and disputes

- Required grain: account, transaction, campaign, treatment, and cost center, with reward earn, burn, liability, servicing events, fraud losses, dispute costs, and collections costs.
- Modules served: rewards liability, marginal contribution, servicing, fraud, disputes, and cost-to-serve.
- Main risk: treating allocated overhead as marginal cost or missing the timing of rewards and servicing cost.

Relationship and deposit records

- Required grain: customer, household, and relationship month, subject to permitted-use, privacy, and governance constraints.
- Modules served: debit substitution, deposit contribution, relationship value, engagement, and cross-product cannibalization.
- Main risk: confounding card-caused spillover with pre-existing engagement. Relationship value should remain separately labeled unless a credible design identifies causality.

C Reference Operating Specification for the Replacement NPV Component

This appendix is implementation contract material. The main body explains why the component needs decision context, valuation semantics, decompositions, statuses, and lineage; this appendix states the reference contract more compactly for build teams, integration owners, and reviewers who need to check field-level implications.

The preceding sections establish what the component must estimate and why. This section translates those requirements into the contract by which existing bank systems call the replacement component. These contracts are not generic platform design. They are controls against the exact failure modes identified above: NPV is latent, value is heterogeneous, populations change through the funnel, counterfactuals differ by decision, and downstream policies change upstream economics. The purpose is to make the NPV component replaceable, testable, auditable, and useful across many callers without letting one ambiguous value field leak into incompatible decisions.

The operating specification therefore has a narrow job. It tells each caller how to declare the decision context, baseline, feasible action set, feature stage, scenario, valuation semantics, and downstream policy assumption. It tells the component how to return value, decomposition, sensitivity, warnings, status, and lineage. It does not tell marketing which customers to contact, underwriting which applications to approve, account management which treatment to execute, or finance which official ledger number to book.

C.1 Decision-context contract

Every request must declare the decision context, baseline action, candidate action or option grid, population, feature stage, scenario, valuation semantics, and downstream policy assumption. This is not administrative ornament. It is the only way to prevent a value model validated for contact-versus-suppress from being reused for approve-versus-decline, line assignment, or account closure.

The component should support at least three groups of consumers.

Acquisition and new-card cross-sell. These calls occur before the account is booked or before a new product is accepted. The available features may be prospect-level, prescreen-level, relationship-level, or application-level. The primary risk is using post-decision data or mistaking new-card spend for incremental all-bank value.

New-to-bank targeting

- Caller: marketing platform.
- Scored unit: prospect or household.

- Baseline and action set: suppress versus contact through one or more approved channels and offers.
- Required output: incremental NPV net of contact and acquisition cost. The caller owns campaign execution.

Existing-client new-card cross-sell

- Caller: CRM or cross-sell system.
- Scored unit: customer, household, or relationship.
- Baseline and action set: no new-card offer versus an approved offer menu.
- Required output: incremental all-bank value after cannibalization, plus diagnostic new-card spend. The caller owns offer execution.

Prescreen or preapproval

- Caller: prescreen system.
- Scored unit: prospect or existing client.
- Baseline and action set: no prescreen versus approved preapproval terms.
- Required output: expected value conditional on prescreen eligibility, response, approval, and booked-account economics.

Application underwriting

- Caller: underwriting engine.
- Scored unit: application.
- Baseline and action set: decline, refer, or approve with feasible terms.
- Required output: risk-adjusted value under approved credit policy. Underwriting owns the final decision and compliance process.

Initial line and offer assignment

- Caller: line or offer engine.
- Scored unit: approved application or account-action pair.
- Baseline and action set: feasible line and terms menu.
- Required output: option-level NPV, usage, EAD, loss, capital, and sensitivity paths. The caller owns the chosen line.

Existing-account treatment. These calls occur after an account exists. The component may use payment, transaction, utilization, delinquency, service, dispute, rewards, and dormancy history that is unavailable earlier in the funnel. That means validation on active accounts cannot be back-projected to prospects without a separate selection argument.

Line increase or decrease

- Caller: account-management or line engine.
- Scored unit: account-action pair.
- Baseline and action set: current line versus feasible line menu.
- Required output: forward NPV, utilization, EAD, PD, LGD, capital, and attrition effects.

Retention or inactivity treatment

- Caller: retention or servicing system.
- Scored unit: account or relationship.
- Baseline and action set: no treatment versus approved fee waiver, reward, line, closure, or contact option.
- Required output: incremental retained value net of concession, servicing, and adverse selection.

Product conversion

- Caller: product or CRM system.
- Scored unit: account-product pair.
- Baseline and action set: organic path versus conversion menu.
- Required output: incremental value after annual fee, rewards, cannibalization, attrition, and customer-state transitions.

Promotional balance or reward treatment

- Caller: offer or balance-transfer engine.
- Scored unit: account-action pair.
- Baseline and action set: no promotion versus approved promotional terms.
- Required output: promotion-state NPV, reward liability, funding, post-promotion attrition, and loss path.

Collections-sensitive forward value

- Caller: servicing or collections system where permitted.
- Scored unit: account.
- Baseline and action set: current treatment path versus approved alternatives.
- Required output: forward value under hardship, collections, recovery, and compliance constraints. The caller owns customer treatment.

Offline finance, risk, and portfolio consumers. Offline consumers should use component outputs for analysis and reconciliation, not as evidence that the NPV component owns finance reporting, CECL, stress testing, or enterprise dashboards.

Campaign budget analytics

- Consumer: marketing analytics.
- Scored unit: portfolio or campaign universe.
- Boundary: the component supplies value and drivers; the budget optimizer remains outside scope.

Vintage profitability review

- Consumer: product, finance, and risk.
- Scored unit: vintage, campaign, product, or segment.
- Boundary: the component supplies decomposed values; official finance reporting remains outside scope.

Scenario and planning analysis

- Consumer: finance, treasury, risk, and planning.
- Scored unit: portfolio and segment.
- Boundary: the component can evaluate approved scenario bundles; stress and planning platforms remain outside scope.

Model-risk review and monitoring

- Consumer: model-risk management.
- Scored unit: use case, model version, and segment.
- Boundary: the component supplies lineage, validation, drift, and reconciliation artifacts; model-risk approval remains separate.

C.2 Valuation semantics and assumption contract

NPV is a valuation convention, not a natural constant. Two systems can both claim to use NPV while differing in horizon, terminal value, discount curve, funds-transfer pricing, capital charge, tax treatment, reward liability, cost allocation, relationship value, cannibalization, macro scenario, and downstream policy. Therefore each request and response should carry a canonical `valuation_semantics_id`. That identifier should resolve to a controlled assumption bundle with an owner, approval record, effective date, and version.

Value basis

- Required definition: incremental versus total value, and pre-tax versus post-tax value.
- Primary approvers: finance and product.
- Risk if unspecified: callers compare incompatible scores as if they were the same economic object.

Horizon and terminal value

- Required definition: explicit horizon, terminal-value formula, decay assumption, and treatment of inactive or closed states.
- Primary approvers: finance, product, and model risk.
- Risk if unspecified: rankings become dominated by hidden long-horizon assumptions.

Discounting and funding

- Required definition: discount curve, funds-transfer-pricing curve, liquidity basis, and date convention.
- Primary approvers: treasury and finance.
- Risk if unspecified: balance-heavy accounts are over- or under-valued.

Capital

- Required definition: economic, regulatory, stress, or business capital charge and how it is allocated to the scored unit.
- Primary approvers: risk capital and finance.
- Risk if unspecified: high-EAD or high-loss growth is ranked too favorably.

Costs and rewards

- Required definition: acquisition, servicing, rewards, fraud, disputes, collections, and allocated-versus-marginal cost rules.
- Primary approvers: finance, product, and rewards finance.
- Risk if unspecified: high spend is mistaken for profitable value.

Relationship value

- Required definition: whether deposit and other-product value is excluded, included, or reported separately.
- Primary approvers: product, finance, and relationship analytics.
- Risk if unspecified: the card offer receives credit for organic household value.

Cannibalization

- Required definition: treatment of old bank card, debit, deposit, outside-card, balance-transfer, and organic paths.
- Primary approvers: product and causal validation owners.
- Risk if unspecified: new-card spend is mistaken for bank-incremental spend.

Scenario and downstream policy

- Required definition: macro scenario, rate path, loss scenario, and downstream account-management policy π .
- Primary approvers: risk, finance, and policy owners.
- Risk if unspecified: upstream value assumes a future operating world the bank will not run.

C.3 Request contract

The production request schema should be machine-readable and contract-tested. The human proposal should specify the semantic obligations, not every eventual JSON field. Each field must have a business definition, type, unit or currency where applicable, cardinality, allowed values, requiredness by decision context and operating mode, and backward-compatibility rule.

Request envelope

- Business definition: `request_id`, `trace_id`, caller, use case, schema version, model version, request timestamp, and feature-cut timestamp.
- Requiredness: required for every call.
- Compatibility rule: new optional fields may be added; changed meaning requires a schema-version increment.

Scored unit

- Business definition: person, household, relationship, application, account, or account-action-pair identifiers and join keys.
- Requiredness: required for every call.
- Compatibility rule: identifier semantics must be stable and lineage-preserving.

Decision context

- Business definition: context, baseline action, candidate action, action grid, action owner, feature stage, and permitted action set.
- Requiredness: required for every call.
- Compatibility rule: new contexts require model-risk and consumer approval.

Product terms

- Business definition: line, APRs, fees, rewards, signup bonus, promotional terms, disclosures, balance-transfer terms, and cash-advance terms.
- Requiredness: conditional on decision context and action grid.
- Compatibility rule: missing terms must trigger no-score, degraded-score, or explicitly approved defaults.

Customer state

- Business definition: bureau, internal relationship, deposits, payroll indicators where permitted, transaction history, payments, balances, utilization, service, fraud, disputes, delinquency, and closure state.
- Requiredness: context-specific.
- Compatibility rule: training-serving availability must be enforced by the feature-cut timestamp.

Scenario and assumptions

- Business definition: macro path, unemployment, rates, inflation, valuation semantics, downstream policy assumption, and scenario version.
- Requiredness: required for every value-producing call.
- Compatibility rule: assumption versions must remain replayable.

The component should reject, warn, or degrade gracefully when callers submit inputs inconsistent with the declared context. A prospect call should not silently use booked-account payment history. A line-assignment call should not score a line outside the approved feasible set. A batch request with mixed feature availability should encode row-level statuses rather than hide no-score rows in aggregate output.

C.4 Response contract and decomposition

The response should be a valuation object with decomposition and lineage. The component should not only return `npv`. At minimum it should return the expected value, horizon values, uncertainty or sensitivity, component cash-flow paths, behavioral paths, status, warnings, and replay identifiers.

Response envelope

- Business definition: request identifiers, schema version, model version, assumption bundle, scenario, timestamp, and lineage hash.
- Unit: not applicable.
- Scope: request and row.

Valuation

- Business definition: expected incremental NPV, total value where approved, horizon values, terminal value, and discount basis.
- Unit: currency.
- Scope: row and option.

Decomposition

- Business definition: incremental PPNR, NII, NIR, NIE, expected loss, capital, tax, relationship value, acquisition cost, rewards, fraud, and servicing.
- Unit: currency.
- Scope: row, option, and horizon.

Behavioral paths

- Business definition: activation, first use, repeated use, spend, payment, balance, utilization, attrition, PD, LGD, EAD, and closure.
- Unit: probability, currency, or ratio depending on field.
- Scope: row, option, and horizon.

Uncertainty and sensitivity

- Business definition: scenario range, confidence or posterior interval where approved, downside metrics, and major driver sensitivities.
- Unit: mixed, with each field typed explicitly.
- Scope: row, option, and segment.

Status and warnings

- Business definition: score status, degraded inputs, out-of-domain flags, unavailable features, validation limitations, and approved downstream handling.
- Unit: enumerated status and text codes.
- Scope: request, row, and option.

The decomposition must sum to the reported NPV within a documented tolerance. If the response includes both new-card spend and all-bank incremental spend, the field names should make the distinction impossible to miss. A report that shows high `new_card_spend_lift` but low `all_bank_incremental_spend_lift` should not be optimized as if the two were the same.

C.5 Score-status semantics

Score status is part of the business contract. A no-score because a use case is unapproved is different from a timeout. A degraded score due to missing merchant-category history is different from an out-of-domain account state. The component should support request-level, row/account-level, and option-level statuses.

Scored

- Trigger: required inputs are present, the use case is approved, the population is in domain, and model execution succeeds.
- Scope: request, row, or option.
- Downstream handling: caller may use the value subject to its own policy and approval.

Degraded

- Trigger: a non-critical feature is missing, stale, imputed, or supplied by a lower-quality fallback.
- Scope: row or option.
- Downstream handling: caller may use the value only if the use-case policy permits degraded values; warnings must be logged.

No-score

- Trigger: required feature missing, unapproved use case, unsupported context, invalid action set, or unavailable valuation semantics.
- Scope: row, option, or sometimes request.
- Downstream handling: caller must use an approved fallback, suppress, or route to manual review. Missing value must not be treated as zero NPV.

Out-of-domain

- Trigger: population, product, macro state, behavior, or policy state is outside the approved model domain.
- Scope: row or option.
- Downstream handling: caller should suppress, route to manual review, or use a conservative fallback according to policy.

Manual review

- Trigger: stakes, conflicts, or policy exceptions exceed thresholds for automated use.
- Scope: row or option.
- Downstream handling: caller routes according to the approved workflow; the component records the reason.

Technical failure

- Trigger: timeout, service error, schema parse failure, or dependency failure.
- Scope: request or batch partition.
- Downstream handling: retry or use technical fallback; do not interpret this as a governed business no-score.

Mixed-status batch and option-grid responses must be explicit. A batch may succeed at request level while individual rows no-score. An option grid may score one line amount and no-score another because the second is outside the approved action set. Downstream systems must receive these statuses, not only a filtered list of successful scores.

C.6 Operating modes and integration patterns

The component should support three modes. Each mode uses the same valuation semantics but has different latency, feature freshness, idempotency, fallback, and replay requirements.

Batch scoring

- Typical consumers: acquisition, prescreen, campaign planning, shadow scoring, and portfolio analytics.
- Request shape: many rows, frozen feature snapshot, and one or more offer options.
- Failure and fallback: mixed row statuses are expected; legacy or conservative fallback is allowed only if approved.
- Audit and replay: full input snapshot, model and assumption versions, row lineage, and run manifest.

Online scoring

- Typical consumers: selected underwriting, servicing, and account-management flows.
- Request shape: one customer, account, application, or small option set.
- Failure and fallback: latency timeout, no-score, manual review, or caller-approved fallback.
- Audit and replay: request/response log, feature-cut timestamp, and deterministic replay where possible.

Option-grid scoring

- Typical consumers: line, offer, promotion, conversion, and treatment menus.
- Request shape: one scored unit with a feasible action grid.
- Failure and fallback: option-level statuses; invalid options should not invalidate valid options.

- Audit and replay: option-level lineage, decomposition, policy constraints, and chosen-action record stored by caller.

Idempotency should be defined by request identifier, feature-cut timestamp, model version, schema version, valuation semantics, scenario, and downstream policy assumption. Replay requires retained feature snapshots or reproducible feature queries, retained code artifacts, model artifacts, and assumption bundles. Where exact replay is impossible because a source system cannot recreate historical inputs, the limitation must be logged and accepted by model-risk and data owners before production use.

D Reference Governance, Migration, and First Production Slice

This appendix records the operating controls and evidence records that support the phased replacement. It is intentionally more checklist-like than the main body because these items have to become integration tests, monitoring reports, model-risk evidence, and experiment manifests.

Migration should be governed as replacement of a production component, not as a greenfield model launch. The bank should run the new component in shadow, compare rankings and decisions against the current component, reconcile aggregate and component economics, analyze disagreements, run downstream contract tests, and define rollback triggers before any production consumer relies on the new value.

D.1 Validation and governance gates

Validation should separate predictive accuracy, causal lift, finance reconciliation, data-quality monitoring, and use-case approval. A model can have good response AUC and still be invalid for NPV if it fails cannibalization, loss, or cost validation. A model can calibrate on booked accounts and still be invalid for prospect targeting if it uses variables not available at targeting time.

Predictive validation

- Primary owners: model owner and model-risk management.
- Veto diagnostics: poor calibration or rank stability for spend, balances, payments, attrition, PD, LGD, EAD, costs, or terminal value.
- Reapproval trigger: new population, product, model, feature set, or material drift.

Causal and treatment validation

- Primary owners: marketing analytics, experimentation, and model risk.
- Veto diagnostics: failed holdout lift, weak overlap, unaddressed selection, unstable CATE estimates, or unidentified cannibalization.
- Reapproval trigger: new offer, channel, action set, or counterfactual.

Finance reconciliation

- Primary owners: finance, treasury, and risk capital.
- Veto diagnostics: PPNR, rewards, funding, capital, tax, losses, or cost decomposition fails tie-out tolerance.
- Reapproval trigger: new valuation semantics, accounting policy, FTP curve, capital method, or cost allocation.

Data and integration monitoring

- Primary owners: data engineering, platform, and model owner.
- Veto diagnostics: missingness, staleness, schema breaks, feature leakage, no-score rate, degraded-score rate, or out-of-domain rate.
- Reapproval trigger: source-system change, schema change, feature pipeline change, or material quality drift.

Use-case approval

- Primary owners: business owner, compliance, model risk, and legal where needed.
- Veto diagnostics: unapproved population, action set, fairness concern, adverse-action mismatch, or unsupported downstream use.
- Reapproval trigger: any new caller, decision context, operating mode, or customer-impacting use.

Experimental admissibility

- Primary owners: business owner, compliance, fair-lending, privacy, legal where needed, and complaints/servicing owner.
- Veto diagnostics: inadmissible control, unapproved contact/disclosure path, impermissible data use, complaint spike, servicing harm, vulnerable-customer concern, or reputation stop trigger.
- Reapproval trigger: any new experiment family, population, offer, credit term, channel, suppression rule, or customer-impact threshold.

NPV explanations are not automatically adverse-action reasons or fair-lending explanations. Caller systems that use NPV in credit decisions must maintain their own approved compliance and adverse-action processes.

D.2 Migration artifacts and cutover gates

The first production slice should be a controlled acquisition or prescreen batch use case. The slice should have one consumer, one offer family, one valuation semantics bundle, frozen input snapshots, campaign holdouts, and shadow scoring against the current component. The first release should not try to cover underwriting, line assignment, existing-account treatments, finance analysis, scenario planning, and account management at once.

Current-consumer inventory

- Purpose: identify every consumer, field, threshold, fallback, report, owner, and latency class that depends on the current component.
- Owners: product, technology, and model owner.

Legacy schema and output map

- Purpose: map old fields to new fields, compatibility fields, retired fields, and downstream contract changes.
- Owners: data engineering and consumer owners.

Valuation-definition comparison

- Purpose: compare old and new horizon, discounting, funding, capital, tax, cost, reward, terminal value, scenario, and policy assumptions.
- Owners: finance, risk, and model owner.

Shadow-score run manifest

- Purpose: preserve data snapshot, command or job, model versions, assumption bundle, scenario, random seeds where relevant, and output paths.
- Owners: model owner and platform.

Ranking and decision-impact comparison

- Purpose: quantify rank correlation, top-decile overlap, decision movement, budget movement, and segment impacts.
- Owners: business analytics and model owner.

Segment aggregate tie-out

- Purpose: compare aggregate NPV and drivers by product, channel, vintage, months-on-book, risk band, geography, and relationship depth.
- Owners: finance, risk, and product.

Decomposition tie-out

- Purpose: verify that PPNR, losses, funding, capital, rewards, fraud, servicing, acquisition cost, tax, and relationship value sum to NPV.
- Owners: finance and model owner.

Disagreement register

- Purpose: document material old-versus-new disagreements and classify them as expected improvements, data issues, assumption differences, or blockers.
- Owners: model owner, business, and model risk.

Downstream contract tests

- Purpose: confirm schema validation, mixed statuses, option-level results, fallback, replay, and rollback interface.
- Owners: platform and consumer system owners.

Rollback plan

- Purpose: define triggers, owner, interface, timing, and data consequences for reverting to legacy or conservative fallback.
- Owners: business owner, technology, and model risk.

Cutover criteria should be numerical and use-case specific. They should include acceptable segment-level aggregate tie-out tolerances, decomposition residual tolerance, ranking and decision-impact thresholds, disagreement-rate thresholds and required disposition, contract-test pass requirements, no-score and degraded-score thresholds, and named rollback triggers. This proposal does not hard-code the tolerances because the right values depend on the legacy component, action stakes, population size, and regulatory sensitivity.

D.3 What this proposal does not approve

This proposal does not by itself approve any production use case beyond a shadow-tested first slice. It does not approve reuse of the score in underwriting, line assignment, account closure, collections, adverse-action reasoning, official finance reporting, CECL, or stress testing. Each such use requires its own decision-context contract, valuation semantics, validation evidence, governance approval, compliance review where applicable, and consumer-system integration test.

D.4 Experimental evidence contract

The proposal should also require an experiment ledger. That ledger is the bridge between the experimental program and the NPV component. It records what was randomized, what was held out, what population was eligible, what counterfactual was used, what horizon was measured, and which output label the result may support. The ledger is not meant to replace the analysis memo for an experiment. It is the minimum machine-readable contract that prevents a caller from converting a response lift, first-use lift, or short-horizon spend result into lifetime causal NPV without the identification, power, horizon, and reconciliation evidence required in Section 6.3.

The minimum evidence contract has five human-readable parts.

Identity and decision context. The ledger must record the experiment name, decision context, treatment, baseline, feasible option set, and admissible action set. This is where the estimand is defined. If the action tested is not an action the bank can actually use, the evidence may be interesting but it is not production evidence for the component.

Population and information timing. The ledger must record eligibility rules, exclusions, randomization unit, assignment probabilities, treatment receipt, take-up, and feature-cut timestamp. It must also record the contact, disclosure, opt-out, and control-cell treatment. These fields prevent the analysis from conditioning on post-treatment variables or transporting evidence beyond the tested population.

Identification, power, and horizon. The ledger must state whether the design is a holdout, encouragement, RD, DiD, event study, adjusted observational design, or scenario bridge. It must name the primary estimand, estimand class, planned power, MDE, outcome horizon, valuation horizon, and tail bridge. This prevents a response-powered or short-horizon test from being promoted as lifetime NPV evidence.

Diagnostics and guardrails. The ledger must specify veto diagnostics: broken randomization, weak overlap, failed pre-trends, leakage, no holdout, bad finance tie-out, contamination, or interference. It must also specify customer-impact and reputation guardrails: complaints, disputes, servicing harm, hardship effects, contact fatigue, branch/social escalations, and pause or stop rules.

Fair-lending and adverse-action boundaries must be explicit whenever the action affects credit availability, terms, discouragement, or treatment.

Allowed use and artifacts. The ledger must name the NPV decomposition, finance/risk reconciliation, allowed-use label, confounders addressed and not addressed, transport boundary, what is not concluded, and artifact location. This is the difference between a useful experiment and a result that gets misused: the component can consume causal incremental NPV, policy-conditional expected NPV, diagnostic NPV, or scenario-only evidence, but it must not blur those labels.

For the first production-aligned slice, the evidence contract should be written first for the acquisition or prescreen holdout and then for the existing-client cannibalization holdout. The latter is the most valuable worked example because it directly tests the difference between new-card spend and all-bank incremental value.

D.5 Worked contract: acquisition or prescreen contact-suppress

The first worked contract is the safest initial production-adjacent design: a randomized contact-versus-suppress holdout in an acquisition or prescreen population. The business question is not “who responds?” It is whether contacting the eligible population with the approved offer creates positive incremental value relative to suppressing contact, after campaign cost, approval selection, activation, spend, balances, rewards, losses, funding, capital, servicing, fraud, and terminal value are included. This design is also the cleanest way to build the first shadow-scored replacement component, because every eligible prospect has a known assignment and the analytic base can preserve nonresponders, nonapplicants, declines, approvals, bookings, inactive accounts, and charge-offs.

“Safest” here does not mean compliance-free. Before randomization, the bank must define the eligible prospect population, the admissible contact and suppression rules, the permissible data sources, the prescreen or solicitation framework where applicable, opt-out handling, required disclosures, contact cadence, and fair-lending review scope. The control group is a valid counterfactual only if suppressing contact is itself an approved customer treatment for that population and channel. The design should therefore have an admissibility memo before launch, not merely a causal analysis memo after launch.

The primary estimand is the intent-to-treat acquisition value:

$$\tau_{\text{NPV,ITT}}^{\text{acq}} = \mathbb{E}[\text{NPV}_i(\text{contact}, o; d_{\text{acq}}, s, \pi) - \text{NPV}_i(\text{suppress}; d_{\text{acq}}, s, \pi) \mid i \in \mathcal{P}_{\text{eligible}}], \quad (200)$$

where o is the approved offer cell. If the campaign has multiple offer cells, the design should estimate contrasts against suppress and against the business-as-usual offer cell. If assignment only encourages application or activation, the component may also estimate a complier effect using Equation (165), but the output must label it as such.

The constructed observed-horizon value should be built for every assigned unit, not only booked accounts:

$$\begin{aligned} \widetilde{V}_{i, H_{\text{obs}}}^{\text{acq}} = & -C_i^{\text{contact}} - C_i^{\text{acq}} + \sum_{h=0}^{H_{\text{obs}}} \delta_h \{ \widetilde{\text{PPNR}}_{ih} - \widetilde{\text{EL}}_{ih} - \widetilde{\text{Kchg}}_{ih} - \widetilde{\text{ServCost}}_{ih} \\ & - \widetilde{\text{FraudCost}}_{ih} - \widetilde{\text{RewardCost}}_{ih} \}. \end{aligned} \quad (201)$$

Nonresponders and nonbooked prospects receive zero realized card cash flows inside the observed window, but they remain in the denominator and retain contact costs where incurred. Declines remain in the denominator because the campaign decision generated applications and underwriting workload. Losses that have not seasoned enough inside H_{obs} should be labeled as modeled loss

bridge, not realized loss. The tail from H_{obs} to H_{val} is a forecast bridge and should be reported separately.

The required input records are campaign eligibility, assignment flag, assignment probability, suppression or holdout reason, contact channel, creative, offer cell, product terms, contact timestamp, prescreen score or selection score retained as of the decision time, application records, approval/decline/refer records, assigned line and APR, booking, activation, first use, transactions, balances, payments, delinquency, charge-off, recoveries, rewards, acquisition cost, servicing, fraud, and macro vintage. All variables in the primary analysis must be available at or before the feature-cut timestamp, except outcomes measured after assignment.

The analysis specification should be fixed before outcome inspection. A typical primary model is Equation (178), with treatment assignment as the main coefficient, pre-assignment covariates for precision, and standard errors clustered at the randomization or interference unit. If assignment probabilities vary by stratum, the experiment ledger should store the probabilities and the analysis should include the corresponding design-based or stratified estimator. The minimum detectable effect should be computed for observed-horizon NPV, not only for response or application rate. If the campaign is powered only for response, it can still supply funnel evidence, but it should not be promoted as an NPV-positive experiment.

Veto diagnostics are broken randomization, lost or overwritten holdout flags, manual suppression correlated with value, re-contact contamination, household or relationship interference, channel capacity changes that differentially affect treatment and control, post-treatment leakage in covariates, missing offer-term lineage, material imbalance after randomization, and failure to reconcile value construction to approved finance and risk definitions. Regulatory and customer-impact vetoes should be recorded in the same ledger: unapproved prospect source, unresolved opt-out treatment, changed disclosure or creative, impermissible data feed, unapproved suppression path, complaint or dispute spike, servicing escalation, fair-lending monitoring concern, or reputation stop trigger. Explanatory diagnostics include response, application, approval, booking, activation, first use, 90-day spend, 180-day spend, balance build, payment rate, utilization, delinquency, rewards cost, and attrition. These diagnostics are useful precisely because they explain *why* NPV moved; they should not replace the NPV estimand.

If the design passes, the resulting evidence can update the response, application, booking, activation, early-spend, early-balance, campaign-cost, reward-cost, and observed-horizon value modules for the tested population, offer family, channel, macro regime, and downstream-policy bundle. It can support causal incremental NPV for the same decision context only if the tail bridge, loss bridge, funding/capital convention, and finance/risk tie-outs also pass. It does not conclude that the same offer is positive NPV for a different channel, a different product, an existing-client cross-sell population, a line-assignment decision, an underwriting decision, or a future downstream policy.

D.6 Worked contract: existing-client new-card cannibalization

The second worked contract addresses the main source of false optimism in existing-client new-card valuation. A customer who already has a bank card, debit card, deposit relationship, or other payment relationship can generate large new-card spend after an offer while creating little incremental all-bank value. The experiment therefore cannot stop at new-card activation or new-card sales volume. It must estimate all-bank spend and contribution under offer and no-offer paths.

The preferred design randomizes eligible existing clients into no-offer and one or more approved

new-card offer cells. The primary spend estimands are:

$$\tau_{S,H}^{\text{newcard}} = \mathbb{E}[S_{i,H}^{\text{newcard}}(o) - S_{i,H}^{\text{newcard}}(\text{no offer})], \quad (202)$$

$$\tau_{S,H}^{\text{allbank}} = \mathbb{E}[S_{i,H}^{\text{allbank}}(o) - S_{i,H}^{\text{allbank}}(\text{no offer})], \quad (203)$$

$$\tau_{S,H}^{\text{oldcard}} = \mathbb{E}[S_{i,H}^{\text{oldbankcard}}(o) - S_{i,H}^{\text{oldbankcard}}(\text{no offer})], \quad (204)$$

$$\tau_{S,H}^{\text{debit}} = \mathbb{E}[S_{i,H}^{\text{bankdebit}}(o) - S_{i,H}^{\text{bankdebit}}(\text{no offer})], \quad (205)$$

with the same population and horizon. A positive $\tau_{S,H}^{\text{newcard}}$ accompanied by a negative $\tau_{S,H}^{\text{oldcard}}$ or $\tau_{S,H}^{\text{debit}}$ is direct evidence of internal substitution. The new-card spend lift should therefore enter the NPV bridge only after the all-bank and product-source decomposition is applied.

The NPV estimand is:

$$\tau_{\text{NPV,ITT}}^{\text{xsell}} = \mathbb{E}[\text{NPV}_i(\text{newcard offer}, o; d_{\text{xsell}}, s, \pi) - \text{NPV}_i(\text{no offer}; d_{\text{xsell}}, s, \pi) \mid i \in \mathcal{P}_{\text{xsell}}]. \quad (206)$$

The cash-flow bridge should use incremental all-bank PPNR and cost, not new-card activity alone. Old-card interchange lost to cannibalization, debit interchange lost where applicable, balance-transfer payoff behavior, reward cost, promotional APR economics, balance build, EAD, PD, LGD, funding, capital, servicing, fraud, disputes, and terminal value should all be represented. Relationship value is excluded unless the same design or a companion design identifies a causal spillover to deposits, debit activity, or other products.

The required records include customer and household linkage, old-card transactions and balances, new-card offer assignment, new-card booking and activation, debit transactions, deposit flows where permitted, card-on-file and digital-wallet provisioning where available, balance-transfer requests, payoff records, bureau refreshes or tradeline aggregates where permitted, merchant category, rewards earn and burn, promotional terms, campaign and acquisition cost, servicing and dispute records, fraud records, delinquency, charge-off and recovery, and macro vintage. If competitor-card spend is not directly observed, it should be treated as partially observed and bounded using balance transfers, payoff behavior, bureau aggregates where available, merchant/network aggregates where permitted, account-aggregation samples, or scenario assumptions. It should not be labeled observed.

Because this experiment uses existing-client relationships, the data-use and customer-impact boundary is more complicated than in a pure prospect campaign. The contract should specify whether deposit, debit, household, servicing, complaint, and digital-engagement fields are permitted for assignment, analysis, aggregate monitoring, or finance-only reconciliation. It should also identify products or situations excluded from randomization, such as hardship, active disputes, sensitive servicing states, complaint escalation, fraud investigation, deceased or protected-status handling, or other bank-defined exclusions. A treatment that appears to increase all-bank contribution by moving balances or spend from one bank product to another may still be inadmissible if it creates customer confusion, avoidable fees, complaint elevation, or unfair treatment concerns.

The primary analysis should be the same randomization-preserving comparison used in the acquisition contract, but with all-bank spend and all-bank contribution as confirmatory outcomes. The analysis should be stratified by existing bank-card ownership, relationship depth, old-card activity, debit activity, risk band, reward-seeking indicators, and months-on-book where those strata were pre-specified or used in randomization. Post-offer variables such as first new-card use are outcomes or mediators, not controls for the primary causal contrast.

The key veto diagnostics are loss of no-offer controls, other simultaneous cross-sell campaigns, product migration not captured in the experiment ledger, old-card portfolio changes during the test, rewards or fee changes not shared between treatment and control where relevant, household contamination, merchant or card-on-file interventions applied unevenly, short follow-up that cannot observe

promotion decay, and failure to reconcile old-card plus new-card plus debit/deposit contribution to finance definitions. Customer and conduct vetoes include complaint or dispute increases, unexpected fee incidence, servicing escalation, opt-out or channel-preference failures, protected-class monitoring concerns where legally appropriate, and any reputation stop trigger defined before launch. A test that shows positive new-card sales volume but fails all-bank reconciliation or customer-impact admissibility should not update the causal NPV module.

If the design passes, it can update the activation, first-use, repeated-use, all-bank spend, internal cannibalization, promotion-decay, rewards-cost, balance, loss, and attrition modules for existing-client new-card offers in the tested population. It can also calibrate the cannibalization-stress sensitivity bundle in Section 8.2.2. It does not conclude that outside competitor wallet was captured unless the external wallet evidence supports that claim. It does not conclude that deposit or relationship value increased unless relationship outcomes were measured in both treatment and control and identified by the design.

E Source-Support Audit

This appendix records source-use boundaries for the proposal. It is not the main exposition. Its purpose is to distinguish what cited literature supports, what follows from project derivation, what is a proposed design convention, and what must be verified with this bank’s data before model use.

The proposal uses literature in three ways. First, it uses primary studies and standard references to justify mechanisms that the component must represent: liquidity and unemployment affect spend, credit terms affect balances, payment adoption differs from payment use, rewards can change behavior, CLV models represent lifetime activity, and causal-inference methods distinguish prediction from treatment effects. Second, it uses supervisory, accounting, and credit-risk sources to justify bank-native decomposition and reconciliation requirements. Third, it uses project derivations to connect those mechanisms to this component’s NPV equations. The literature does not provide this issuer’s activation rate, competitor-wallet capture, cannibalization rate, loss elasticity, reward elasticity, funding charge, capital charge, or relationship spillover. Those quantities are empirical bank objects.

The proposal also uses official regulatory and public-data sources as boundary-setting sources. Citations to Regulation B, Regulation Z, the Fair Credit Reporting Act permissible-purpose provisions, the UDAP statutory boundary, and the CFPB complaint database support the claim that experimental designs must record admissible populations, actions, disclosures, opt-outs, data-use limits, complaint monitoring, and customer-impact stop rules [27–29, 82, 83]. They are not used as legal advice, as approval for a specific experiment, or as evidence of any treatment effect. Final interpretation and approval remain with the bank’s legal, compliance, fair-lending, privacy, servicing, and business-control owners.

The panel-grade evidence package should therefore add four ledgers before model-governance submission. A *technical-source ledger* should record for each cited paper the inspected sections, equations, tables, appendix material, empirical design, publication status, and any retraction or erratum check. A *claim-support ledger* should map each major proposal claim to a source section, project derivation, or bank-data requirement. A *snowball ledger* should record backward and forward citation checks for foundational and direct-method sources, including important omitted papers and the reason for omission. A *model-use evidence ledger* should map each component module to its internal experiment, quasi-experiment, observational diagnostic, or scenario assumption. Until those ledgers are completed, the document is suitable as a design proposal and technical roadmap, not as a completed model-validation package.

The claim-support rule for future model work should be simple: if a statement is used to justify a model requirement, it should be backed either by checked technical literature, by bank data or an experiment, or by an explicit design convention. Literature can justify the mechanism; internal bank evidence must justify issuer-specific rates, elasticities, and cannibalization; design conventions can justify output tags, status semantics, and contract fields. Anything else should be labeled as assumption or open gap, not quietly promoted to fact.

The wallet-movement claims in this proposal are deliberately treated as project derivations and data requirements. Public literature and payment-choice evidence motivate adoption, usage, rewards, and treatment-response mechanisms, but they do not observe this issuer’s competitor-card wallet or identify the offer-specific movable fraction. Those quantities require internal anchors, experiments, quasi-experiments, or explicitly labeled scenario assumptions.

The dynamic-state-space design uses another source class: formal dynamic-modeling and bank-value references. State aggregation, MDP, POMDP, state-space, filtering, and factor-model sources support the language of state abstraction, belief state, conditional-kernel factorization, common macro shocks, and fixed-policy simulation [10, 12, 13, 30, 32, 51, 57, 68, 75, 77, 89]. Credit-card and relationship-pricing dynamic models support the relevance of state-based profitability and customer-policy analysis in nearby banking settings [61, 76]. Bank-value and CLV sources support the relevance of relationship economics [11, 33, 71, 87]. None of these sources proves this bank’s chosen state aggregation, Markov sufficiency, compressed belief vector, factorization, relationship spillover, or optimization threshold. Those remain proposal derivations, validation questions, or bank governance choices.

The claim-support ledger should be framed by reviewer questions rather than by source names. The following cards state what currently supports each claim and what remains to be proven, approved, or estimated.

Why optimize incremental value rather than response alone?

- Current support: direct-marketing profit targeting, CLV, causal treatment-effect methods, and the project synthesis in Equations (7)–(10).
- Remaining bank evidence: validate incremental NPV with holdouts or credible quasi-experiments for each use case.

Why make spend, balances, payments, and losses state dependent?

- Current support: unemployment, rebate, liquidity, credit-card limit/APR, CARD Act, and macro-shock evidence.
- Remaining bank evidence: estimate issuer-specific elasticities and monitor macro, employment, and liquidity drift.

Why separate issuance, activation, first use, repeated use, and wallet share?

- Current support: payment-choice and rewards studies plus the proposal’s adoption/use and state-transition derivations.
- Remaining bank evidence: estimate transition probabilities and treatment effects by product, channel, segment, and feature stage.

Why require identification designs for treatment-response modules?

- Current support: causal-inference, program-evaluation, IV, RD, DiD, double/debiased ML, and uplift literatures.

- Remaining bank evidence: document identification, overlap, balance, leakage controls, and allowed use for each module and decision context.

Why measure cannibalization and spend source?

- Current support: project derivations in Equations (47)–(56) and Equations (44)–(46), with causal/uplift methods supporting the identification strategy.
- Remaining bank evidence: estimate all-bank incremental spend, old-card shift, debit/deposit substitution, competitor-card shift, promotion timing, and measurement uncertainty with internal data, anchors, or experiments.

Why distinguish outside wallet, movable wallet, causal response, and NPV?

- Current support: project derivation in Equation (45) and the decision-context logic in Equation (1).
- Remaining bank evidence: tag evidence grade and uncertainty for outside-wallet and movable-wallet estimates; large outside wallet should not be promoted to captured wallet or positive NPV without causal and financial validation.

Why use state and duration modeling for lifetime value?

- Current support: CLV, customer-base, and duration literature support active-state, frequency, monetary, and attrition components.
- Remaining bank evidence: validate transactor, revolver, promotion, dormant, conversion, and charge-off paths, including terminal-value sensitivity.

Why decompose losses into PD, LGD, and EAD?

- Current support: regulatory, credit-risk, and accounting sources support PD/LGD/EAD and lifetime expected-loss framing.
- Remaining bank evidence: validate calibration, line response, vintage seasoning, principal/non-principal subledgers, and stress behavior.

Why make PPNR, rewards, funding, capital, tax, costs, and relationship value explicit?

- Current support: supervisory PPNR/stress sources, CLV contribution logic, and project cash-flow identities.
- Remaining bank evidence: finance, treasury, risk, product, and model-risk owners must approve semantics and reconciliation tolerances.

Why require valuation semantics and downstream-policy assumptions?

- Current support: project derivation from latent long-horizon NPV and dynamic policy-value literature.
- Remaining bank evidence: owners must approve horizon, discounting, funding, capital, tax, terminal value, relationship value, scenario, and policy-bundle conventions.

Why should the component value options but not execute actions?

- Current support: design convention and governance boundary supported by decision-specific counterfactual analysis.
- Remaining bank evidence: each caller must maintain its own action policy, compliance process, adverse-action process where applicable, and integration approval.

Why use public and licensed data as scenarios, controls, benchmarks, and priors?

- Current support: official public-data and LSEG/Refinitiv-source descriptions support their role as external macro, credit, market, and product-term data sources.
- Remaining bank evidence: confirm entitlements, permitted use, lineage, real-time vintages, join keys, and whether each source is a scenario, control, benchmark, prior, or causal input.

The source-support ledger should remain more granular in the model-validation package. For this proposal, the readable source map is:

Direct marketing and CLV foundations

- Sources: Bult and Wansbeek [21], Berger and Nasr [11], Rust et al. [71], and Venkatesan and Kumar [87].
- Use in this paper: support campaign expected-profit logic and discounted-contribution framing.
- Boundary: the credit-card NPV equation here is a bank-specific extension with losses, capital, funding, fraud, rewards, and cannibalization terms.

Household finance, liquidity, and macro shock evidence

- Sources: Ganong and Noel [41], Johnson et al. [50], Gross and Souleles [42], Agarwal et al. [3], and Baker et al. [8].
- Use in this paper: justify state-dependent spend, balance, payment, fee, disclosure, revenue-channel, and macro/category-mix modules.
- Boundary: these sources do not supply this bank's card routing share, loss response, attrition response, or customer-level NPV lift.

Payment choice, rewards, and behavioral usage

- Sources: Koulayev et al. [53], Agarwal et al. [2], and Prelec and Simester [67].
- Use in this paper: support the distinction between adoption, activation, conditional use, rewards response, and payment-medium effects.
- Boundary: they do not measure this issuer's wallet capture, cannibalization, or all-bank incremental value.

Customer-base, duration, and attrition models

- Sources: Schmittlein et al. [72], Fader et al. [35], Fader et al. [36], and Bolton [20].
- Use in this paper: support frequency, recency, monetary, active-state, duration, dormancy, and terminal-value components.
- Boundary: they do not by themselves supply card-specific loss, capital, funding, rewards, fraud, or regulatory accounting economics.

Credit-risk, accounting, and supervisory decomposition

- Sources: Basel Committee on Banking Supervision [9], Financial Accounting Standards Board [39], Board of Governors of the Federal Reserve System [15], Board of Governors of the Federal Reserve System [16], Hand and Henley [43], Thomas [79], Lessmann et al. [59], and Schuermann [73].
- Use in this paper: support PD/LGD/EAD, expected-loss framing, PPNR decomposition, stress/scenario language, and the need to separate balances, revenues, costs, and losses.
- Boundary: these are architecture and decomposition supports, not customer-acquisition estimators.

Causal, uplift, and program-evaluation methods

- Sources: Rosenbaum and Rubin [69], Lo [62], Wager and Athey [88], Chernozhukov et al. [25], Imbens and Rubin [47], Angrist and Pischke [5], Imbens and Wooldridge [48], Hernán and Robins [45], Angrist et al. [6], Lee and Lemieux [58], Callaway and Sant’Anna [22], Sun and Abraham [78], Heckman [44], Arellano and Bond [7], Blundell and Bond [14], and Manski [65].
- Use in this paper: support estimands, identification assumptions, overlap, sample selection, dynamic panel caution, heterogeneous effects, and treatment rules by expected payoff.
- Boundary: method papers do not provide bank-specific treatment effects.

Dynamic decision and policy-value methods

- Sources: Puterman [68], Murphy [66], and Rust [70], with additional POMDP, filtering, and OPE sources cited in the dynamic section.
- Use in this paper: support representing acquisition, underwriting, line assignment, and account management as linked sequential decisions.
- Boundary: they do not prove this bank’s state aggregation, belief compression, or safe optimization threshold.

Public and licensed data sources

- Sources: Board of Governors of the Federal Reserve System [17], Board of Governors of the Federal Reserve System [18], Board of Governors of the Federal Reserve System [19], Federal Reserve Bank of New York [37], U.S. Bureau of Labor Statistics [85], U.S. Bureau of Economic Analysis [84], U.S. Census Bureau [86], Federal Reserve Bank of St. Louis [38], Consumer Financial Protection Bureau [26], London Stock Exchange Group [64], and London Stock Exchange Group [63].
- Use in this paper: support external scenario construction, controls, benchmarking, real-time-vintage backtesting, competitive product-term context, and licensed market/macro indicators.
- Boundary: they do not replace internal causal evidence for activation, spend response, cannibalization, PD, LGD, EAD, reward elasticity, or relationship spillover.

No paper in this survey is used to conclude a universal activation rate, spend elasticity, or NPV parameter. The activation and cannibalization parameters for a large U.S. retail bank should be estimated from internal experiments or credible quasi-experiments.

Sensitive Source Boundaries

The most likely citation challenge is not that the cited papers are irrelevant. It is that some papers are easy to overextend. The boundaries below state what the proposal may and may not infer from the sources most exposed to that risk. The main body should follow these boundaries. If future model documentation needs a stronger claim, the claim should be supported by checked technical text, issuer-specific data, or a project derivation.

Ganong and Noel [41] and Johnson et al. [50]

- Allowed use: motivate income, unemployment, transfer, liquidity, and timing variables in spend models; support heterogeneous short-run spending-response mechanisms.
- Forbidden or not-yet-supported use: do not identify this bank’s card routing share, payment-rate response, loss response, attrition response, or customer-level NPV lift.

Gross and Souleles [42]

- Allowed use: support credit-card balance response to limits and interest rates, especially for liquidity-constrained accounts; motivate line/APR as policy variables in balance models.
- Forbidden or not-yet-supported use: does not by itself identify full NPV, expected loss, funding, capital, activation, or offer-specific acquisition response.

Agarwal et al. [3]

- Allowed use: support fee, disclosure, repayment-allocation, and revenue/payment-channel relevance; motivate regulated product terms as governed economic objects.
- Forbidden or not-yet-supported use: does not directly support broad claims about activation, rewards liability, servicing costs, attrition, or complete transition-kernel behavior.

Agarwal et al. [2]

- Allowed use: support booked-account reward effects on spend, debt, and reactivation-like outcomes; motivate reward treatment modules and reward-cost accounting.
- Forbidden or not-yet-supported use: does not supply a universal activation parameter or direct evidence for new-card acquisition-stage activation without issuer-specific validation.

Koulayev et al. [53] and Prelec and Simester [67]

- Allowed use: support adoption-versus-use distinctions and payment-instrument behavioral mechanisms.
- Forbidden or not-yet-supported use: do not measure this issuer’s wallet capture, cannibalization, or profitable all-bank incremental value.

Public and licensed external data

- Allowed use: support scenarios, controls, benchmarks, priors, monitoring, and real-time-vintage discipline.
- Forbidden or not-yet-supported use: do not replace internal causal evidence for activation, spend response, cannibalization, PD, LGD, EAD, reward elasticity, or relationship spillover.

CLV, customer-base, and duration sources

- Allowed use: support discounted contribution, active-state, frequency, monetary, and duration modeling components.
- Forbidden or not-yet-supported use: do not by themselves supply card-specific losses, funding, capital, rewards, fraud, cannibalization, or regulatory accounting economics.

Causal-inference method sources

- Allowed use: support estimand definitions, identification assumptions, diagnostics, and evidence grades.
- Forbidden or not-yet-supported use: do not provide bank-specific treatment effects; valid design and bank data are still required.

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